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The abstracts represent both oral communications and posters, and are listed alphabetically by author

Morphological and Immunohistochemical Study of the Zebrafish (*Danio rerio*) Oral Cavity

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The oral cavity in vertebrates plays a very important role in food intake mechanics and also contains chemical sensors for food and environmental water. Comparative studies revealed morphological and functional inter-specific differences that can be correlated to evolutionary events and might represent the adaptation to environmental conditions present in different habitats (Iwasaki, 2002). Particularly these events are strictly dependent on the food intake mechanisms determining the success of the adaptation of vertebrates to their environment and therefore their survival (Roth and Wake, 1989). Since data about the morphology of the zebrafish oral cavity are scarce the aim of the present research was to study it by light, scanning and transmission electron microscopy and immunohistochemistry for some growth factor receptors, i.e. *trkA* and *trkB*. The zebrafish was chosen because as a vertebrate it has valuable features as a model for study of vertebrate development (Withfield, 2002). Adult zebrafishes were anesthetized with MS222 and the samples were taken and routinely processed for light and electron microscopy and for indirect peroxidase immunohistochemistry. The topographical and ultra-structural study demonstrated the presence of three different zones. The outer zone is characterized orally by the presence of a papilla and by an epithelium with numerous taste buds that are chemosensory endorgans consisting of modified epithelial cells. Their function is to sample potential food and selecting it according to its edibility (Hansen et al., 2002). Both in upper and lower intermediate zones, the presence of internal valves with a semi-lunar shape was evident, probably able to regulate the mouth closure. The most internal lower zone corresponds to the tongue that in zebrafish is only a thickening of the pavement mucosa with a stratified epithelium. Immunohistochemistry demonstrated both in upper and lower zones the presence of numerous taste buds that were immunoreactive to *TrkA* and *TrkB*-like neurotrophin receptors, confirming the role of these receptors in taste buds sensory cells (Germanà et al., 2004). The present results give a contribution to the

morphological knowledge of the zebrafish oral cavity, an animal model commonly used for developmental studies of vertebrates.

Biometrical Studies on the Scleral Shelf in Some Domestic Animals

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This work was carried out on the scleral shelf of the eyeball of the donkey, cow, buffalo, camel, sheep, goat, pig and dog. The results of this work shows that the length of the shelf varies in different regions in all studied animals. The dorsal shelf is the longest followed by ventral one, the lateral shelf is the smallest followed by the medial one. The camel has the longest dorsal and ventral shelf followed by donkey, while the buffalo and cow have the smallest ones among the large animals. Among the studied small animals, the dog has the smallest dorsal and ventral shelf followed by pig. However, the small ruminants have the longest ones. On the other hand, the medial and lateral shelves are longest in donkey and smallest in camel. The dog and the small ruminants have the longest medial and lateral shelf of the studied small animals. The camel has the highest value of dorsal and ventral shelf and lowest value in the medial and lateral ones. Therefore the outline of the outer surface of the cornea is highly oval, the reverse results were observed in donkey; in which the cornea has highest degree of circularity followed by cow and buffalo.

Early Embryonic Development of the Camel Lumbar Spinal Cord Segment

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The lumbar spinal cord segment of the camel embryo at CVRL 2.4 to 28 cm was examined. Major changes are occurring in the organization of the lumbar spinal cord

segments during this early developmental period. At the CVRL 2.4, 2.7 and 3.6 cm the three primary layers, ependymal cells layer, mantle cells layer, marginal cells layer in the developing lumbar spinal cord segment were demonstrated. The mantle layer is the first to show striking differentiation, while the marginal layer is represented by thin outer rim. Proliferation and differentiation of the neuroepithelial cells in the developing spinal cord produce the thick lateral walls, thin roof and floor plates. The spinal ganglion and dorsal root of the spinal nerve are differentiated. At 2.7 cm CVRL differential thickening of the lateral walls produces a shallow longitudinal groove called *sulcus limitans*, which separates the dorsal part (alar plate) from ventral part (basal plate). The ventral root of the spinal nerve, the spinal cord and ganglion are embedded in loose mesenchyme, which tends to differentiate into spinal meninges. At 3.6 cm CVRL the basal plate, which is the future ventral gray horn, seem to be quite voluminous and the dorsal and ventral roots unite to form the beginning of the spinal nerve. At 5.5 cm CVRL the alar plates enlarge forming the dorsal septum. At 8.4 cm to 10.5 cm CVRL the basal plates enlarge, and bulge ventrally on each side of the midline producing the future ventral medium fissure, and the white and gray matters can be recognized. At 28 cm CVRL the lumen of the spinal cord is differentiated into the central canal bounded dorsally and ventrally by dorsal and ventral gray commissures, and therefore the gray matter takes the appearance of a butterfly. The lumbar spinal nerve and their roots are well distinguished.

Vascularization and Production of Mitogenic Factor(S) in Domestic Cat (*Felis catus*) Testes: Preliminary Results

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Some male seasonal breeders are known to undergo testicular growth and regression throughout the year. The periodic and fast physiological development and regression processes, with simultaneous fast changes in blood flow, appears to be stimulated by angiogenic substances and inhibited by anti-angiogenic factors. The objective of this study was to evaluate the effect of season on microvascularization and production of angiogenic factors by testicular tissue, using the male cat as a model. Monthly, testes were collected *post mortem* from domestic stray cats, for histology and for tissue culture. Testis explants were cultured for 18h and conditioned

media were tested for their ability to stimulate mitogenesis of bovine aortic endothelial cells (BAEC). Vascular endothelial growth factor (VEGF) was used as a positive control. Proliferative response of BAEC to samples was evaluated by determining the number of cells in each well using a Neubauer chamber. Vascular density was assessed by a computerized image analysis ('Scion Image, NIH, USA') based on the total histological area occupied by the vascular lumen. Percentage data subjected to arcsine transformation were analysed by one-way ANOVA and post-comparison tests. There was a significant increase in microvascular areas in November ($P < 0.001$), when compared to the remaining months (LSD test). All cat testicular tissue showed the capability to increase BAEC proliferation, when compared to negative controls, throughout the study. However, there was an increase in BAEC mitogenesis in February ($P < 0.05$) and November ($P < 0.001$), when compared to the other months. Our data show that cat testicular tissue from different periods of the year stimulates angiogenic factor(s). Furthermore, changes in angiogenesis may play a role on vascular growth and regression of the testes during the breeding and non-breeding season in the male cat.

Leishmaniasis in *Canis familiaris* – a Report About Microvascular and Cellular Architecture of the Infected Spleen

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Leishmaniasis are a globally widespread group of parasitic diseases from the group of anthroponoses. They are caused by an intracellular protozoan parasite that causes a wide spectrum of diseases in humans and dogs worldwide. In fact, the dog is considered one of the most important reservoir. The spleen is one of the several haematopoietic and immunocompetent organs involved. As this organ is a blood filter, the authors looked for the expression of this infection over the microvascular environment and modifications of the spleen cell population related to immunological responses to this parasitic condition.

Tools: Scanning electronic microscopy over intact tissue and corrosion casts, transmission electronic microscopy, histology and immunohistochemistry (MAC 387 and CD3).

Results: Our results show three important modifications concerning the microvascular architecture of the spleen when compared to the normal pattern. A striking scarcity

of the sinusoidal system sheath that surrounds the central arteries of the white pulp, a huge development of pulp venules and veins, differentiation of typical features of high endothelial venular cells.

Remarks: According to our results it seems that independent of the virulence of the parasite involved and the kind of immunity prevalent in a particular host, the spleen develops blood dynamic conditions that allows a sluggish blood flow so that cells involved in immunological processes can proliferate and differentiate and it also contributes to the trapping of lymphocytes to the area through the differentiation of typical characteristics of HEV endothelial cells.

Microvascularization of the Third Eyelid in Dogs

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The third eyelid protects the eye, contribute to the aqueous portion of the pre-ocular film and its distribution over the corneal surface, as well as removal of foreign material from the anterior surface of the eye. It is rather vulnerable during fights and clinicians often have to perform surgery for third eyelid neoplasia. In dogs, this lid has a triangular shape, being widest at its free margin. It is vascularized by branches of the malar artery. Being of such importance as an adnexa of the eye, the authors concentrated their attention on its vascularization, which is not well defined.

Methods: Scanning Electronic Microscopy of vascular corrosion casts and intact tissue.

Observations: It was quite interesting to notice the presence of two patterns of vascularization inside the third eyelid. One can say that there are two segmental levels of vessels. In one of these there is a predominance of arteries and veins, while in the other a meshwork of third order arteries, capillaries and post-capillary venules are present. In fact, the branch of the malar artery that reaches the base of the third eyelid divides into smaller branches that cross almost the complete length of the lid giving only a few collateral vessels, in a monopodical way, and only near the free margin of the lid is possible to observe dichotomical ramifications that became rather anastomotic (being anastomoses of type I and II are present) and contribute to the microvascular meshwork. This microvascular net is drained by post-capillary venules and veins that empty as a brush into bigger veins that occupy the same segmental level as the third eyelid artery.

Remarks: This uncommon angioarchitecture is probably an adaptation to the fact that in the medial canthus of the eye this lid has a smaller space and the way of obtaining an efficient vascularization is by going straight to the top of the lid.

CT and Cross-sectional Anatomy of the Normal Head of the Loggerhead Sea Turtle (*Caretta caretta*)

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The objective of this study is to describe the CT anatomy of the normal loggerhead sea turtle (*Caretta caretta*) head using three loggerhead sea turtle heads. CT imaging was performed using the following parameters: K.v.: 120 mAs: 220, and a soft-tissue window (WW 1880, WL 465): Transverse and sagittal CT images were obtained. Detailed anatomy of the head was acquired with the sagittal and transverse series. The heads were frozen and then sectioned using an electric saw, to compare them with the CT images. In CT, the grey scale is directly related to the radiation attenuation of the head structures. The skull and hyoid bones, and the lower jaw were easily identifiable due to the high CT density image compared to adjacent or surrounding structures such as the respiratory system, oral cavity, oesophagus and cranial cavity that appeared as a low opacity. The nervous structures, salt gland, eyeball, ramphoteca structure and different muscles of the head had an intermediate CT density and appeared grey. CT images of the loggerhead sea turtle head provided excellent detail of clinically relevant anatomy and correlated well with corresponding gross specimens. CT of the head has considerable advantages over other techniques: CT provides excellent spatial resolution and good discrimination between bone and soft tissue, and the structures are viewed without superimposition. The planimetric or sectional anatomy of the head allows a correct morphologic and topographic evaluation of the anatomic structures, which is a useful tool for the identification of the CT images. With developing technology, CT imaging may soon become more readily available for exotic animals imaging. In the same way, we consider it quite useful to be able to establish some references on head, in order to scan only selected parts during a clinical or experimental approach. The information presented in this communication should serve as an initial reference to evaluate CT images of the loggerhead sea turtle head and to assist interpretation of lesions of this region.

Anatomy of the Thorax Structures in a Newborn Dromedary Camel by Computed Tomography

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The anatomical and clinical studies of computed tomography (CT) in Camelidae are scarce. The use of CT in large animal medicine is currently limited by the logistic problems of acquiring computed tomographic images. Several CT studies exist on adult llamas, but not in camels. Accurate interpretation of the planimetric CT normal anatomy is necessary for the study and evaluation of pathological tissues. The purpose of our work was to evaluate the thorax of the newborn camel and related structures by means of CT images and macroscopic sections. One newborn camel of one week was used. It was euthanized for medical reasons unrelated to disease of the thorax. CT images were obtained and detailed anatomy of the thorax was acquired. Different CT windows, soft-tissue and pulmonary windows, were applied in order to obtain detailed attenuation shades of the thoracic structures. The camel was frozen and sectioned using an electric saw, and we obtained high quality images of the thorax compared with CT images. Clinically relevant anatomic structures of the thorax cavity were identified and labelled in the corresponding CT and gross-section photographs. The information presented in this paper should serve as an initial reference to evaluate CT images of the newborn camel thorax.

Temporo-spatial Expression of the Inducible Nos-isoform During Development of the Murine Enteric Nervous System

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In the enteric nervous system, nitric oxide (NO) is regarded as an important messenger for the non-adrenergic and non-cholinergic neurotransmission. Synthesized mainly by the constitutive nitric oxide synthase (NOS) isoforms NOS I and NOS III, this molecule executes pre-junctional inhibitory effects in the submucosal plexus as well as relaxation of the enteric smooth muscles. In order to

elucidate the role for NO during enteric development, we looked for the expression of all three NOS-isoforms in the enteric nervous system during mouse development from E8 to E20 using immunohistochemistry. Starting around midgestation, a transient expression of the NOS-II isoform during the very early development of enteric neurones was detected in parallel to that of HNK-1 exclusively in the myenteric plexus. Similar to findings for other neuronal systems, NOS-I and NOS III isoforms could be traced starting significantly later to increase towards the end of embryonic development when NOS II immunoreactivity faded and a strong expression of the vasointestinal peptide (VIP) could be detected. In contrast to the NOSII expression, the constitutive isoforms can also be detected in the submucosal plexus. Altogether, these findings suggest NOS-II to be exclusively involved during early steps of enteric nervous system development. Absence of downstream signalling elements, such as sGC and cGMP both in neurons and in enteric muscle until the end of the second third of gestation, may indicate different effects executed by NO during development, expressed by Ca²⁺ dependent and independent NOS isoforms.

Bone Marrow Mesenchymal Stem Cells: From Characterization of Neural Differentiation to a Possible Therapeutic Use in Neurodegeneration

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Bone marrow derived stromal cells are of mesenchymal origin and precursor cells for skeletal tissue components such as chondroblasts and osteoblasts. Furthermore, under experimental conditions, a differentiation potency into myogenic and neuronal cells could be demonstrated. Because of their multipotency these cells represent a population of non-haematogen stem cells that can be regarded as an alternative to human embryonic stem cells for future autologous cell replacement therapies. For a closer look at the differentiation capacity of these cells, rat and human bone marrow stromal cells were isolated from the femur bone and kept in the cell culture applying different cultivation protocols. In a cultivation medium with a serum content of 20%, the majority of these cells express a variety of neuronal markers such as β -III Tubulin and NeuN as well as the astrocyte marker GFAP, while a minority of about 20% express the marker for neural precursor cells nestin. Cultivation in a chemically defined serum free medium results in the differentiation of a

markedly higher percentage of nestin positive neural precursor-like cells. Using bFGF in combination with B27 these cells can be forced to form three dimensionally organized spheres. In order to elucidate a possible therapeutic potency of the bone marrow derived cells the synthesis of neurotrophic factors such as BDNF and NGF were analysed using the ELISA technique. Furthermore, they can be infected using a third generation adenoviral vector with high efficiency and show migratory activity *in vitro*. After injection of bone marrow derived mesenchymal stem cells into the lateral ventricle of adult rats they adhere to the ependymocytes and pass the ependymal barrier in order to settle in the subventricular space.

GP IIb-IIIa Receptor Blockers Minimize Vascular and Perivascular Damage in the Hippocampus after Cardiopulmonary Bypass Management

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Brain injury remains a significant and potentially devastating outcome of cardiopulmonary bypass (CPB) under circulatory arrest. These outcomes caused by a microvasculature embolization are associated with increased mortality, longer hospital stays and increased use of intermediate or long term care facilities. The administration of heparin in heart surgery during deep hypothermic cardiopulmonary bypass is the basic prophylactic strategy for reducing or even preventing, microvasculature embolization. Unfortunately, an incidence of neuropsychological impairments (NPI) is found in as many as 25 % of the survivors. As it is suspected that these impairments are correlated with morphological alterations, in our study we use the GP IIb-IIIa receptor blocker Eptifibatide for the inhibition of platelet aggregation, in order to look for a reduction of tissue damage compared to the standard treatment. Two groups of 11 piglets (mean body weight of 15 ± 5 kg) underwent 10-minute normothermic bypass, 40-minute cooling on cardiopulmonary bypass, 60-minutes deep hypothermic circulatory arrest (DHCA) at 15°C, and 40-minute rewarming to 37°C. Group 1 was treated only with unfractionated heparin (UFH), whereas Group 2 was medicated with Eptifibatide, in addition to the UFH-treatment group 1. After rewarming, all animals underwent bilateral carotid perfusion with 4% paraformaldehyde. Histological investigations of semi thin sections reveal a marked decrease of hippocampal alterations

by using the GP IIb-IIIa receptor blocker in addition to standard UFH treatment. We detect a reduction of degenerative areas in perivascular (vessels with 10–30 μ m in diameter) tissue. These semi-quantitative data are confirmed by ultrastructural findings.

Scanning Electron Microscopic and Immunohistochemical Study of Uterine and Placental Vascularization in the Rat During Different Stages of Pregnancy

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During gestation, the uterus of eutherian mammals is subject to extensive modifications, which are most prominent in species with a haemochorial placenta. Pregnancy-associated changes in haemochorial placental types includes a rearrangement of endometrial blood vessels together with a rapid invasion of the endometrium by trophoblast cells, inducing the alteration of the phenotype of stroma cells, which is called decidualization. The aim of this study is to re-evaluate these modifications with special reference to the development of the placental vascularization during gestation of the rat. 15 Wistar rats were sacrificed on days 4, 6, 9, 12, 16, 19 and 20 of gestation (2 per day) and during estrous cycle (1) and perfusion-fixed. Implantation chambers and/or placental tissues were excised and paraffin-embedded. Additionally vascular corrosion casts were produced on the same days of gestation from another series of rats, and examined scanning electron microscopically. Paraffin-embedded sections were used for immunohistochemistry with antibodies against cytokeratin, α -smooth muscle (sm) actin, and histochemistry with lectin (*Lycopersicum esculentum*). The histological findings were compared to the scanning electron microscopical results. Cytokeratin labelled epithelial cells, namely uterine luminal and glandular epithelium as well as most types of trophoblast cells. α -sm actin was observed in smooth muscle cells, maternal vascular endothelium and decidua cells. *Lycopersicum esculentum* lectin was detected in endothelial cells and in yolk sac epithelium. The uterine microvasculature in the course of gestation showed specific alterations, like the development of spiral arteries and a central artery running straight to the fetal placental surface where it branches into arterioles. These arterioles feed the placental labyrinth through blood channels, which are lined by syncyt-

iotrophoblast instead of endothelial cells. In parallel, the vascular diameter increases distinctly, and comparison to histology confirms the presence of protruding endovascular trophoblast cells as indentations on vascular casts. The present findings allow to establishing a reference defining the principal stages of the constitution and development of the chorioallantoic placenta, particularly with regard to the modifications concerning the morphology of the placental and uterine vessel tree, and the development of the cellular composition in different parts of the placenta.

The *Corpus fibule* in Sheep (*Ovis ammon f. aries*) – its Ontogeny and Reduction or Persistence from Fetal Age to Adulthood

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The *corpus fibulae* (CF) in sheep is represented by a cartilaginous anlage during early embryogenesis, which may calcify and ossify with ongoing gestation and finally towards term is supposed to be replaced by a connective tissue cord called *ligamentum fibulare* (LF). A recent study in goat (Boos A and T Bartels, 2002), however, revealed that 58 % of the animals examined exhibited a pin-shaped bone within the LF. This indicates that a complete transformation of the anlage of the CF to a connective tissue cord is not performed in all goats raising the question of existing parallels in sheep. Forty-two fetuses and lower legs of 18 lambs and 100 sheep being elder than 2 years were collected at the slaughterhouse. Radiographs and macerated or enzymatically cleared and stained lower legs were assessed for the expression of a CF. In 28 of 42 fetuses (66%), 11 of 18 lambs (61%) and 48 of 100 adult sheep (48%) at least one cartilaginous – only fetal period – or osseous CF could be detected. Differences between groups were significant (Chi-square test, $P < 0.05$). In 2, 3 and 15 animals, respectively, only a unilateral expression was evident. In early fetuses, the fibula either consisted of a long anlage having the length of the tibia or of a long proximal and a short distal part paralleling the tibia. In somewhat larger fetuses the CF was composed only of the proximal part or even completely lacked (crown rump length 7.5 ± 1.5 vs. 11.2 ± 2.4 and 12.2 ± 2.0 cm, respectively). Pre-term fetuses, lambs and adult sheep were devoid of a CF ($n = 5/7$, $7/11$ and $52/100$, respectively) or exhibited a pin-shaped bone of varying length within the LF. This osseous CF was either in close contact to the *caput fibulae* ($n = 6$ animals) – always separated by

connective tissue – or located more distally. In the latter cases the centre of the osseous pin was usually situated at a level between the proximal and intermediate thirds of the tibia. Results suggest that the *corpus fibulae* in sheep may persist in many individuals (50%) as a pin-shaped bone throughout life.

Reference

Boos, A., and T. Bartels, 2002: *Ann. Anat.* 184, 289–293

Double Labelling Immunohistochemistry on the Paravertebral Ganglion Related to the Smooth Vasa Musculature of the Swine Cremaster Muscle

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Retrograde neuronal tracing and double labelling immunohistochemistry methods were used to study the co-existence of tyrosine hydroxylase (TH) with vesicular acetylcholine transporter (VACHT), neuronal nitric oxide synthase (nNOS), calcitonine gene related peptide (CGRP), leucine-enkephaline (LENK), neuropeptide Y (NPY), substance P (SP) and vasoactive intestinal polypeptide (VIP) in the peripheral paravertebral neurons projecting to the smooth vasa musculature of the swine cremaster muscle (CM). The fluorescent tracer Fast Blue (FB) was injected into the left CM of 4 50 Kg non-emasculated pigs. After 7-day survival time, the ipsilateral paravertebral L4 (PaGL4) ganglion, the most positive among the autonomic ganglia, was collected. 12 μ m-thick cryostat sections were observed under a fluorescent microscope. In the PaGL4, there were 575.5 ± 152.93 FB+ neurons. On 206 ± 51.07 FB+ cells tested for TH/ VACHT, $24.52\% \pm 5.6\%$ were TH+/VACHT+, $33.65\% \pm 7.33\%$ TH+/VACHT-, $7.46\% \pm 2.77\%$ TH-/VACHT+, $34.38 \pm 2.66\%$ TH-/VACHT-. 211.5 $\pm$ 47 FB+ neurons were tested for TH/ nNOS: $24.72\% \pm 5.43\%$ were TH+/nNOS+; $29.35\% \pm 7.71\%$ were TH+/nNOS-; $7.23\% \pm 3.22\%$ were TH-/nNOS+; $38.7\% \pm 2.88\%$ were TH-/nNOS-. On 213.25 ± 71.82 positive neurons tested for TH/CGRP, $29.4\% \pm 4.36\%$ were TH+/CGRP+, $29.09\% \pm 7.61\%$ TH+/CGRP-, $11.45\% \pm 4.43\%$ TH-/CGRP+, $30.06\% \pm 3.7\%$ TH-/CGRP-. On 248.5 ± 73.82 FB+ neurons tested for TH/LENK, $37\% \pm 4.25\%$ were TH+/LENK+, $40.52\% \pm 5.96\%$ TH+/LENK-, $3.54\% \pm 1.37\%$ TH-/LENK+, $18.94\% \pm 2.09\%$ TH-/LENK-. 210 $\pm$ 53.22 FB+ cells were tested for TH/NPY: $50.72 \pm 2.8\%$ were TH+/NPY+; $15.38\% \pm 2.55\%$ TH+/NPY-; $11.12\% \pm 4.32$ TH-/NPY+; $22.79 \pm 2.78\%$

TH-/NPY-. 218.75 ± 65.02 of positive cells were tested for TH/SP: 26.76 ± 4.73 were TH+/SP+; $36.82\% \pm 7.58$ TH+/SP-; $3.78\% \pm 2.43\%$ TH-/SP+; $32.64 \pm 3.75\%$ TH-/SP-. On 202.5 ± 47.65 FB+ neurons tested for TH/VIP, $30.95\% \pm 8.2\%$ were TH+/VIP+, $29.51\% \pm 3.87$ TH+/VIP-, $8.38\% \pm 2.33\%$ TH-/VIP+, $31.16\% \pm 6.04\%$ TH-/VIP-. The obtained results allow us to document that the neurochemical content of the paravertebral neurons projecting to the CM differ considerably. The contemporaneous presence of adrenergic, peptidergic, nitrergic, and cholinergic neurons shows the complex role that these autonomic ganglia have in the control of smooth vasal musculature.

The Development of the Metanephric Kidney in the Pig

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Aims: The metanephric kidneys of the pig are used as xenotransplants in human medicine. In order for transplants to fit within the host organisms, the subcapsular blastema and blood vessels are crucial for the development of new nephrons to sustain the organ functions. The aim of this study is to obtain data concerning the post-natal development of metanephric nephrons in the porcine kidney.

Materials and Methods: The metanephric kidneys of six porcine fetuses with a crown-rump length ranging from 40 mm to 220 mm of eight piglets aged between 6 to 10 weeks and of three adult pigs were studied. Eight lectins as well as anti-actin and anti-myosin antibodies were used for lectin- and immunohistochemistry to study the subcapsular metanephric blastema, to visualize the blood-urine barrier in the nephrons and collecting tubules, and to study the blood vessels in both the renal cortex and marrow.

Results and Conclusions: A subcapsular metanephric blastema was still present in the kidney of 10-week-old piglets. Dense condensation of mesenchymal cells surrounded the terminal branches of the collecting ducts and showed first signs of mesenchymal-epithelial transformation. Characteristic comma-shaped and s-shaped bodies were found in and underneath the subcapsular blastema. In the fibrous renal capsule of six-week-old piglets, a first faint binding reaction of anti-actin was visible and intensified in the fibrous renal capsule in ten-week-old piglets and in adult pigs. In addition, the smooth-muscle layers of the blood vessels were stained by the anti-actin and anti-myosin antibodies. The lectins showed various affinities to the endothelium of blood vessels and to the epithelial cells lining of the capsules of the metanephric renal corpuscles, the various parts of the renal tubules, as

well as the collecting tubules and the renal pelvis. The affinity of the epithelial cells to a specific lectin varies in neighbouring cells, indicating different cell activities or cell cycles.

The Gap Junctional Protein Connexin(Cx)43 in Testicular Cancer: its Loss Marks Progression from Carcinoma *In Situ* to Invasive Germ Cell Tumour

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Carcinoma-in-situ (CIS) of the testis is known to be the pre-invasive stage of most human germ cell tumours (seminoma and non-seminoma), but the mechanisms leading to an increased pubertal proliferation of CIS cells after a long latency and to progression of CIS to an invasive malignancy are still not known. Additionally, CIS and seminoma have also been reported in equine testis (Veeramachaneni and Sawyer, 1998). The gap junctional protein and tumour suppressor gene connexin(cx)43 represents the predominant cx in human, canine and rodent testis so far and it is expected to play a key role for the regulation of both proliferation and differentiation of germ cells (spermatogonia and spermatocytes), and its gene- and protein-expression pattern is typical for the pubertal terminal differentiation of somatic Sertoli cells. Using cDNA-microarray analysis, in-situ hybridization (ISH), RT-PCR from tissue homogenate and semi-quantitative RT-PCR from well defined microdissected tubules with normal spermatogenesis, CIS, intratubular seminoma (ISe) and from seminoma cells from invasive seminoma we found a downregulation of cx43 starting in intratubular CIS, leading to a complete loss in most invasive seminoma cells. This indicates that regulation of cx43 expression takes place at transcriptional level confirming and expanding earlier studies of protein expression (Brehm et al., 2002). This reduction of cx43-expression suggests that an early intratubular derangement in cx43-gene expression and disruption of inter-cellular communication between Sertoli cells and/or Sertoli cells and pre-invasive tumour cells via cx43-gap junctions may play a role in the proliferation of CIS cells and seminoma cells and in the progression phase of testicular seminoma development.

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Development and Fine Structure of the Reticulate Scales of the Turkey Foetuses with Regard to Occurrence of the Food Pad Lesions in Turkey Poult

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Introduction: Most of the morphological studies on the skin of birds refer to chickens. Little can be found about the specific fine structure of the foot-pad epidermis of turkeys and almost nothing is known about the developing reticulate scales in turkey foetuses. This study deals with the question as to whether the development and morphology of the reticulate scales of turkeys are similar to those of chickens. Since the first signs of food-pad dermatitis can be observed in turkey poults at the age of four days, it was additionally examined if these alterations can be found in turkey foetuses already.

Methods: Foot pad samples of heavy strain BUT Big 6 turkeys were collected at different developmental stages (day 20, 23, and 28 of incubation). The specimens were compared by light- and electron-microscopic methods.

Results: On the 20th day of incubation, the reticulate scales cover the food pads as small knob-like structures. Between the embryonic epidermal cells wide inter-cellular spaces are visible, and in the intermediate cells multigranular bodies (MGBs) are detectable. Peridermal cells cover the reticulate scales. They are characterized by peridermal granules and protuberances of the outer cell membrane. On the 23rd day of incubation, many lipid droplets are scattered within the intermediate cells. Because a thin *stratum corneum* has formed, the periderm is almost completely sloughing off. On the 28th day of incubation, the turkey poults hatch. The epidermis has almost reached maturity and a broad *stratum corneum* is detectable.

Conclusions: The development and morphology of the reticulate scale epidermis of turkey is comparable to that of chicken, although little differences were found. Besides former investigations, which described the peridermal

protuberances as microvilli, scanning-electron microscopy proved that they also show a fingerprint-like surface structure that has not been observed for birds previously. Up to the hatching day, no signs of food pad alterations are seen in developing turkeys.

Innervation Pattern of Dermal Blood Vessels in the Bovine Claw Shows Functional Adaptation in Different Areas

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Introduction: The blood circulation in the claw plays an important role in the pathogenesis of claw diseases. The blood vessel diameter is regulated by innervation with different nerve fibre qualities. In case of inflammation or unphysiologically high mechanical load, the delicate balance of vasodilative and vasoconstrictive neurotransmitters seems to be disturbed and may aggravate the alterations. Knowledge about the innervation pattern of the dermal blood vessels could therefore help to understand some aspects in the pathogenesis of claw diseases.

Material and Methods: The lateral claws of the hind limb of cows from a local abattoir were examined. Tissue samples were taken from different segments of the claw, fixed and processed for immunohistochemistry. Antibodies against protein gene product 9.5, tyrosine-hydroxylase, choline acetyltransferase, substance P and calcitonin gene-related peptide were applied and colocalization studies were carried out. Results were evaluated with a confocal laser scanning microscope.

Results: Dermal blood vessels in the claw are surrounded by nerve fibres containing different neurotransmitters. Occurrence and density differ depending on the respective segment. In the bulbar segment, more vasodilative fibres are demonstrable while in the solar segment vasoconstrictive nerve fibres prevail. The highest innervation was found in all segments around specialized sphincteric arteries in the dermis just beneath the dermal surface modification.

Conclusion: The innervation pattern in the claw depends on segment-specific functional demands. The fine-tuning of the blood flow in the dermal vascular system is mediated by a combination of different nerve fibres that emit vasoactive substances.

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Computed Tomography as a New Tool for Investigating Leg Weakness in Turkeys

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Introduction: Leg weakness causes high economic losses in commercial poultry. The unspecific term includes, amongst others, diseases of the skeletal system caused by genetic, nutritional or microbiological factors. The analysis of computed tomography images may be a new tool to discriminate and evaluate skeletal abnormalities in turkeys and can therefore provide valuable hints in the treatment of turkey stocks suffering from leg weakness.

Material and Methods: Legs of healthy turkeys and those showing clinical signs of leg weakness were examined using a Tomoscan M/EG/Compact (Philips Medical Systems) CT system. The slice thickness and distance were set to 2 mm. In addition to the evaluation of the obtained images, the software system 3D-Doctor was used to create three-dimensional objects of the investigated samples.

Results: Differences between physiological and abnormal parts of the leg bones are distinguishable in the CT-images and three-dimensional reconstructed objects. Bone density can be calculated and thus provides a base to estimate possible nutritional demands or deficiencies. The three-dimensional reconstruction compensates for the difficulties in understanding the detailed anatomical CT cross-sections by which the examiner is confronted while studying the CT images.

Conclusion: Although computed tomography of poultry is too expensive for routine herd diagnosis, and far less for single-animal diagnosis, it is a valuable tool for investigating skeletal abnormalities and is very useful in examining samples related to leg weakness.

A Potential Role for Platelet-activating Factor (PAF) During Bovine Placentation

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Platelet-activating factor (PAF) is an important phospholipid messenger involved in diverse physiological and

pathological conditions. Actions of PAF are mediated by a specific PAF receptor (PAF-R) and the inhibition of biological activity by PAF acetylhydrolase (PAF-AH). To elucidate the potential role of PAF in stimulating angiogenesis and trophoblast giant cell migration during gestation, we examined the expression and localization of PAF-R mRNA and protein as well as PAF-AH mRNA in the bovine placenta. RT-PCR, *in situ* hybridization and immunohistochemistry were applied to placentomes from days 20, 80, 150, 200, 240, 270 of gestation, and immediately pre-partum (around day 280). PAF-R mRNA and protein were colocalized in immature trophoblast giant cells (TGC), uterine epithelial cells, and endothelial cells of fetal and maternal vessels. There was no apparent difference in placental PAF-R mRNA or protein levels until day 270. In contrast, in placentomes taken immediately pre-partum the localization of the immunohistochemical and hybridization signals switched from immature TGC and maternal epithelium to cells of the maternal stroma, particularly endothelial cells of maternal vessels. mRNAs for PAF-AH isoforms Ib (β - and γ -subunit), II, and plasma-isoform were detected by RT-PCR in placentomes throughout gestation. In tissue sections mRNA of the PAF specific PAF-AH-Ib β -subunit was predominantly found in TGC, while the intensity of the signal was weaker in maternal epithelium and endothelium of fetal and maternal vessels. Corresponding to the PAF-R expression, mRNA of PAF-AH Ib was detected mainly in maternal endothelial cells prior to parturition. The constitutive expression and localization of PAF-R and PAF-AH in epithelial and endothelial cells during gestation, together with the parturition-related switch to the maternal stroma suggests a functional role of PAF for the interaction of TGC with the maternal epithelium, and thus may be associated with TGC migration, placental angiogenesis and pre-partum tissue programming.

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Occlusion of the Middle Cerebral Artery: a New Method of Focal Cerebral Ischemia in Rats

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The study in Wistar rats attempted to improve the occlusion technique of the middle cerebral artery (MCA) as a precise method for initiating stroke. In a first part it was necessary to study the exact anatomy of blood vessels of the brain in seven rats of 170-410 g body weight by corrosion cast. The lengths and diameters of defined locations of the blood vessels were measured. The temporary as well as the permanent methods were refined or replaced. The first one was completed in main training the physiological blood flow

after temporary occlusion, while the permanent occlusion was performed by positioning a silicone cap in the MCA. A filament guide was introduced from the common carotid artery (CCA) via internal carotid artery (ICA) to guide the silicon cap at the branch of the MCA. Histological sections of the brain of rats showed 24 h after the permanent occlusion a reproducible infarct in the brain. This area corresponded very well with the supply of the MCA. The new occlusion method with a silicon cap was compared with the occlusion methods of CCA route and external carotid artery (ECA) route. The total infarct volume was significantly larger in the CCA route and ECA route groups than in the silicon cap group (means: CCA route 261 mm³; ECA route 191 mm³ vs. 128 mm³ silicon cap group; $P < 0.05$). It could be demonstrated that the new silicon cap occlusion technique imitates the pathological situation of a cerebral infarct in man. Moreover it is less invasive for the animals and more precise and reproducible regarding the infarcted area in comparison to the other occlusion methods. Based on anatomical measurements of the blood vessels the described silicon cap method can be recommended for rats of a body weight between 340–370 g.

Computed Tomography of the Lungs of the Dog by a Six-generation CT Scanner, Intravenous Contrast Medium and Different Windows

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Computed tomography (CT) is a modern technique of image diagnosis particularly recommended in human medicine to evaluate the existence of pulmonary pathological changes such as neoplasms, metastasis, interstitial infiltrates, etc. In veterinary medicine, however, few anatomical and clinical CT studies in the dog have used apparatus of the latest generation, including injection of intravenous contrast and application of regional specific CT windows with different window width (WW) and window level (WL) to evaluate the lungs, the pulmonary vessels and the bronchial structures. This methodology allows the obtaining of clear CT images with high capacity of tissue discrimination and different shades of attenuation. In this work we have planned a tomographic study of the lungs of the dog by using a six-generation spiral CT scanner (Toshiba Ex Vision), belonging to the private Medical Institute of Radiology 'Irion' of Porto Alegre, Brazil. Four mixed-breed mature dogs (4–6 years,

15–20 kg) were used, two males and two females. The dogs were anaesthetized and kept in a maximum inspiration when obtaining the images. Dogs were placed in a stretcher in a ventral or sternal recumbency. Previously, the contrast urografin® was injected in the cephalic vein. Different CT windows were applied in order to increase the quality of the images: pulmonary window (WW 928; WL -680), high-resolution pulmonary window (WW 1085; WL -750), and soft tissue window (WW 652; WL -34). The use of intravenous contrast, different CT windows and a modern CT apparatus produced excellent images of the pulmonary parenchyma, the pleural cavity, the pulmonary veins, the lobular rami of the pulmonary artery and the lobular bronchi.

Computed Tomography of the Mediastinum of the Dog by a Six-generation CT Scanner, Different Windows and Peroral and Intravenous Contrast Medium

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The anatomical and clinical studies of computed tomography (CT) in the dog are still scarce. Particularly, there are few CT studies combining the use of rapid apparatus of the latest generation, the supply of contrast medium (digestive or intravenous) and the application of regional specific CT windows with different window width (WW) and window level (WL). The use of this CT methodology is particularly recommended in the evaluation of mediastinal masses, heart pathologies, etc. In this work the mediastinum of the dog was studied by using a sixth generation spiral CT scanner (Toshiba Ex Vision), belonging to the private Medical Institute of Radiology 'Irion' of Porto Alegre, Brazil. Four mixed-breed mature dogs (4–6 years, 15–20 kg) were used, two males and two females. The dogs were anaesthetized and kept at a maximum inspiration when getting the images. Previously, the urografin® contrast had been injected by peroral and intravenous (cephalic vein) via. Dogs were placed in a stretcher in sternal recumbency. Different CT windows were applied in order to increase the quality of the images and to obtain detailed attenuation shades of the mediastinum organs: soft tissue window (WW 652; WL -34), pulmonary window (WW 928; WL -680) and mediastinal-vascular window (WW 352; WL 220). The mediastinal space was studied craniocaudally from the

level of the cranial cava vein, brachiocephalic trunk and left subclavian artery, to the caudal end of the lumbodiphragmatic recess. The obtained images allowed a very good tomographic evidence of the great vessels, the heart, the thoracic portion of the trachea, the principal bronchi, the thoracic portion of the oesophagus, etc.

The Anatomy of the Kidney of the Soleidae (Teleostei: Pleuronectiformes): the Importance of Plastination and Interest for the Phylogeny of Flatfishes

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Pleuronectiformes or flatfishes are teleostean fishes well known for their external asymmetry, with the two eyes on one side of the body, right or left according to the species. Their internal asymmetry has not been studied in detail in various species and we report here some original observation and manipulation on the kidney of some flatfishes. In most flatfishes, the kidney is an uneven and symmetrical organ present in the dorsal part of the abdominal cavity, lying on the ventral side of the vertebral column, between the haemal arches. Dorsally, the kidney bends ventrally and the whole organ is boomerang-shaped. However, in two flatfish families, the structure of the kidney appears to be different. Kobelkowsky (2000) shows that the kidney of the American soles (family Achiridae) is asymmetrical with a more developed right posterior part extending backwards between the haemal spines and the right epaxial musculature. Such an asymmetry has been mentioned in the common sole (*Solea solea* [Soleidae]) by Audigé (1910) and forgotten in the subsequent literature. However, it is the left side posterior part that is more developed in the common sole. This peculiar structure has been observed in dissected and plastinated specimens of several soleid species. This anatomical feature seems to be unique to flatfish species belonging to the Soleidae, a more developed left posterior of the kidney appears to be a synapomorphy of the animals belonging to this family and confirms the monophyly of this group. We hope to specify this point in a near future. The anatomy of kidney should then be taken in

account in future works dedicated to flatfishes systematics. Conclusions: (1) The anatomy of soft organs is mostly unknown for flatfishes, even for those of economic importance, and examination of the structures of these species is highly needed. (2) The study of the soft anatomy reveals a new set of data for the research of flatfish phylogeny. (3) In order to make a better comparison between the different species, a process of good conservation, like plastination, is needed for preservation of the structures (Grondin and Olry, 2000; Dieckwisch, 1990). In fish, plastination is well known to preserve the internal organs (Asadi, 1998).

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Anatomical Location and Histology of the Tonsilla Paraepiglottica in Sheep

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The location, size and histological structure of the ovine paraepiglottic tonsils were determined in 20 sheep. Macroscopic examination of the pharyngeal tonsils was performed in 10 sheep of 9 months of age, of which the heads were fixed in acetic acid 2% for 24 h, and in non-fixed heads of five sheep aged between 6 and 9 months. The paraepiglottic tonsil is located lateral to the epiglottis in the ventromedial segment of the pharyngeal sidewall, i. e. at the floor of the piriform recess. The tonsil appears as a cluster of small nodular elevations, which are white after fixation in acetic acid and are called hereafter tonsillar nodules. The number of these nodules per tonsil ranged from 1 through 8. The length of the tonsil varied from 5 till 14 mm, and the width ranged from 2 till 5 mm. The tonsils of 5 sheep aged between 6 and 9 months and of five sheep aged between 9 and 12 months were excised and fixed in 3.5% formaldehyde. The blocks were processed routinely for histology and embedded in paraffin wax. Serial sections (8 µm thick) were stained with eosin and haematoxylin. Histology revealed that each tonsillar

nodule consisted of several lymphoid follicles (*noduli lymphatici*) surrounded by parafollicular tissue. Most lymphoid follicles were secondary follicles consisting of a germinal centre, a dark zone and a mantle zone. Primary follicles devoid of germinal centres were most obvious at the periphery of the tonsillar nodules. The stratified squamous pharyngeal epithelium covering the lymphoid tissue was frequently infiltrated by numerous lymphocytes, forming a follicle-associated epithelium. Crypts were absent, but in some cases the pharyngeal epithelium was slightly infolded into the lymphoid tissue. The tonsillar nodules were almost completely encapsulated by dense connective tissue from which septa extended into the larger nodules.

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Teaching of Anatomy of Genital Organs in the Large Animals

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At the University of Missouri-Columbia, USA, teaching the anatomy of genital organs in large (and in small) animals is clinically oriented.

In the male horse, ox, and pig, the descriptive anatomy of the structures is taught in that order as they are listed in *Nomina Anatomica Veterinaria*. Clinical correlates are immediately mentioned in relation to each of the following structures. The testicle, epididymis, *ductus deferens*, spermatic cord, and tunics of spermatic cord and testis are correlated to the castration, cryptorchidism, ectopic testicles, and inguinal herniae, as well as to the landmarks and approaches to different clinical techniques. The penis and male urethra are correlated to the prolapse of the prepuce, ulcerative posthitis, balanitis, penile deviation, penischisis, persistent penile frenulum, short retractor penis muscle, catheterization of the urethra, the contagious equine metritis (CEM), etc. In the female horse, ox and pig, following the similar order as in the male species, the ovary and the salpinx are correlated to the diagnosis of pregnancy by rectal exploration, ovarian hypoplasia, ectopic pregnancy, and ovariectomy. The uterus is correlated to the different aspects of metritis and endometritis, to the retained placenta, pyometra, uterine torsion, uterine prolapse and eversion, Caesarian section, diagnosis of pregnancy and different stages of oestrus by rectal exploration, double external ostium of the cervix,

distocia, and the assessment of possible difficulties in the birth process. The vagina and vestibulum vaginae are correlated to the gaertneritis, vaginal and cervical prolapse, vaginitis, catheterization of the urinary bladder, and pelvimetry. The vulva and clitoris are correlated to vulvitis, balanitis, distocia, episiotomy, and transmissible genital diseases. Different kinds of udder diseases (mastitis) are based on the thorough knowledge of the anatomy of the udder including the blood and nerve supply and the lymphatic system. The two techniques of epidural anesthesia (Magda and Farquharson in ruminants) and subsacral anaesthesia (Popescu) for diagnosis and treatment purposes are explained in detail.

Teaching Developmental Anatomy of the Genital System to VM1 Students at the University of Missouri-Columbia, USA

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Developmental anatomy is integrated with gross anatomy. It consists of 12 lectures and is divided into body systems. The development of each system is correlated to its adult structures and species differences. The urinary and genital systems are developmentally and anatomically associated and originate together from the intermediate mesoderm. The urethra forms from the distal part of the urogenital sinus. Starting with the gonadogenesis, and continuing with the duct systems and external genitalia, the lectures are focused on the male and then on the female structures. At the end, the anomalies of the genital organs are exposed in detail, with suggestions for students, specialists and breeders to prevent and treat teratological conditions. The gonadogenesis is first presented in the indifferent stage. The differential stage is explained for the testis and then for the ovary. The genital duct system is exposed in the indifferent stage. The differential stage for the male includes the formation of the efferent ductules, the epididymal duct and the ductus deferens, while in the female, the focus is on the formation of the uterine tubes, uterus (including the horns, the body and the cervix), and the vagina. The indifferent stage of external genitalia is a complex stage, which originates from the mesodermal swellings adjacent to the cloacal membrane. In the male, the differential

stage includes the formation of penis and penile urethra. In the female the genital tubercle forms the clitoris, the urethral folds form the labia, while the genital swellings degenerate. The hermaphroditism, gonadal dysplasia, the cryptorchidism, the persistent vestigial structures, and the penischisis (hypospadias and epispadias) as anomalies of the genital system conclude the lecture.

Distribution and Activity of Fatty Acid Amide Hydrolase (FAAH) in the Gastrointestinal Neurons of Mouse

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The endocannabinoid anandamide may regulate intestinal motility through activation of CB1 receptors. Anandamide is then inactivated by fatty acid amide hydrolase (FAAH), a membrane bound enzyme. Under pathological conditions, inactivation of such enzymatic activity may lead to inhibition of the intestinal motility. Here, preliminary reports on the distribution of Fatty Acid Amide Hydrolase (FAAH) immunoreactivity in the mouse gastrointestinal neurons, and the pharmacological effects of N-arachidonoylserotonin (AA-5HT), a selective inhibitor of FAAH, are reported. FAAH was revealed by an indirect immunofluorescence. Laminar preparations containing the myenteric or the submucous plexus adhered, were peeled off after the whole gut wall had been stretched out and fixed in 4% paraformaldehyde. They were subsequently incubated with a polyclonal anti-serum directed against a region near the N-terminus of the human FAAH and revealed by a FITC-conjugated goat anti-rabbit secondary anti-serum. FAAH-immunoreactive neurons were observed within the myenteric ganglia throughout the GIT. The positive nerve cells varied in size and density of immunoreactivity. Stomach and large intestine showed the highest neuronal density. AA-5HT significantly reduced both gastric emptying and gastrointestinal tract transit. Such inhibitory effect was reduced by the C1 receptor antagonist SR141716A. Both morphological and pharmacological results suggest that FAAH may play a critical role in controlling gut anandamide levels.

Comparison of the Conformation of 20 International Level and the 20 Low Level Jumping Horses Using a 3-D Video Morphometric Measurement Method

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A new method of morphometric measurement has been developed (method presented as a poster at the 2002 EAVA congress). Its main originalities are (1) to use video (each horse is filmed while walking instead of being photographed at standing), (2) to be in 3-D (4 digital cameras are used; they film the horse from its lateral side, and also from the front and rear), (3) not to require skin markers (which makes the recording procedure quicker). This method is presently applied in the context of the breeding program of the Selle Français (SF) horse, through two large sample studies. Before starting the analysis of the corresponding data, a preliminary study was performed in order to settle the analysis methodology and to select the most pertinent morphometric parameters to characterize the "model" (conformation) of sport horses. Twenty international level SF horses (jumping performance index (ISO) ≥ 150 , group A), and 20 low-level (class D) jumping horses (group B) were compared using the 3-D method. Out of the 400 morphometric parameters calculated for each horse, about 132 revealed significant differences (Student test, $P < 0,05$) between the 2 groups; P was < 0.001 for 26 of these parameters. The height of the main anatomical landmarks of the pelvic limb, acting as essential lever arms for propulsion muscles (ex: point of the hock (*tuber calcanei*), ischial tuber), and the height – absolute and relative (to the withers height) – of the croup, the length of the croup and thigh, the length of the fore cannon, the shortness of the forearm, and the relative shortness of the trunk, are all conformational traits that characterize group A compared with group B.

Localization of Estrogen Receptor Beta (ER β) mRNA within Different Bovine Ovarian Follicles

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The oestrogen receptor beta (ER β) is largely distributed in the ovary of many species but data for the bovine ovary are scarce. Therefore, the expression of ER β mRNA in the different follicles of the bovine ovary was studied using *in situ* hybridization. Ovarian tissue sections of three cows with different plasma progesterone concentrations were used (cow 1: 3.50 ng/ml; cow 2: 1.00 ng/ml, cow 3: 0.35 ng/ml). A 602 bp fragment of ER β mRNA was cloned, sequenced and digoxigenin (DIG)-labelled. Subsequently, *in situ* hybridization was performed by incubating the sections with the DIG-labelled RNA anti-sense probe. For the semi-quantitative evaluation of ER β mRNA expression the ER β mRNA score (SER) was determined for the different follicular cell types using the formula: SER = 0.n0 + 1.n1 + 2.n2 + 3.n3 with n0, n1, n2, n3 indicating the percentage of cells exhibiting a staining intensity 0 (absent), 1 (weak), 2 (moderate) or 3 (strong), respectively. High ER mRNA levels were noticed in primordial and primary follicle cells, and suggest a role of ER mRNA in early folliculogenesis. A lower SER was observed in the granulosa cells of secondary and tertiary follicles. This significant difference in the SER of follicle cells during follicular growth may be associated with cell proliferation. In obliterated and cystic atretic follicles high SER were observed, although ER β mRNA levels in obliterated follicles showed much inter-individual variation. This is suggestive for ER β mediated oestrogen action in atretic follicles. In the *corpora lutea* moderate ER β mRNA levels were noticed. Our findings are in accordance with studies in the ewe in which *corpora lutea* cells synthesize estrogen. These preliminary findings will be further elaborated in a higher number of cows to examine the role of ER β in the ovary throughout the oestrus cycle.

Detection of Steroid Hormone Receptors in the Canine Circumanal Gland

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The morphology and function of the circumanal glands in dogs is under current discussion. They are clinically

important because they rank fourth in frequency as the site of canine skin tumours. The tissue is known to be governed by steroid hormones, as the gland exhibit sexual dimorphism, castration reduces the size of the gland and lead to tumour regression. The aim of our study was to demonstrate the hormone receptors in healthy circumanal glands as well as adenomas and adenocarcinomas in intact and castrated female and male dogs of different age groups. Immunohistochemical staining was performed using antibodies against androgen receptor (AR, Affinity Bioreagents, rabbit pc, 1:150) and oestrogen receptor (ER, Zymed Lab., rabbit pc, 1:200) after antigen retrieval with microwave cooking. Positive and negative nuclei were counted using Eclipse software and subsequently statistically evaluated. ER and AR-positive nuclei could be detected in healthy as well as in diseased tissue sections of the circumanal gland. The number of positive nuclei was higher in the flattened peripheral cells compared to the large polygonal cells in the centre of the lobule. There was no significant difference between male and female tissue and between healthy glands and tumours although in tumours there was a tendency to a higher number of positive cells. In tumours, the distribution of hormone receptor positive cells was more irregular and varied between single lobules. Castrated animals of both sexes showed a slightly higher number of positive nuclei than intact animals, but the difference was not significant. Our results demonstrate for the first time the presence of steroid hormone receptors in the canine circumanal gland. We suggest that an evaluation of steroid hormone receptors is justified before general hormonal or anti-hormonal treatment of patients.

The *ligamentum olecrani* of the Elbow Joint in Dogs and Cats

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The olecranon ligament (*ligamentum olecrani*) is described as an elastic ligament of the elbow joint in carnivores that tenses the caudomedial part of the joint capsule. The aim of the study was to compare the course and the microscopic structure of the ligament in dogs and cats. The elbow regions of 25 dogs and 15 cats were dissected to examine the topography of the ligament in extension and flexion. Furthermore, the olecranon ligaments of five dogs and five cats were studied using routine histological methods. Additional sections were stained with Resorcin-Fuchsin and Orcein to detect elastic fibres. In both species the olecranon ligament originates at the lateral surface of the *epicondylus medialis humeri* and inserts at the cranial crest of the olecranon extending distally to the roof of the *processus anconeus*. Tension of the ligament only occurs

when the elbow joint is flexed maximally. This tension is increased by a slight supination of the forearm, which takes place automatically in this joint position. In dogs the ligament is long (30–40 mm in medium sized breeds) and relatively slim (approx. 2–4 mm). In cats the ligament is short (10–12 mm) and relatively strong (5–8 mm). The histological examination of the olecranon ligament shows all signs of a tight collagenous ligament with a negligible amount of elastic fibres. The olecranon ligament helps to limit the maximal flexion of the elbow joint. In addition, it controls a slight lateral movement of the processus anconeus during the automatic supination of the antebrachial bones in extreme flexion of the elbow joint.

Specific Localization of the Gap Junctional Protein Connexin 32 in Equine Gastric Mucosa

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Gap junctional inter-cellular communication (GJIC) is known to be important in the maintenance of tissue homeostasis. Evidence is accumulating that GJIC plays an important role in the gastric mucosal defence system and that the loss of GJIC is associated with gastric ulcer formation. The prevalence of gastric ulceration in foals and horses is high. However, there are no studies about the presence of gap junctions in gastric mucosa of horses. The purpose of this study was to investigate the distribution pattern of the gap junction protein 32 in normal gastric mucosa of horses. Immunocytochemical observations were made with light microscopy on cryosections of fresh frozen gastric mucosa from the fundic region, pyloric region, *margo plicatus* and *pars nonglandularis*. Immunohistochemical staining was performed by standard immunoperoxidase techniques using an anti-connexin 32 polyclonal antibody. *In Situ* Hybridization, Western Blot Analysis and RT-PCR were performed to confirm the presence of connexin 32. In normal horse gastric mucosa, connexin 32 was abundant in the surface epithelium and in a decreasing staining gradient extending down to the neck of the glands. Gastric surface mucous cells are formed in the neck of the glands and migrate along the foveola toward the mucosal surface. So, immature forms of surface mucous cells are found in the generative zone and in the deep part of the foveola, whereas well-matured ones cover the upper part of the foveola and the luminal surface of the mucosa. Therefore, these findings are in line with other studies, which described larger and more numerous gap junctions in mature surface mucous cells than in immature cells. These results show that gap

junctions develop during the maturation of surface mucous cells and suggest that GJIC between gastric surface mucous cells plays an important role in the regulation of cell differentiation and in tissue homeostasis. Further studies are required to investigate the distribution pattern of connexin 32 in the gastric mucosa of horses with gastric ulcer.

Expression and Role of Connexin Proteins in Bovine Claw Epidermis

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Gap junction mediated inter-cellular communication plays a crucial role in maintaining tissue homeostasis in skin. Gap junctions consist of hemi-channels (connexons), which in turn are established by transmembrane proteins (connexins, Cx). The pattern of Cx- protein expression is characteristic for tissues and depends on the differentiation and functional state.

Aim: As part of our investigations of cell biological causes for lameness in dairy cows we wanted to assess the expression of Cx-proteins in relation to differentiation in bovine claw epidermis.

Methods: We examined hoof tissue from five defined regions of the claws of 10 cows from the abattoir by immunohistochemistry, Western blot and by transmission electron microscopy. The following monoclonal antibodies were used: anti- Cx26, Cx31, Cx37 (ADI, USA) and anti-Cx43 (Chemicon, USA). Immunohistochemistry was carried out on frozen and paraffin sections.

Results: Cx43 was expressed in the bovine claw epidermis in a pattern described for skin. Cx43 expression was strongest in the stratum basale in all regions. Cells of the stratum spinosum expressed lower levels, and in terminal differentiating cells Cx43 was disappearing. Expression of Cx26, 31 and 37 showed a high variation between individuals and regions. Protein expression was verified by Western blotting. The formation of gap junctions in the respective layers was confirmed by electron microscopy.

Conclusion: Our results provide evidence that Cx 43 is expressed in bovine claw epidermis in a distribution similar to skin. Cx 43 is progressively downregulated with the progress of differentiation. This suggests a role of Cx43 in cellular coordination particularly in the early stages of differentiation. Results for Cx 26, 31 and 37 will be confirmed in further studies also on the mRNA level to get deeper insight in the regulation of Cx expression in claw epidermis. We hypothesize that Cx 43 plays a similar role in controlling differentiation as in skin and may be involved in disruption of claw horn formation in claw disorders.

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S100 Expression in the Lateral Line System of Zebrafish from Juvenile to Adult Stages

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The lateral line system (LLS) is a sensory organ present in fishes and in the larval stage of aquatic amphibians, which provides the animal with information about its surrounding environment. Mechanoreceptive sensory cells in the LLS of fishes are grouped into neuromasts that are considered the morphological unit of this system. Structurally the neuromast consists of a central cluster of hair cells with a ring of support cells with some lateral mantle cells. The LLS is localized in both the body surface (superficial neuromast or pit organs) and in a system of canals distributed along the head, trunk and tail (deep neuromast) (Cernuda-Cernuda et al., 1996). The study of hair cells in the neuromast of fishes is becoming of a great interest, since they are similar in structure and function to the sensory cells of the inner ear and show some identical immunohistochemical properties (Foster et al., 1993). Recently, we have demonstrated the occurrence and the presence of S100 protein in the neuromast hair cells of some teleostean species (Abbate et al., 2002; Germanà et al., 2004). The zebrafish has become an ideal model for the study of developmental genetics, and the neuromast has been used as a model for the study of vertebrate auditory system (Withfield, 2002), so the need for cell selective markers has increased dramatically. Therefore, in this study we have investigated by tissue and whole-mount immunohistochemistry the presence of S100 protein in both deep and superficial neuromasts in the zebrafish larval, juvenile and adult stages. The results of this work demonstrated that the localization of S100 protein was restricted only to the hair cells of the neuromast at all the examined ages. The mantle cells and supporting cells within the neuromast were unreactive. Nevertheless, heterogeneity has been detected in the S100 protein expression by hair cells, independent of their localization within the neuromast. Our data suggest that the S100 protein is a specific marker for sensory cells in the neuromast of zebrafish at post-embryonic different stages of development, and it could

be important for the analysis of development, regeneration and turnover of hair cells.

Anatomical Basis of Physiotherapeutic Treatment of Lumbar Spinal Nerve Disorders

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In contrast to human medicine, the significance of appropriate post-operative care in companion animals is often neglected or even unknown. Physiotherapeutic exercises, also called physical therapies, are the golden standard of rehabilitation in human orthopaedics. In dogs and cats, such supportive therapies following surgery are rarely performed. Presently, physical therapeutic exercises, which are well established in man, are transferred to companion animals. However, there is no generally accepted scheme of procedures adequate to cats and dogs. Furthermore there is no study dealing with the exact effects of the various exercises performed with these species. Due to this paucity of information, the effect of several physical exercises concerning the lumbar spinal nerve roots transferred from man to the dog were assessed macroscopically. In ten euthanized healthy beagles up to eight years of age, the lumbar spine was surgically separated from adhering dorsal muscles. The lumbar nerve roots were visualized by removing the connective tissues surrounding the *foramen intervertebrale*. Nerve roots, i.e. spinal ganglia, were marked immediately outside the vertebral column applying a dot of yellow paint while the cadaver was in lateral recumbency. In man, for maximal sciatic nerve root movement the straight leg raising (SLR) (Maitland, 1986) exercise is applied on the patient lying on the back. For *nervus femoralis* movement the passive knee bending exercise is performed. In the dog, however, performing only the transferred SLR has a poor effect on L7. It can greatly be enhanced by a supplemental inward rotation of the hip joint by fixing the stifle joint and moving the lower leg medially. Concerning L4-L6 nerve roots, the hip joint is maximally extended and the effect can be improved by an outward rotation of the leg. The results of the present study indicate that the simple transfer of physical exercises from man to companion animals does not result in identical results in the latter species. Modifications considering different anatomical conditions should be kept in mind if clinical results are poor.

Reference

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Morphological and Ultrastructural Study of Vaccine-induced Granuloma in Atlantic Salmon (*Salmo salar*)

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Aim: The aims of this study were immunohistochemical (IHC) and ultrastructural characterization of cells involved in vaccine granuloma formation in Atlantic salmon.

Methods: Fish suffering from severe reaction 6 to 9 months after vaccination were selected. We investigated granuloma from peritoneum, at the injection site. Morphological staining techniques, IHC to visualise MHC+ cells and ultrastructural analysis were performed.

Results: Microbial investigations revealed no infective agents. Histological and ultrastructural studies revealed Russel bodies, macrophages, multinucleated giant cells, melanomacrophages (MMP), mast cells (MC) and Splendore-Hoeppli material in the lesions.

Conclusion: Signet ring cells identified after vaccination have been hitherto described as swollen mast cells. This study shows that these cells fit the Russel body characteristic, consistent with humoral stimulation following vaccination and foreign material precipitate, Splendore-Hoeppli material. MHC+ cells were located circumferal to granuloma centers. MMP were present containing melanosomes in different developmental stages.

Using Plastinated Specimens in Teaching Veterinary Anatomy

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Teaching veterinary anatomy continually presents new challenges. Cadaver procurement and usage comes under question from students, administration and the community. Student expectations rise while their participation wanes. Grades are hopelessly inflated. Time has been sliced away from many basic science courses. What can we do? We can run away or devise a plan. We use a tool that can make all these problems seem worse, it will take more of your time and your Dean's money, it will make some students angry with you, but

it is a great hobby for those who enjoy preparing anatomical specimens. Plastination! Our lab has been using plastination for 20 years. Plastination was invented by Dr. Günter von Hagens 26 years ago at the University of Heidelberg. The silicone plastination process has four major steps. For a laboratory to be successful and remain even after you retire or get fired, the first step in the process must be done meticulously. Step 1: Specimen preparation and fixation - plan what the specimen should demonstrate. Hollow organs must be dilated and fixed in a dilated form. Specimens must be positioned in an anatomical arrangement. Step 2: Dehydration: Cold (-25°C) Acetone is preferred, however alcohol may be used. Step 3: Impregnation: Replacing the volatile solvent (acetone) with a curable polymer. Step 4: Curing, hardening, or cross-linking the polymer - also a critical step. You must know what the final product is to look like. Once cured, the specimen will remain in that position. Silicone plastination yields somewhat life-like, dry, durable, aesthetically pleasing specimens that can be used in most any circumstance and any audience. Labelling specimens and/or photographing the specimens and labelling the photograph will enhance student use. Usage: A. Anatomical instruction - Two formats have been used. 1. Plastinated specimens used with no wet dissection of that region and 2. Specimens used as a supplement and review after dissection has been completed. B. Client or public education. C. Anatomical review for Veterinarians. D. Display of anatomy area for tours. Benefits: A. Medical students and Practitioners - Specimens are dry, therefore they are convenient to use at any time even on short notice. Items that are difficult to dissect and/or visualize are good adjuncts to teaching. B. Client/Public education - Great public relations as well as seeing the involved anatomy. C. Anatomy program - The use of prosected, plastinated large animal specimens allows students access to competently prepared, long lasting study material. By not having to dissect the entire large animal cadaver, students do not rush through dissection of the equine and bovine species limbs to compensate for the reduction in laboratory time. This approach allows sufficient time for students to study the limbs of various species within the time frame dictated by the curricular revision. Disadvantage: Some students may not do a complete dissection and/or not use their wet specimens for study/review. Some students want everything to be plastinated so they can memorize one example and forget about variation. Polyester and epoxy polymers may also be used for plastination. These polymers yield are used for preparing thin body slices, which are not flexible. With the advent of MRI, CT and ultrasonography an understanding of sectional anatomy is necessary, these thin slices are an ideal tool to aid understanding images of these modalities. Plastination is a fabulous tool, which is recommended as an adjunct for teaching anatomy.

The Great Vessels and Tributaries of the Heart of the Ringed Seal (*Phoca hispida*)

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The ringed seal (*Phoca hispida*), as well as other seals, exhibit some unique anatomical properties when compared to their terrestrial counterparts. In the ringed seal, the most conspicuous adaptation is the aortic bulb, a large dilatation of the ascending aorta, which is comparable to that found in other seal species and marine mammals. Coronary arteries are similar to those of terrestrial mammals. The branches of the ascending aorta (brachiocephalic trunk, left common carotid artery and left subclavian artery) are similar to those of higher primates and man. The pulmonary trunk originates from the right ventricle near the ventral midline of the thorax. The peculiarities of the venous system are three pulmonary veins, a pericardial venous plexus, a caval sphincter, a hepatic sinus with paired caudal vena cavae and a large extradural venous system. Generally, three pulmonary veins (right, left, middle) empty into the left atrium. The right and left pulmonary veins drain the cranial and middle lung lobes of their respective lung, while the middle pulmonary vein drains both caudal lung lobes and the accessory lobe. The pericardial venous plexus lies on the pericardial pleura on the auricular (ventral) surface of the heart. The azygous vein is formed from the union of right and left azygous veins near the 5th thoracic vertebra. The caval sphincter surrounds the caudal vena cava as it passes through the diaphragm. Caudal to the diaphragm, the vena cava is dilated excessively (the hepatic sinus) and near the kidneys it is biphid. Cardiovascular physiological studies have shown some of these anatomical variations, especially of the venous system and the ascending aorta, to be modifications for diving.

Biometrical Studies on the Cornea in Some Domestic Animals

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This work was carried out on the cornea of the eyeball of the donkey, cow, buffalo, camel, sheep, goat, pig and dog. The following dimensions were measured: the outer and inner dorsoventral and mediolateral dimension, also the

plain, weight and the volume of the cornea and fibrous coat were recorded. The morphometric studies include also thickness of the cornea and its different layers centrally and peripherally. Generally, the thickness of the cornea is about 0.5–0.8 mm, the thickest cornea was observed in cow followed by pig and donkey, the thinnest cornea was observed in sheep followed by camel, goat and buffalo, while the cornea of the goat and buffalo has nearly the same thickness. The biometry of the cornea has been useful for assessment of certain pathologic and congenital abnormalities such as microcornea, megalocornea moreover this measurement are of utmost important when superficial keratectomy and keratoplasty are indicated.

New Concepts for the Functional Maturation of the Bovine Claw: Endothelial Biology, Angiogenesis and Ageing

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Introduction: Endothelial dysfunction and impaired angiogenesis are connected to ageing. However, the mechanisms involved in age-related alterations are not fully elucidated, whereby reduced VEGF-expression and decreased mobilization of circulating endothelial progenitor cells are suggested as possible causes. When cultured *in vitro*, endothelial cells often enter a growth arrest phase called 'cellular senescence' that is consistent with many changes in age-related vascular disease. Based on the decreasing lifespan observed in high yielding dairy cows, a hypothesis has been formulated that these cows, due to their enormous metabolic performance, show an accelerated process of ageing, resulting in typical production-related diseases, particularly lameness. As a basis for further research into the role of endothelial cell biology in high producing cattle, the aim of this study was to set up a model of endothelial ageing. As a first step, the role of angiogenesis in the functional adaptation of the microvasculature of the bovine pododerma was examined.

Material and methods: Scanning electron microscopy of vascular corrosion casts of healthy juvenile and adult claws as well as pathologically altered claws, and routine light microscopy of pododermal serial sections was compared to an *in vitro* model of angiogenesis based on cultured bovine microvascular endothelial cells. For detection of angiogenic molecular control mechanisms effective in both, the pododermal system and our cell culture model, lectin- and immunohistochemistry as well as RT-PCR were carried out.

Results and discussion: In the adolescent and adult claw, sprouting and intussusceptive angioadaptation was demonstrated. Evidence of angiogenic processes was also detected in the pathologically altered claws. Immunolocalization of angiogenic factors within the dermo-epidermal system of the claw was established. The detected angioadaptation may be visible expression of the increased metabolic demands of the claw caused by the growing body-weight-load. Increase of the pododermal vascular density via angiogenesis was also the connectional reparative principle in pathologically altered claws.

Vascular Mimicry of Granulosa Cells: a New Concept of *Corpus Luteum* Development?

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So far, it was generally accepted that newly formed blood vessels are exclusively comprised of endothelial cells, and complemented by pericyte and myocyte recruitment during vessel maturation. Accordingly, participation of non-endothelial cells in the formation of blood vessels has rarely been suggested. Recently, evidence supporting the existence of tumour vessels lined by non-endothelial cells has emerged. Consequently, the concept of the inherent capacity of non-endothelial cells to behave like endothelial cells has been discussed for tumours, and this pathomechanism has been termed vascular mimicry. The *corpus luteum* is one of the most intensely vascularized tissues, and angiogenesis in the *corpus luteum* is more effective than in highly malignant tumours. Our results indicate active involvement of granulosa cells in luteal angiogenesis, and the aim of this study was to shed more light on this exciting prospect. The study was based on cultured granulosa cells isolated from the bovine ovary in different stages of follicle maturation. Morphology of angiogenic granulosa cells was studied by phase contrast, transmission electron and scanning electron microscopy. Expression of angiogenesis-regulating factors and their receptors was demonstrated by polymerase chain reaction (RT-PCR). Cultured granulosa cells underwent changes reminiscent of endothelial angiogenesis, i.e., migration, proliferation, differentiation and three-dimensional organization, and expressed angiogenesis-regulating factors and their receptors. Our results suggest a tight regulatory and structural association of endothelial and granulosa cells in luteal angiogenesis, suggesting physiological vascular mimicry in the ovary.

The Epithelial Layer of the Term Placenta of the Asian Elephant (*Elephas maximus*): a Light and Scanning Electron Microscopic Study

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This combined light and scanning electron microscopic (SEM) analyses are the first investigation of the fine morphology of the fetal membranes of the placenta in the Asian elephant. We used two term placentas with gestation periods of 634 days (Zoological Garden Leipzig) and 657 days (Zoo Hagenbeck) and selected unequal sites as specimens, particularly - free membrane of the allantois, - lateral edge of the placenta band, the 'haemophagous zone', - allanto-chorion near the placental band, and allanto-chorion on the end of the chorioallantoic sac. Microscopically the free membrane of the allantois shows a simple, cuboidal epithelium with apical domains of microplicae and microvilli (fig. 1, inset). In the SEM analyses we documented a cobblestone-like architecture of the epithelial cells with various cell sizes, small, middle and giant (fig. 1.). Furthermore, we found pear-shaped cells with long pedicle, attached to the basement membrane and cell openings, presumably due to cell desquamation (fig. 2.). The haemophagous zone was characterized with a lamellate system between the long finger-like chorionic villi and the maternal blood lacunae. The simple, squamous and columnar trophoblast cells are here bathed in the maternal blood. In some chorionic villi brown-yellow material is deposited in the fetal connective tissue. A special result was the presence of strip-like microplicae in the middle of chorionic villi. The allanto-chorion near the placental band sometimes indicated simple and ramified chorionic villi with smooth, gyrus-like trophoblast cells and foamy cytoplasm. The attached allantois possesses a simple, columnar epithelium with microvilli. Furthermore, we observed villi-like projections rising up to the allantoic cavity. The white 'pustules' on the outer surface of the end of the chorioallantoic sac consist of stratified squamous epithelia. Numerous mitoses were documented in the basal domain of the pustules. A surprisingly result was the presence of intra-epithelial capillaries. They invade the trophoblast cells to presume a nutrient exchange also outside of the placental band with its important functional metabolism. This study revealed unknown and interesting features of the epithelial layer of different locations in the placenta of the Asian elephant. Perhaps, with a detailed knowledge of the morphology of the epithelium we could convey a better understanding of functional cohesions of elephant placentation.

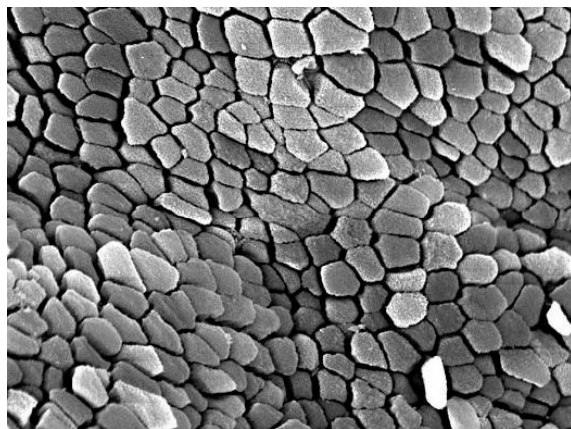


Fig. 1: The free membranes of the allantois of the term placenta of the Asian elephant (*Elephas maximus*) shows a cobblestone-like architecture of the simple cuboidal epithelial cells with various sizes. The surface morphology was determined through microplacae and microvilli.

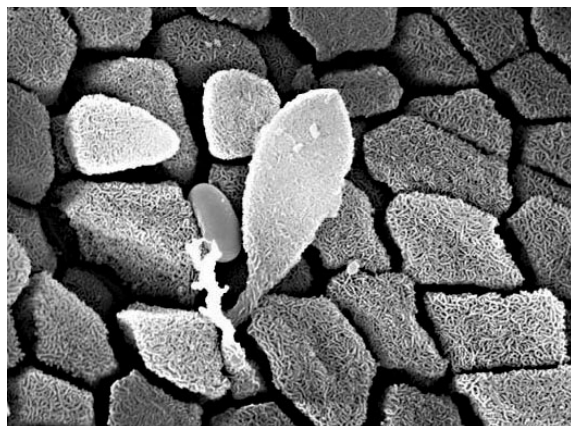


Fig. 2: Numerous pear-shaped cells with long pedicle exist among the cuboidal epithelial cells, attached on the basement membrane. The cells possessed modifications of their surface, particularly microplacae. Some pear-shaped cells show only the pedicle, perhaps that is a loss through preparation or its presumable due to cell desquamation. One erythrocyte is localized between these cells.

Effects of Ozone (O_3) on Survival and Development of Chick Embryos After Gas Exposure *In Ovo*

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We investigated the embryotoxicity of ozone (O_3), after gas exposure *in ovo*, to chick embryos of the Lohmann-Weiss

strain. The main aim of the study is to reduce the microbiological content, particularly of *Salmonella enteritidis* on the egg surface by the use of ozone in different concentrations and response times. The aim of our morphological study is to analyse the influence of gas exposure to the embryonal development of chick. Ozone has proved its worth as a disinfectant without the harmful adverse effects of formaldehyde. The chick embryo provides a useful and well-defined model for analysis of toxicity of chemicals, particularly with regard to an alternative disinfectant: ozone. Embryotoxicity of ozone in developing chick embryos was studied by examining different parameters: deformities and pathohistology of heart, liver, kidney and spleen, weight, length and survival of embryo. After oviposition the eggs were transferred to a chamber filled with ozon-oxygen-mixture in a defined concentration (1% - 5.22%). After achievement of the ozone concentration the eggs were incubated over various response times of 20, 30, 60 min or 17 h. After the incubator times of 18, 19 or 20 days we studied the development of the gas exposed embryos and unexposed controls. The organs were fixed in 4% formalin, embedded in paraffin, cut and stained with hematoxylin-eosin. High ozone concentration over 5% and high response time of 1 hour provided low graded survival rates of 3.3–13.3%. There was a significant increase of the survival rate after decline of ozone concentration and response time of 90–100%. We found also a remarkable decrease of weight and length in fewer surviving gas exposed embryos with high ozone concentration and response time (Fig. 1.), compared with the normal development of gas-exposed embryos with study of high survival rate (Fig. 2.). We calculated a 60% increase in the survival rate of embryos exposed to high ozone concentration without response time. In low ozone concentration (1.8%) with high response time of 17 hours the survival rate decreased to 10%. We found no signs of deformities and pathohistology of organs in our embryological examination. In conclusion, ozone has a dose-dependent effect. The response time constitutes an important factor in the survival and growth of chick embryos after exposure to ozone *in ovo*.



Fig. 1: High ozone concentration over 5% and high response time of 1 hour result in a low survival rate of 3.3% and the surviving embryo shows a decrease in body weight and length (right) in comparison to the biggest (left) and lowest (middle) unexposed embryo.

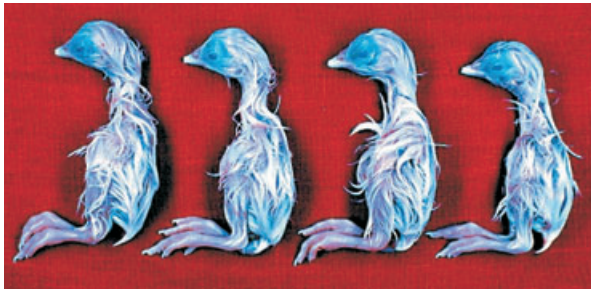


Fig. 2: Height comparison between the unexposed controls (both embryos on the left side) with gas exposed embryos (both embryos on the right side) in a study with low ozone concentration of 1% and low response time of 20 minutes. There are no differences in body weight and length between the unexposed and exposed embryos.

Long-term Cultivation of Bovine Hoof Cells in a Novel Perfusion Chamber System

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The aim of this study was to establish a co-culture of bovine hoof keratinocytes and fibroblasts in different types of perfusion chambers under defined conditions. The perfusion chamber PCS3c (Oligene, Berlin) was used to culture dermal fibroblasts and epidermal keratinocytes separated by a Millicell® Insert (Millipore, Schwalbach). In addition novel perfusion chambers developed by Dirk Hoffmann were used in combination with S & S membrane filters (Schleicher & Schuell, Dassel). First, fibroblasts were seeded onto one side of the insert/membrane and allowed to grow for 4 days. Subsequently keratinocytes were seeded onto the other side and allowed to adhere for one day. Then the chambers were connected to the tubing system with an attached 8-channel pump and a gas permeable media bag. We worked with different media and flow rates ranging from 0.035 ml/min to 0.35 ml/min. The system was run for up to 20 days. After a few days in culture the cells had grown to confluence. Then keratinocytes began to differentiate and built up stratified colonies. Within these colonies the cells showed the characteristic morphology of a stratified squamous epithelium. The use of perfusion chambers allows three-dimensional cultures to grow and survive for weeks, because of an unlimited medium supply and gas exchange. Additional, perfusion chamber systems enable the co-cultivation of fibroblasts and keratinocytes separated by membranes, which permit the exchange of molecules like growth factors. Furthermore, it is possible to perfuse the two different parts of the perfusion chamber with various and even different types of media. This pro-

vides the opportunity to add a growth factor to the medium for only one cell type and to study the effect of this factor to the other cell type. This work was supported by the European Communities under the Lamecow project QLK5-CT-2002-00969. The authors are solely responsible and the work does not necessarily represent the opinion of the European Communities.

How to Open the Mouth: the Kinetic Skull, and the Jaw and Branchial Apparatus in Sharks

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Aims: Mammals cannot swallow if their necks and, hence, their *mm. sternohyoidei et sternothyroidei* are stretched. In order to understand the functional morphology of mastication and swallowing in mammals, it is useful to understand its evolutionary origin.

Material and Methods: Five formalin-preserved specimens of the Spiny Dogfish (*Squalus acanthias*) were used for dissection or to prepare sagittal or transverse serial sections.

Results and Conclusions: The feeding apparatus of the shark *Squalus acanthias* as a representative of the cartilaginous fishes serves as an excellent model of that of an ancestral vertebrate with a kinetic skull, in which the chondrocranium is movable relative to the jaw apparatus. To open the mouth, the cranium is raised by the epibranchial musculature, the palatoquadrate cartilage (i.e., upper jaw) is entrained, and the mandibular cartilage is held in place and prevented from following the upper jaw by the coracomandibular muscle. At the same time, the branchial basket and oropharyngeal cavity are expanded and held in place by the coracohyoid and coracoarcual muscles, which prevent the hyoid and branchial arches from being entrained cranially by the raised chondrocranium. The mandible is not retracted during the opening of the mouth in sharks because the branchial region would have to be compressed instead of expanded. In the course of the evolutionary modifications leading to mammals, the heart and respiratory apparatus became decoupled from the feeding apparatus through the formation of a neck. The skull became akinetic through the integration of the upper jaw into the base of the cranium. The mouth is not opened any longer by lifting the cranium, but by depressing the mandible through a special muscle, the *m. depressor mandibulae*. The coracomandibular muscle lost its function as a stabilizer of the mandible during mouth opening and became modified into the lingual and geniohyoid muscles.

Angiogenesis and Vascular Regression in the Corpus Luteum Periodicum

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Introduction and Aim: The *corpus luteum* is one of the most intensely vascularized tissues. Luteal angiogenesis is strictly controlled and blood vessels regress completely within a short period of time. The aim of this study was to investigate vascular dynamics in relation to cellular and molecular mechanisms of luteal angiogenesis and anti-angiogenesis.

Material and Methods: Endothelial cells of blood vessels in paraffin sections of bovine *corpora lutea* from different stages were examined by labelling with the lectin Bandeiraea simplicifolia agglutinin I. Angiogenesis was studied by morphometry of the capillaries, and immunolocalization of the angiogenic factor VEGF and VEGF-receptor 2. Presence of apoptotic luteal and endothelial cells was investigated using the TUNEL test and transmission electron microscopy.

Results: During development of the *corpus luteum* (day 3–8 of the oestrous cycle) a dense capillary network (8–12% area ratio) is established and maintained until day 17. Early regression (day 18–24) is characterized by a remarkable decrease of capillaries (1% area ratio). In the regressing *corpus luteum* the number of apoptotic luteal cells is closely correlated ($r = 0.9$) to the number of apoptotic endothelial cells. VEGF is immunolocalized in luteal cells (day 3–17), smooth muscle cells and endothelial cells of arterioles of the regressing corpus luteum. During late luteal regression, a moderate increase of capillaries (2.5% area ratio) is obvious.

Conclusions: The dynamic changes of the capillarity during development and regression of the cyclic corpus luteum correlate with VEGF and VEGF-R2 activities. In contrary to expectations the late stage of luteal regression is accompanied by angiogenesis. One reason for this phenomenon may be an increase in metabolic activity resulting in re-organization of blood vessels already regressed.

Angiogenesis and Interstitial Pressure in a Rat Tumour Model

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Introduction and Aim: Angiogenesis, the formation of new blood vessels, is a crucial process in physiological and pathological growth. Pathological angiogenesis is responsible for growth and metastasis of solid tumours,

and, when blocked, improves prognosis. As a result of the angiogenic cascade in solid tumours an irregular, leaky capillary network develops. The aim of the present study was to define malignant tumours' vascular characteristics and reveal functional anatomy by quantification of the microvasculature and interstitial pressure (IP) in relation to tumour fluid dynamics as visualized by contrast enhanced magnetic resonance imaging (MRI).

Material and Methods: Dynamic MRI and measurement of the IP was performed in 21 rats implanted with colon carcinomas subcutaneously. Angiogenesis was studied by morphometry of the capillaries, and immunolocalization of the angiogenic factor VEGF and VEGF-Receptor 2.

Results and Conclusions: Histology, immunohistochemistry and MRI confirmed concentric arrangement of 4 tumour zones. The tumour margin included loose connective tissue with abundant mononuclear cells. Many large microvessels were seen in this most intensely vascularized zone. IP measurement in this zone was adjusted to the zero level. Diameter of the peripheral zone of vital cells measured 1.3 mm. Capillaries were smaller and sparse. Dynamic MRI revealed peripheral washout of the contrast agent in this zone. After an initial increase of the signal intensity a hypo-intense rim was formed within a few minutes. The intermediate region was characterized by islands of vital tumour cells containing 3% capillaries (hot spots). The innermost area, the necrotic zone, took 35% of the total tumour area with less than 0.5% vessels. The IP increased from the periphery to the centre. VEGF and VEGF-receptor 2 was found in the vessels of the tumour margin and vital tumour cells of the peripheral zone. From this can be concluded that the peripheral washout phenomenon seems to be correlated with elevated interstitial pressure and increased capillary density and therefore may be a reliable sign of malignancy.

Morphology as a Tool in Embryo Technology

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The increased use of embryo technologies, in particular in cattle, has emphasized the need for assessing embryo quality by morphological techniques, such as transmission electron microscopy, immunohistochemistry, immunocytochemistry for confocal laser scanning microscopy, and fluorescence *in situ* hybridization. By a combination of these techniques, we have demonstrated that (1) the rRNA gene activation in the 8-cell bovine embryo as monitored by nucleolar development is comparable in embryos developed *in vivo* and produced *in vitro*, whereas embryos reconstructed by cloning by nuclear transfer

display deviations in nucleolar development, (2) generating bovine embryos by *in vitro* production and reconstruction by nuclear transfer is associated with increased occurrence of apoptosis, in particular in the inner cell mass of the blastocyst, (3) the same two embryo production techniques are associated with increased occurrence of mixoploidy, i.e. embryos presenting a major population of normal diploid cells and a minor population of abnormal haploid or polyploid cells, and (4) upon hatching from the *zona pellucida*, the inner cell mass of the bovine blastocyst differentiate into hypoblasts and epiblasts, the latter of which forms a laminin enclosed pluripotent cell niche staining positive for the pluripotent transcription factor Oct-4; a morphology that resembles that maintained by bovine ES-like cells cultured through several passages *in vitro*.

Angioarchitecture of the Branchial Arterial System of Carp (*Cyprinus carpio* L.)

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The arterial system of the gills of carp and its histological structure were studied light and electron microscopically by making Mercor or Neoplane Latex corrosion cast preparations. Four pairs of afferent and efferent branchial arteries, and a pair of afferent and efferent pseudobranchial arteries were identified in the branchial arterial system. The 1st and 2nd afferent branchial arteries are given off directly from the ventral aorta, and the 3rd and 4th afferent arteries originate from their common trunk, which is branched off from the ventral aorta caudal to the origin of the former branchial arteries. Numerous afferent filamental arteries are connected to the lamellar blood capillary networks in the gill lamellae via afferent lamellar arterioles, and efferent filamental arteries followed the efferent lamellar arterioles are converged into four efferent branchial arteries that are connected to the dorsal aorta. To the pseudobranchia, afferent pseudobranchial arteries are connected with the ventral branches of the 1st efferent branchial arteries to provide arterial blood to the organ through the afferent mandibular arteries. Afferent pseudobranchial lamellar arterioles originating from the afferent pseudobranchial filamental arteries are connected with the blood capillary networks in the pseudobranchial lamellae, and blood in the capillary networks is drained into the efferent pseudobranchial filamental arteries via 2-4 pseudobranchial lamellar arterioles. Branches of the efferent pseudobranchial filamental arteries are connected with the arteries to the eyeballs and provide blood to choroid of the vascular tunic of

them. Pseudobranchial cells surrounding lamellar capillaries in the pseudobranchia are furnished with abundant mitochondria and tubular structures, and the histological findings suggest the cells may share an ability to exchange physiological materials between the cells and the blood in the capillary networks of pseudobranchia.

Fish immunology. Fish Erythrocytes Release Cytokine-like Factors that Modulate Autologous Macrophage Function

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According to our previous findings (Passantino et al. 2002a,b; 2004) we discovered that fish erythrocytes (FE) act as immune cells in that they bind and engulf *Candida albicans* (CA). Then, these activated FE interact with macrophages (MØ) so that a so-called rosette results from such an interplay. A rosette consists of one or more MØ containing ingested CA, centrally located and surrounded by CA activated FE. Quite interestingly, we found that supernatants of CA-activated FE contain cytokine-like factors able to modulate MØ function. In particular, active supernatants are able to inhibit MØ migration under agarose and this activity was identified as macrophage inhibiting factor (MIF) – like function, by analogy to human MIF. On the other hand, same supernatants could enhance MØ phagocytosis of CA, thus contributing to fungus killing. Taken together, these results suggest that FE act as immune cells by secreting cytokine-like factors, which contribute to the rosette formation through modulation of MØ.

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The Distribution and Chemical Coding of Intramural Nerve Structures Supplying the Porcine Stomach

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The present study investigated the arrangement and chemical coding of intramural nerve structures supplying the porcine stomach. Tissue samples comprising all layers of the wall of the ventricular fundus were collected from juvenile female pigs ($n = 4$), which were first deeply anaesthetized and then transcardially perfused with buffered paraformaldehyde. The cryostat sections were processed for double-labelling immunofluorescence to study the distribution of the intramural nerve structures (visualized with antibodies against protein gene-product 9.5) and their neurochemical characteristics using antibodies against vesicular acetylcholine transporter (VACHT), nitric oxide synthase (NOS), galanin (GAL), vasoactive intestinal polypeptide (VIP), somatostatin (SOM) and substance P (SP). The study confirmed the presence of three distinct nerve plexuses within the wall of the porcine stomach including one myenteric plexus and two, outer and inner, submucous plexuses. The outer and inner submucous plexuses (OSP and ISP, respectively) were similar in respect to the chemical coding of neurons they contained. Most of the neurons expressed immunoreactivity to SP (ISP 58%; OSP 60%) or to VACHT (ISP 56%; OSP 56%), some of them stained for GAL (ISP 18%; OSP 15%) and solitary nerve cells were SOM-positive (in ISP only). No neurons in the submucous plexuses displayed immunoreactivity to VIP or NOS. In the myenteric plexus, some neurons stained for NOS (20%), VACHT (15%), GAL (10%), VIP (8%) or SP (8%) while no neurons immunoreactive for SOM were encountered. In both submucous and myenteric plexuses, many varicose nerve fibres expressed immunoreactivity to VACHT, GAL or SP, while VIP-, SOM- or NOS-positive nerve terminals were less numerous. The comparison of the present results with those obtained

by other authors has revealed distinct inter-species differences regarding the arrangement and chemical coding of nerve structures supplying the mammalian stomach.

Vasculogenesis by Endothelial Progenitor Cells *In Vitro*

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Introduction and Aim: Vasculogenesis is the *de novo* formation of blood vessels from embryonic precursor cells. The formation of additional vessels is called angiogenesis, the sprouting from pre-existing vessels by migration, proliferation and differentiation of endothelial cells. Until recently vasculogenesis was considered to be restricted to embryogenesis, however, evidence suggests that progenitor cells exist in the adult, able to participate in neovascularization. In order to study cellular and molecular control mechanisms without animal testing we have established *in vitro* featuring the three-dimensional formation of capillary-like structures from microvascular endothelial cells. The aim of this study was to identify the specific subpopulation of cells that triggers the primary process of vasculogenesis *in vitro*.

Material and Methods: Endothelial cells were isolated from the bovine *corpus luteum* and examined by phase contrast and transmission electron microscopy. Immunolabelling was done for the most important pro-angiogenic factor (VEGF) and its endothelium-specific receptors (VEGF-R1, -R2) as well as for endothelial cell surface markers (CD 31, CD51/61, CD29) and presumptive stem cell marker CD 34. Additionally VEGF and VEGF-R1, -R2 expression was demonstrated by polymerase chain reaction (RT-PCR).

Results: The beginning of vessel formation *in vitro* was initiated by confluence of endothelial cells. A linear and circular side-by-side arrangement of cells and the formation of capillary-like structures with an internal lumen could be observed. Specific clusters of cells were identified. These cells were arranged in a star-like formation, i.e., one particular round cell in the centre surrounded by several cells with extensions. Immunolocalization of VEGF, VEGF-R1, -R2, CD 31, CD51/61, CD29 and CD34 was positive. VEGF-R2 and CD 34 were found to be co-localized in specific cells exclusively.

Conclusions: Results for the first time reveal the specific subpopulation of endothelial cells that trigger the start of vasculogenesis *in vitro*. These progenitor cells are characterized by a specific morphology, by the presence of endothelial cell markers and by the expression of stem cell marker *in vitro* models

Morphological Studies on the Development of the Bronchial Epithelium of the Fetal Camel Lung

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Light microscopic observations revealed that in camel foetuses of 25 mm crown-to-rump length (CRL) the primordial tubular system of the prospective lung was formed of several tubules lined by undifferentiated columnar epithelium. Intra-epithelial neuroendocrine cells were the first elements to be differentiated in the lining epithelium of the primordial tubular system of the prospective lung as early as 25 mm CRL. On reaching 50–67 mm CRL, the primordial tubular system started to differentiate into two systems of primordial tubules, the prospective bronchial or light tubules and the future respiratory or dark tubules. The lining epithelium of the prospective bronchial tubules revealed a clear evidence of ciliogenesis as early as 80 mm CRL. From 800 mm CRL onwards, the bronchial epithelium demonstrated ciliated and non-ciliated secretory cells. The non-ciliated secretory cells of the bronchial epithelium of fetal camel lung showed moderate reaction to AB/PAS technique, for the first time, in fetuses reaching 600 mm CRL.

Epithelium of the Bursa of Fabricius Mucosa

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The structure and topography of the Bursa of Fabricius (BF) differs from other central immune organs. Structural components, providing BF function are located in mucosa, which is constantly influenced by antigen stimulation. Therefore the reports, which have appeared recently, indicate that BF may function as a peripheral immune organ. The latter is known to be situated in the alimentary canal mucosa (caecal tonsils, Peyer's patches) and possess peculiar epithelium. The aim of our study was to investigate the structure and cell composition of BF epithelium in chicken, turkey, quails, ducks and pigeons using light and electronic microscopy. The studies have shown that epithelium overlying BF folds is pseudostratified columnar, but between them and in their base it might be cuboidal or columnar. Pseudostratified epithelium consists of columnar cells with a few microvilli on

their apical surface, goblet cells, infrequent endocrine cells and macrophages. BF epithelium is spongy due to its infiltration by lymphoid cells. The large quantity of the latter is found in subepithelial connective tissue. Infiltration by lymphoid cells is considered to be typical for the alimentary tract mucosa, where the aggregated lymphoid nodules are situated. Thus, our study supports the thoughts that BF may have function of central and peripheral immune organ.

On the Development of the Feline Skeleton

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To find precise data about the development of the skeleton of the domestic cat (*Felis catus*) embryological sections cleared specimens stained for bone and cartilage and radiographs were used (for details see our upcoming publication). The pre-natal development is characterized by three main phases: the mesenchymal skeleton, already formed around stage 5 (Knospe, 2002) of the cat's development, with the notochord, somites, pharyngeal arches and limb buds, the cartilage skeleton arising during stage 14, and the bony skeleton formed by intramembranous and endochondral ossification from stage 16/17 on. The mesenchymal skeleton survives as periosteum in different ligaments, joint capsules, syndesmoses and intervertebral disks. The cartilage skeleton is reduced up to birth to the epiphyseal and tympanic cartilages, and survives as laryngeal, tracheal, joint, rib, sternal, nasal cartilages and synchondroses. In the newborn cat all primary centres are present, except the carpal and distal tarsal bones, secondary centres aren't established radiographically yet. The secondary and tertiary centres are formed post-natal from the first week on up to the second year. The results are presented in nine slides showing histological pictures with the main pre-natal events and schemes of the post-natal development with the skull, trunk and limb centres and their first appearance.

Morphometric and Structural Examination of the Collateral Ligaments of the Canine Elbow Joint

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The aim of this study was to describe morphological differences between the lateral collateral ligament (LCL) and the medial collateral ligament (MCL) of the canine elbow

joint. Forty forelimbs of 22 dogs (adult German shepherd dogs and shepherd mongrels) were dissected. The length and width of defined segments of the ligaments (i. e. total length, areas of origin and insertion, 'free part', 'humeral part', 'antebrachial part') were measured in extension (160°) and flexion (30°). Statistical analyses of the data were performed using the data analysis software SAS 8.0. In addition the collateral ligaments of ten forelimbs of five shepherd mongrels were studied histologically. The LCL differs from the MCL in the following morphological items: a) the areas of origin and insertion are larger and in consequence the 'free part' of the lateral ligament is shorter; b) the caudal crus of the LCL is fan-shaped with a broad insertion area at the ulna whereas the caudal crus of the MCL remains slim; c) in the LCL the 'humeral part' varies in length depending on extended or flexed position of the joint, which is due to the spiral shape of the capitulum humeri; d) the microscopical structure of the LCL is organized less tightly; the collagen fibre bundles cross in varying angles. Referring to these details the LCL seems to allow and to limit a slight rotation of the forearm when the elbow joint is extremely flexed.

Functional Morphology of Equine Pre-ovulatory Cumulus-oocyte Complexes

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Knowledge of the functional morphology of equine cumulus-oocyte-complexes (COCs) shortly before ovulation is scarce. Therefore the aim of our studies was to characterize COCs matured *in vivo* by light microscopy, transmission electron microscopy (TEM), scanning electron microscopy (SEM) and immunohistochemistry (ICC). Pre-ovulatory COCs were obtained by ultrasound-guided transvaginal aspiration of pre-ovulatory follicles of 26 pony mares. The morphology of pre-ovulatory COCs was compared to that of immature COCs obtained by slicing of ovaries from the slaughterhouse. As shown by light microscopy and SEM, immature COCs possess numerous round, densely packed cumulus cells, which contact each other and possess merely single short processes. Contrary, in pre-ovulatory oocytes the cumulus cells are widely separated but send long cytoplasmic processes to the *zona pellucida* (ZP). All the cumulus cells are embedded in a huge amount of homogenous extracellular matrix. As shown by alcian blue staining and Periodic Acid Schiff Reaction (PAS) with and without amylase digestion, this matrix contains glycoproteins and acidic and sulphated glycosaminoglycans. In pre-ovulatory COCs both the oocyte

and the cumulus cells produce glycosaminoglycans, whereas immature COCs are negative for alcian blue. Similarly, glycoproteins are synthesized in pre-ovulatory, but not in immature COCs. As shown by ICC, hyaluronic acid is one of the most abundant mucopolysaccharide in the pre-ovulatory COC. The high synthetic activity of the cumulus cells in the pre-ovulatory COC is confirmed by TEM showing densely packed endoplasmic reticulum (ER) and accumulation of glycogen in the cumulus cells. Our results imply that in the equine *in vitro* maturation of the oocyte is characterized by the cumulus cells synthesizing an extracellular matrix of glycoproteins and acidic and sulphated glycosaminoglycans. The extensive production of extracellular, water-binding matrix in the pre-ovulatory COC ensures mechanical protection and nutrition of the oocyte.

Regulation of Cumulus Cell Numbers in the Course of the Canine Estrous Cycle

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In the canine, success rates of assisted reproduction techniques are very low. Since cumulus cells are known to play an essential role during the *in vitro* maturation (IVM) of oocytes, the aim of this study was to investigate the regulation of cumulus cell numbers in the course of the canine estrous cycle. Eighty-two cumulus-oocyte-complexes (COCs) were obtained from 16 bitches undergoing elective ovariectomy during proestrus, estrus, metestrus and anestrus. First, mean cumulus cell numbers during the different stages of the cycle were compared. Then the incidence of programmed cell death (apoptosis) was detected by terminal deoxynucleotidyl transferase mediated dUTP labelling (TUNEL). Scanning electron microscopy (SEM) was used to confirm the presence of apoptotic cumulus cells. Additionally, mitosis was detected by using antibodies directed against the nuclear cell proliferation-associated antigen Ki-67. The mean number of cumulus cells in the COCs increased from proestrus to estrus (168.2 ± 9.2 to 265.5 ± 32.7 , $P < 0.01$) and decreased from estrus to metestrus (265.5 ± 32.7 to 139.1 ± 10.7 , $P < 0.01$) till anestrus (150.0 ± 8.3 , $P < 0.01$). The percentage of apoptotic cumulus cells was generally very low during proestrus and oestrus (3.3% and 12.6%, $P < 0.01$) and distinctly increased during metestrus (83.1 %, $P < 0.001$). The predominance of apoptosis in the cumulus cells during metestrus was confirmed by SEM, which showed a high

number of cells revealing cell shrinkage, loss of membrane integrity and cell disruption. Mitosis was highest during proestrus (2.7%) and oestrus (2.6 %) and decreased in the course of metestrus (1.8%) and anestrus (0.2%, $P < 0.05$).

Our results imply that the distinct increase of cumulus cell numbers during proestrus and oestrus is due to increased mitosis and reduced apoptosis. The distinct reduction of cumulus cells during metestrus is predominantly regulated by apoptosis.

Embryo-maternal Communication during the First Days of Embryonic Life

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The mechanisms of embryo-maternal communication during the first days of embryonic life are largely unknown. Using the bovine as a model, the aims of our study were to morphologically characterize the interaction between the pre-implantation embryo and the epithelium of the maternal ampulla, isthmus and uterotubal junction by light and scanning electron microscopy. For this purpose, oviducts were removed from cows revealing a functional *corpus luteum* on day 3 after insemination. These were compared to oviducts removed on day 3 (metestrus) of the estrous cycle. Three days after insemination, the majority of the epithelial cells in the ampulla were secretory cells distinctly protruding into the oviductal lumen. Contrary the ampulla of cows on day 3 of the cycle predominantly revealed ciliated cells in the oviductal epithelium. As shown by Periodic Acid Schiff reaction (PAS) with and without amylase digestion, the secretory cells of the ampulla synthesized merely glycoproteins during metestrus, but large amounts of glycogen during pregnancy. In the isthmus no morphological differences were seen between pregnant and cyclic cows. The most conspicuous finding during pregnancy was seen in the uterotubal junction: Vital cumulus cells embedded in between epithelial cells had developed short cytoplasmic processes intensely contacting the epithelial uterine cells. The embryos obtained *ex vivo* were regularly covered with a thick layer of homogenous extracellular matrix. Contrary embryos produced *in vitro* – both with and without coculture with oviductal cells – revealed a clearly visible *zona pellucida* with spongy appearance and numerous pores. Our results imply that already during the first days of life there is intense interaction of the pre-implantation embryo and the maternal genital tract part of which may be mediated by cumulus cells.

Laser Micro-dissection and Quantitative RT-PCR as a Tool for Cell Identification

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We have applied a laser micro-dissection microscope followed by RT-PCR analysis of micro-dissected tissue to identify gene expression of various immune genes in the Atlantic salmon, *Salmo salar*. A major problem in teleost anatomy is the lack of specific cell markers to identify different leukocyte populations. For instance, the involvement of T-cells in inflammatory reactions is unknown. We used frozen sections from the gut of salmon fed to induce localized inflammatory reactions and micro-dissected regions showing proliferative changes. RNA was extracted from the micro-dissected specimens and subjected to quantitative RT-PCR using primers and TaqMan probes for T-cell receptor, MHC class II and elongation factor, which served as a house-keeping gene. The results showed expression of T-cell receptor, MHC class II and elongation factor. The expression of the elongation factor corresponded invariably to the area micro-dissected tissue. In conclusion, T-cells are present in teleost inflammatory reactions.

Incompleteness of the Structure of Liver Veins and Tissue Components in Neonatal Animals

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Introduction: The formation of the liver structure at mammal embryogenesis is interconnected by drawing up of venous mains. In parallel to the transformation of the main veins, there is a reduction of the blood formation function and the liver transformation from the primary organ of embryogenesis into the secondary one, but the exception is immaturity and underdeveloped maturity mammals (Smirnova, 1967; Mechtiev, 1968; Bresler, 1993). The purpose of research is to establish morpho-functional criteria of the incompleteness in the blood vessels and in tissue components of the liver in neonatal animals.

Material and Methods: The afferent and efferent liver veins of calves, piglets and puppies at one day of age were investigated. The complex of morphological, roentgenological, ultrasound and statistic methods was used.

Results: It was established that the liver veins in newborn animals portoumbilical (afferent) and caval (efferent) collectors. In 1-day-old calves, the afferent collector is connected with caval way of the large vein flow of the main type. Relative weight of the liver in the calves is 2.31–2.85%, hepatocytes form balks and they are not always oriented radially to the central venuels. There are no seats of haemopoiesis. Sonography of the liver and its veins proves inconstancy of the sizes depending on acts of breath. In 1-day-old piglets, many treelike anastomoses follow with the left hepatic vein. Relative weight of the liver in the piglets is 2.27–4.64%. The parenchyma of the organ is spongy with the seats of haemopoiesis (0.04–3.61%). In 1-day-old puppies afferent and efferent collectors combine through the venous duct. Additionally, in the abdominal cavity of puppies, there are the vitelline artery (starts from a. mesenterica) and vein (runs to v. porta near the afferent collector), following to the umbilical ring. Relative weight of the liver in the puppies is about 5.18–6.04%. Hepatocytes form the chaotic congestion with haemopoietic points (2.04–7.14%).

Conclusion: Thus, incompleteness of morphology of the liver is shown in preservation of provisional vein structures and tissue components of the embryonic type, which is one of the factors complicating diseases of digestive organs in neonatal animals.

Comparative Morphology of the Arterial Circle of the Brain in Some Species of Mammals

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Investigations were performed on 10 adult individuals of chinchilla (*Chinchilla laniger*), midday gerbil (*Gerbillus gerbillus*), rabbit (*Oryctolagus cuniculus*), guinea pig (*Cavia porcellus*) and Egyptian spiny mouse (*Acomys cahirinus*). The animals were killed with ether anaesthesia and injected with coloured acryl latex through the abdominal aorta or left heart ventricle. The heads of the animals were fixed in formalin and after decalcification in 5% HNO₃, injected arteries of the base of the brain were exposed. The different types of arterial circle of the brain in investigated species were observed. The open type with morphological asymmetry in the anterior part was found in chinchilla. It depended on an unusually strongly developed anterior cerebral artery at

the right side. In the midday gerbil the most common form was the closed type of investigated circle, with a lack of typical communicating artery posterior. The morphological asymmetry of the anterior part of the circle without fully developed anterior cerebral artery was found in the rabbit. Typical, the closed type of investigated structure with little individual variability was observed in guinea pig. The most individual variability connected with anterior and posterior part of the Willi's circle was found in the Egyptian spiny mouse where it was difficult to describe the typical arterial circle of the base of the brain.

A Novel *In Vitro* Model of Canine Malignant Hemangioendothelioma

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Introduction and Aim: Canine malignant haemangioendothelioma is an aggressive neoplasia that affects mostly older dogs of large breeds with a strong predilection for the spleen, liver, heart and skin. The tumour originates in the vascular endothelium and consists of transformed cells forming large and leaky vessel-like structures. Prognosis is poor because surgery and chemotherapy have limited success in prolonging survival times and increasing quality of patients. A new strategy to treat this malignancy could be anti-angiogenic therapy based on the inhibition of proliferation, migration and three-dimensional organization of transformed cells. In order to reduce animal experiments, *in vitro*-models are required to test the safety and efficacy of anti-angiogenic drugs. So far only few models of angiogenesis are available using mostly human, rodent and bovine cells. Therefore, the aim of our study was to establish an *in vitro* model of canine haemangioendothelioma.

Materials and Methods: Tumours were collected from dogs during surgery or immediately after euthanasia. Isolation of cells was done from different areas of the tumours and by enzymatic digestion of the tissue. Cells were incubated in culture media with and without endothelial growth factors. Cells were characterized by lectin histochemistry using *Dolichos biflorus* agglutinin, *Ulex europaeus* agglutinin and *Bandeiraea simplicifolia* agglutinin I. Moreover, RT-PCR (polymerase chain reaction) was employed to investigate the expression of vascular endothelial growth factor (VEGF) and its endothelium-specific receptors VEGF-R1 and -R2.

Results and Conclusions: Different populations of cells were isolated and cultured successfully from canine malignant haemangioendothelioma. Cells show characteristics of microvascular endothelial cells of an angiogenic phenotype, i.e. the formation of spheroids and

tube-like structures as well as strong labelling for *Bandeiraea simplicifolia* agglutinin I. Thus, morphological and glycohistochemical results confirm the vascular character of the cells isolated. RT-PCR showed expression of VEGF. However, endothelium-specific VEGF receptors were not expressed. Loss of typical receptors is common in cancer and may correlate with increased tumour dedifferentiation.

Veterinary Anatomy Education in Korea

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In Korea, there are 10 veterinary schools and about 500 new veterinarians are produced annually. Veterinary education is 6-year-system composed of pre-veterinary (2 yrs) and veterinary course (4 yrs). Total of 28 to 33 subjects are taught in professional curriculum, including 21 of 23 subjects recommended by the World Veterinary Association (WVA) as minimum requirements for veterinary education. Only economics and botany are excluded. Currently, 6648 veterinarians are active in the profession in Korea; about 31% of these in clinical practice and 69% in non-clinical areas. Korea has 3.5 million pets, 10.4 million livestock, 105 million chickens, and 0.5 million other animals. In general, 630 livestock units (LU) are covered by each veterinarian. With improvement of economic status and life style change, the number and variety of species of companion animals are expanding, accordingly. Also the importance of molecular biology is recognized and included in the professional curriculum. In these changing times, it is believed that gross anatomy, histology and embryology are essential integral component of found in veterinary education. But the teaching hours of histology, embryology and gross anatomy have reduced 15% during last 5 years due to new subjects. Total teaching hours in professional curriculum is about 4100 hours (pre-clinical 29%, paraclinical 32%, clinical course 38%). Average 200 hours are allocated for gross anatomy, histology and embryology lecture and average 250 hours for laboratory practice. The number of students per class ranges 40–80 depending on schools. The ratio of male and female students is 6–4, and the teaching faculty and students 1–22. The Korean Association of Veterinary Anatomists (KAVA) recommend one cow, five dogs, two pigs, five goats and 20 chickens per 20 students to be used for the dissection. The students are evaluated using various types of exam such as assignments, written and oral tests. Recently some schools are beginning to adopt block lectures and/or PBL.

Calcium-Binding Proteins and Neurotransmitters in the Stomach Myenteric Plexus of the Korean Native Goat

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The expression of calbindin D-28k (CB), calretinin (CR), substance P (SP) and calcitonin gene-related peptide (CGRP) in the stomach myenteric plexus of the Korean native goat stomach was investigated by immunohistochemistry. The results demonstrated the presence of nerve fibers and cell bodies immunoreactive (IR) to CB, CR, SP and CGRP. In tissues of rumen, reticulum, omasum and abomasum, some distinct neuronal populations could be distinguished according to their morphologic and neuronal chemical properties: Dogiel type I cells which have irregular lamellar dendrites and a single axon, Dogiel type II cells which have large ovoid cell bodies and several long axon-like processes, and small filamentous interneurons. CB-, CR-, SP- and CGRP-IR neurons and fibers were observed in the myenteric plexus of stomach, and varicose nerve fiber immunostained to SP and CGRP also were found in the muscle layer. In myenteric plexus of the stomach, CB- and SP-positive neurons were characterized by Dogiel type II and CR-IR neurons were classified Dogiel type I with lamellar dendrites, and immunoreactivity of CGRP was very weak in the somata. SP- and CGRP-IR nerve fibers formed dense networks within the myenteric ganglia. SP-IR cell bodies and their fibers were found in the myenteric plexus, and the immunoreactivity and number of cell bodies were more than CB-, CR-, and CGRP-IR neurons. These results suggest that SP, CGRP, CB and CR in the myenteric neurons of Korean native goat stomach may have play an important role in the dynamic movement. (Support contributed by: Korean Research Foundation 2003-015-E00195).

Effect of Flooring Systems on Microarchitecture of the Bovine Hoof Horn

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Introduction and aims: Claw diseases in dairy cows not only lead to high economic losses, but also cause pain and

reduce animal welfare. Former studies pointed out that flooring properties influence the microstructure of hoof horn. In this study, we looked for correlations between different flooring systems and horn quality. We obtained horn samples from an experimental farm in Alnarp/Sweden from cows that were first kept on slatted floor and subsequently on mastic asphalt respectively on rubber mats.

Methods: Untreated horn samples were glued onto wooden blocks: 7 μm sections were prepared with a Polycut-S (Leica) microtome and afterwards stained with the PAS reaction. For morphometrical analysis we used a Zeiss Axioskope, a digital camera (Progress 3012) and the Lucia 32 G software (Nikon). These data were exported into EXCEL (Microsoft) and statistical analysis was carried out with SPSS (SPSS Inc. [[Au: please provide manufacturer details for 'SPSS': city, state code (if USA) and country]]). We measured area, minimal and maximal diameter of medulla and cortex of the horn tubules. Statistical evaluation included number of tubules and tubular area per visual field, total area of medulla and cortex and the limiting diameters.

Results: We found significant alterations of the hoof horn after moving the cows from hard to soft floor. The diameter of the horn tubules decreased significantly. The area of medulla and the ratio of medulla to cortex decreased as well.

Conclusions: Our findings indicate a higher quality of claw horn and an increased resistance against mechanical, chemical and microbial environmental influences in animals kept on soft floors. The functional implications of these observations are discussed with regard to claw health as well as the arising consequences for dairy housing systems. This work was supported by the European Communities under the Lamewcow project QLK5-CT-2002-00969. The authors are solely responsible and the work does not necessarily represent the opinion of the European Communities.

Pressure Distribution Under Paws of Chinchilla (*Chinchilla Laniger*)

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The purpose of this study was to estimate the pressure distribution under the paws of chinchilla (*Chinchilla laniger*) using static pedobarography. We included in our study three healthy chinchilla aged 1–3 year, both sexes, weighted 480–570 g. We performed static pedobarography (PEL 38 podometer) in each chinchilla and we took

into account standing on four legs or two hind legs. We found that on standing on four legs, two types of contact area with base in chinchilla exist: first type – the ring shape of contact area with base of forefeet with maximal pressure in forefeet at their central part and elliptic shape of foot contact area with base of hind legs (paws) with longer diameter about twice longer than the shorter one and with maximal pressure in hind legs at their posterior part (greater than pressure under forefoot paws), that a value was 98 g/cm². In the second type of foot contact area with base a pattern of hind legs was similar to that of forefeet. During standing on two hind legs we distinguished the another type (third) of foot contact area with base that was similar to the first type with the difference made by the fact that animal was standing only on hind paws. In each animal we found also the contact area with base of a tail – the contact area was similar to that of forefeet but the value of pressure was twice decreased. Concluding, the most frequent manner of contact area with base in chinchilla is 4 points contact with hind paws contact area twice greater than contact area of forefeet. Moreover, the fifth point of support can exist – a tail, which is mainly important during standing on the hind paws.

Visualization of Surface-specific Antigens in Various Strains of Enterotoxigenic *E. coli*

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Proteinaceous surface antigens of enterotoxigenic *E. coli* (ETEC) appear as pili, and are important virulence factors as they allow bacteria to attach to the small intestinal mucosa. Surface antigens are classified as colonization factor antigens (CFA) and coli surface antigens (CS). Known groups include CFA/I, CFA/II (consisting of CS1, CS2 and CS3), CA/III and CFA/IV (consisting of CS4, CS5 and CS6). The goal of the present study was to examine the morphology of pili by transmission electron microscopy (TEM) and to localize specific surface antigens by immunolabelling. Using different strains of *E. coli* grown under various culture conditions, pili were visualized by negative staining and corresponding surface antigens were demonstrated by immunogold-labelling using both polyclonal and monoclonal antibodies. Expression of pili was dependent on culture conditions and sample handling. In contrast to CFA/I and CS3, CS6 pili were not detectable after negative staining. Selected antibodies, however, allowed surface antigens to be demonstrated

unequivocally. These results will be of value in investigating the expression of colonizing factors in genetically modified bacterial strains.

3D-Determination of the Position of the Distal Phalanx Relative to the Claw Capsule in Cattle

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Introduction: On one hand, the distal phalanx in cattle is suspended by the wall segment of the dermis, on the other hand it is supported in the palmar/plantar region by the bulb with its specialized subcutaneous tissue. Recent results of various studies suggest that changes in tissue composition of the bulb and slackening of the suspension can lead to an unphysiological position of the distal phalanx within the horn capsule. This leads to an altered distribution of pressure and mechanically influences the dermal blood supply. In the worst case these changes may lead to sole ulcer.

Aims: The aim of this study was to develop a method to allow measuring the minimal distance between the surface of the distal phalanx and the inner surface of the horn capsule under various loading.

Material and Methods: The right front and the left hindclaws have been taken from seven healthy beef bulls (German Simmentals, average body weight 580 ± 25 kg) from the local abattoir. After functional claw trimming the feet have been scanned in a computed tomograph (veterinary surgical clinics) in unloaded and the loaded state [distribution of body weight: 55% (front legs): 45 % (hind legs)]. Each distal phalanx and claw capsule have been reconstructed in both datasets via thresholding ('marching-cube-algorithm'). The minimal distance between both objects has been measured by the 'nearest-neighbour-algorithm'.

Results: No significant differences could be observed between front and hind claws and lateral and medial claws respectively. In the unloaded state there was a distance of 4–6 mm in the dorsal, abaxial and axial aspect of the distal phalanx, which almost remained constant in the loaded state. On the solear surface values increased from 3 mm at the tip of the coffin bone to 7–8 mm at the flexor tubercle where significant distance changes (approx. 3 mm) could be observed.

Discussion: A tilting movement of the distal phalanx becomes obvious with loading. Thus, especially the region

of transition between the soft and the hard bulb experiences heavy compressive stress. This coincides topographically with the incidence of sole ulcers.

Biomechanics of the Fractured Medial Coronoid Process and the Isolated Anconeal Process in the Canine Elbow Joint

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Introduction: Elbow dysplasia is one of the most important orthopaedic diseases of the canine elbow joint. The medial coronoid process (MCP) and the anconeal process (AP) are involved with a high incidence.

Aims: The aim of this study was to clarify whether these processes are especially loaded resulting in osteoarthritis.

Material and Methods: Elbow joints were examined from dogs of various breeds, with a body weight over 20 kg and an age ranging from 1 to 12 years. Only joints without damage to the articular cartilage have been included in this investigation. Articular surfaces have been evaluated macroscopically, subchondral bone density (long-term loading) and split-lines (long-term tensile loading) have been determined.

Results: In the humeral fossa olecrani, there was a distinct impression on the inner side of the lateral compared to the medial epicondyle. In the ulna, the MCP was much greater than the lateral coronoid indicating that the medial process has to support the humeral condyle to a higher extent. Subchondral split lines with a transverse orientation in the fossa olecrani gave evidence of long-term transverse tensile loading in this area. Split lines on the MCP were oriented radially as if the lateral edge was bent downwards. Subchondral bone density in the fossa olecrani was higher towards the lateral epicondyle in comparison to the medial. Furthermore, there was a bone density maximum on the medial part of the humeral condyle opposite of the MCP with its very high density.

Discussion: Gait analyses showed that there is a transverse, medially oriented force of up to 4% bodyweight acting on the paw during midstance. As the carpus is stable when slightly hyperextended during midstance loading there is a long lever arm from the ground up to an assumed rotation centre in the depth of the trochlear notch. The medially directed ground reaction force slightly rotates the forearm inwards causing a bending

moment about the elbow joint, which leads to an increased pressure of the AP and the MCP. This bending in addition to sagittal loading is the reason for the high susceptibility of the MCP and AP.

Functional Anatomy and Biomechanics of the Canine Hip Joint

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Introduction: In many breeds osteoarthritis of the hip joint is the most typical orthopaedic disease. Numerous studies concerning etiology, pathogenesis and therapy of this disease still consider an exact congruity to be the basis of a physiological joint function. No data, however, are available as far as the distribution of load in the articular surfaces is concerned.

Aims: It was the aim of this investigation to describe the biomechanical features of the canine hip joint as exactly as possible with respect to long-term and momentary loading.

Material and Methods: Hip joints from 43 dogs (body-weight >20 kg) with an average age of 6 years were investigated. The subchondral bone density was determined with the method of CT-osteodensitometry. With help of cartilaginous and osseous split lines the preferential orientation of the collagen fibres could be determined. In an *in vitro* experimental setup the contact areas and the contact pressure distribution within the joint have been measured.

Results: In the acetabulum, maxima of subchondral bone density were distributed tricentrically: cranially, dorsally and caudally. In the femoral head, only the area close to the attachment of the ligament was highly mineralized. Cartilaginous and osseous split lines within the acetabulum gave evidence of a long-term tensile stress. In contrast to these findings, there was no clear preferential orientation in the caput ossis femoris. With load increasing initially isolated contact areas extended and became confluent at a loading force of 75% body weight. Values of maximal joint pressure (maximal joint reaction force applied: 400% bodyweight) reached 8–10 MPa.

Discussion: The findings clearly show that the articular surfaces of the hip joint are loaded inhomogeneously. For the first time, it was possible to demonstrate that the healthy canine hip joint is physiologically incongruent. Furthermore, areas of cartilage degeneration within the joint surface coincide with regions that are loaded to a higher extent *a priori*. From the biomechanical point of view it is recommended that all kinds of acetabular fractures – even those affecting the caudal third of the acetabulum – should be surgically repaired in an adequate way.

The Development of Parabronchi in the Duck

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This work was carried out on the parabronchi of ducks during the pre- and post-hatching periods. For this purposes, the samples were taken from the embryo of incubated eggs from the 10th to 28th day of incubation, at regular interval, and from adult ducks under anaesthesia. The parabronchi first appeared at 14th day of incubation as out-pouches, from the secondary bronchi, with solid free end invading the surrounding mesenchym. These pouches was lined by high cuboidal cells rested on PAS positive basal lamina, the cells showed free luminal quarter, which indicated an apocrine secretion. At 17th to 22nd day of incubation, irregularities invading the underlying tissue were observed. This invading part was club-shape forming the future atrium and air capillaries. At 24th to 28th day of incubation the parabronchi were observed at the center of a polygonal shape lobule (lung lobule), it was lined with squamous cells at the head of the clubs, while atrium was lined with cuboidal cells at both side of the stalk (interatrial septum), and the air capillaries was lined with attenuated squamous cells. Because of the breathing of the duck started before hatching the structure of the parabronchi was similar in post-hatching period to that of late stage of pre-hatching period. The wall of parabronchi was pierced by a number of openings leading to the atrium. The lining epithelium was followed by a layer of smooth muscles, which formed a sphincter around the opening of the atrium regulating the amount of air passed to the atrium and air capillaries to avoid hyperventilation. The E.M. picture revealed that, the epithelium of parabronchi was squamous and displayed a scarce number of microvilli, while that of the atrium was cuboidal with a large number of microvilli, the cells of both contained osmiophilic multilamellar bodies that extruded to the surface. Under the epithelium of parabronchi, there were: smooth muscle cells, macrophages, mast cells, collagen fibers, and fibroblasts, while in the interatrial septum between the epithelium on both side there were collagen fibers and fibroblasts only.

The Blood-Vessel System of the Equine Periodontal Ligament: a Mesoscopic Study on Corrosion Casts

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Several investigators have suggested that the blood-vessel system in the equine periodontal ligament does not only

serve nutritional needs, but also specific functional needs, e.g. in terms of a shock absorbing system. In order to supplement the previous data, corrosion casts of the periodontal blood vessels were studied with special attention to the spatial arrangement and connections of the vessels. The heads of eight warm-blooded horses (age 2.5–23 years, seven female, one male), which had been killed for medical reasons, were dissected. The arteria alveolaris inferior and the a. infraorbitalis were perfused with 100 ml Heparin (20 000 IU) followed by 100 ml tap water. After storage at 4°C for 24–96 h, 10–200 ml methacrylate resin (BiodurTM E20) were injected by use of a cartridge pistol for approx. 1 h. After polymerization for 24 h at 60°C, the jaws were frozen and cut with a steel band saw in a horizontal, sagittal or transversal plane to obtain segments that exposed the periodontal space. The specimens were macerated in detergent at 60°C until the soft tissues had vanished. Care was taken to keep the periodontal vessels complete and in topographical integrity. The vascular casts were mesoscopically examined by use of a dissection microscope (magnification 5.8×–40×). The blood vessels in the periodontal space connected with vessels in three other sites:

- (1) occlusal – with the blood vessels of the gingiva;
- (2) apical – with blood vessels next to the apical foramen;
- (3) at the sides – with blood vessels that penetrated the lamina cribiformis.

The blood-vessel system in the periodontal space consisted of main vessels in a parallel alignment with interconnecting branches with smaller diameter. There were indications in some of the specimens that these blood vessels were arranged in layers: one layer next to the cementum, the other next to the alveolar bone.

Expression of Agouti-related Protein (Agrp) and its mRNA in the Hypothalamus and the Adrenal Gland of the Duck (*Anas platyrhynchos*)

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Introduction: Agouti-related protein (AGRP) is a neuropeptide involved in the control of body weight. Morphological and pharmacological studies have shown that AGRP is implicated in the central control of feeding behaviour acting as an endogenous antagonist of the alpha-melanocyte stimulating hormone (α -MSH), a potent satiety-inducing factor, at the melanocortin 3 (MC3)- and four (MC4)-receptors.

Aim: The aim of the present study was to investigate the expression of AGRP and its mRNA in the hypothalamus

and adrenal gland of the duck and, in particular, to establish which type of adrenal tissue is involved in the AGRP synthesis.

Methods: Immunohistochemistry, western blotting, reverse transcriptase (RT)-polymerase chain reaction (PCR).

Results and Discussion: AGRP-immunoreactivity was observed in neurons and nerve fibres in a restricted area of the hypothalamus. AGRP-ir neurons were located in the nucleus infundibularis and distributed ventromedially to the third ventricle in the hypothalamic tuberal region. These neurons were round or, with a lesser extent, elongated in shape. AGRP-ir fibres were seen to project to the median eminence (ME) and anterior periventricular hypothalamus. The AGRP ir-fibres in the ME were distributed in the external layer in close vicinity to the capillaries of the hypothalamo-hypophysial portal system. In the avian adrenal gland, AGRP immunoreactivity was observed in the medullary tissue. A partial sequence of AGRP cDNA was identified using RT-PCR cloning and sequencing. This sequence was highly homologous to the corresponding fragment of the chicken AGRP gene. The western blotting analysis of adrenal gland and hypothalamus tissue extracts showed a well-defined single band with an electrophoretic mobility consistent with the molecular weight of the avian AGRP protein. These results demonstrate that AGRP is expressed in the hypothalamus and adrenal glands of the duck and suggest an involvement of this peptide in the regulation of the melanocortin system in birds.

Effects of Castration on the Expression of Neurotrophic Factors in the Vas Deferens and Accessory Genital Glands of the Rat

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Introduction: Neurotrophic factors constitute a group of growth factor families, which have important effects on survival and differentiation of neuronal cells. The neurotrophin family is composed of nerve growth factor (NGF), brain derived neurotrophic factor (BDNF), neurotrophin (NT)3 and NT 4/5. Neurotrophins act by means of high (TrkA, TrkB and TrkC) and low (p75) affinity receptors on numerous neuronal populations of central and peripheral nervous system. The family of glial derived neurotrophic factor (GDNF) includes, besides the GDNF, neurturin (NTN), persephin (PSP) and artemin (ART). They bind to a common receptor Ret, but the binding specificity is due to a group of proteins (GFR α 1–4). These factors show a trophic effect on

dorsal ganglia, motor neurons and autonomic nervous system. The aim of the present study is to evaluate the expression of NGF, BDNF and GDNF in the vas deferens and accessory genital glands of normal and castrated rats.

Methods: Immunohistochemistry, enzyme linked immunoassay (ELISA), reverse transcriptase (RT)-polymerase chain reaction (PCR).

Results and Discussion: Immunoreactivity to NGF, BDNF and GDNF was observed in all the investigated tracts. Generally, this immunoreactivity seemed to increase in castrated rats. ELISA and RT-PCR were performed to evaluate the levels of BDNF protein and its mRNA. In the normals, the greatest concentration of BDNF was observed in the vesicular gland, the lowest in the prostate. In the castrated, the BDNF concentration significantly decreased in the vas deferens. Conversely, it increased in the vesicular gland and in the ventral and dorsal prostate. BDNF transcripts were detected in both normal and castrated rats. These results suggest that neurotrophic factors are produced by the vas deferens and accessory genital glands and, in normal conditions, they are downregulated by androgens.

Anatomical Features and Histology of the Nasal Cavity of One Humped Camel (*Camelus dromedarius*)

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This study was conducted on nine heads of normal adult one-humped camels. The specimens were collected from Cairo slaughterhouse. The nasal cavity in the freshly collected samples were dissected and photographed. The specimens for microscopic studies were fixed in different fixatives and prepared to examine by light and scanning electron microscope (SEM). The nasal cavity of the camel was studied grossly and by using of light and scanning electron microscope. Specimens from different regions of this cavity were subjected to different histological stains and also demonstrated by the acid and alkaline phosphatases. Gross morphological examination of this cavity showed its three parts: rostral part (the nasal vestibule) covered with skin of usual structure then it lined with smooth mucosa. The middle part (respiratory) had dorsal, middle and ventral nasal conchae, but the caudal part (olfactory) contained the ethmoidal concha. The lining mucosa of the camel nasal cavity was similar to that of other mammals, but there were some differences: the respiratory epithelium showed a small number of goblet cells and there were a mixture of acidic and neutral mucins inside the

epithelial and glandular mucous cells. Strong acid and alkaline phosphatase reaction was observed in the lining epithelium of the nasal cavity. By SEM, it showed the surface epithelial layer of the nasal cavity mucosa in three regions (vestibule, respiratory and olfactory) and resulted that it was stratified cuboidal to columnar epithelium, ciliated pseudostratified columnar epithelium with few goblet cells or olfactory mucosa containing neurosensory olfactory cells. This study aimed to investigate the anatomical features, the histological and histochemical structures of the nasal cavity in one humped camel. The findings of this study were discussed with the previous works in this field with the other domestic animals.

Perineal Nets of Proteoglycans and Glycoproteins in Albino Rat Hippocampus

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Perineuronal nets (PNs) of condensed extracellular matrix (ECM) have been shown to characterize the microenvironment of individual neurons and the chemoarchitecture of some brain regions. In the present study, PNs in the hippocampus were visualized with a cationic iron colloid method for sulphated proteoglycan content and a plant lectin from *Vicia villosa* agglutinin (VVA) for N-acetylgalactosamine containing glycoconjugates. The ECM molecules were organized in reticular coats (PNs) around non-pyramidal cells in the Ammons horn (Corneu Ammons, CA) and subicular region, in addition to pyramidal neurons located in CA2 and CA3 regions. CA2 stratum pyramidale exhibited the most intense staining of its PNs and a diffuse intervening neutrophil labelling, while CA3 region showed a graded fashion of staining intensity. The subiculum displayed intensely stained perineuronal coats. Notably, the hippocampal perineuronal nets revealed overlapped staining characteristics with both staining methods. However, cell coats of the subicular neurons showed various degrees of labelling characteristics with both markers. It is suggested that the PNs in the hippocampus are correlated with the fast spiking inhibitory GABAergic neurons and their target pyramidal cells, and this is important to keep the excitatory elements under control and therefore control the information processes within the hippocampus.

Ultramicroscopical Study on Thyrocalcitonin Cells in the Camel (*Camelus dromedarius*)

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Thyrocalcitonin cells (C cells) were predominantly demonstrated in the ultimobranchial (UB) structures of the thyroid gland of one-humped camels. The UB population was observed as clear cells, either singly distributed or as groups of cells between the component cellular elements lining the UB structures. They displayed various shapes: oval, pear-like, rounded, cubical, columnar and or ellipsoid. Those populating the cell nests or solid cellular masses were almost polygonal or ovoid in shape. The cytoplasm contained well-developed golgi complexes, several mitochondria and characteristically presented rounded or oval membrane-bound electron dense granules of different sizes. The latter granules were irregularly distributed within the cytoplasm especially at the vascular pole. The nucleus of C cells was relatively large, indented, occupying a more or less central position in the cytoplasm. It presented various chromatin contents; the heterochromatin variety was predominantly demonstrated at the marginal zone. However, the euchromatin occupied a more central position within the nucleus.

Quantitative Histochemical Study of Perineural Nets in the Retrosplenial Cortex of Albino Rats

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Reticular or net-like perineuronal coatings enriched with proteoglycans (PGs) and/or glycoproteins (GPs) were demonstrated to ensheath cell surfaces of certain neuronal circuits in the central nervous system of mammals, reptiles and fishes. In this investigation the coated or ensheathed neurons in the retrosplenial cortex (RSC) of adult albino rats were quantified using a computerised analysing system. They were distributed throughout the II–IV cortical layers and comprised about 24.25% of total neurons. The perineuronal coatings were expressed with a

cationic iron colloid (CIC), CIC/Bodian staining and with certain plant lectins from *Vicia villosa* agglutinin 'VVA', *Wisteria floribunda* agglutinin 'WFA' and *Glycine max* agglutinin 'SBA'. The mean number of coated neurons per unit area (UA, 60.15 μm^2), as labelled by CIC or CIC/Bodian, VVA, WFA and SBA staining was 4.01 ± 0.42 , 4.43 ± 0.37 , 6.45 ± 0.87 , 5.70 ± 0.73 and 3.91 ± 0.53 , respectively. In addition, labelling indicated by the aforementioned stains were 10.55%, 11.55%, 16.97%, 15.00% and 10.23%. Notably, the labelled retrosplenial neurons underwent a non-significant increase with progression of animal age at the first post-natal year. However, they declined thereafter toward senility. Qualitatively, there were three types of coated neuronal circuits at the RSC of rats: PGs-, GPs- and PGs/GPs-coated cells, which comprised 9%, 64% and 27%, respectively. This data indicated that the GPs- and PGs/GPs-coated neurons in the RSC of albino rats were more prevalent than the PGs-coated ones.

Magnetic Resonance Imaging of the Harderian Gland in Piglets

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The main purpose of the present study was to investigate the suitability of functional and morphological magnetic resonance imaging (MRI) to assess the extent of brain injury in a hypoxia-ischemia piglet model and further to validate that the ischemic injury was successfully induced. MR imaging was performed at 1.5 T (Siemens Vision, Siemens Medical, Erlangen, Germany) in piglets ($n = 10$; age 12–36 h) under general anaesthesia. Hypoxia was induced by ventilation of a gas mixture containing 8% O_2 in N_2 (AGA, Oslo, Norway). Ischemia was induced by bilateral clamping of the common carotid arteries (CCA). HI was maintained for 30 min, followed by re-oxygenation with ambient air (21% O_2) for 2.5 h. Relative cerebral flow (rCF) was estimated using a single shot SE-EPI sequence (TE = 85 ms, voxel size $1 \times 1 \times 5 \text{ mm}^3$), measuring the first-pass response following bolus injection of an intravascular iron oxide nanoparticulate contrast agent (NC100150 Injection; Amersham Health, Oslo, Norway) at a dose of 4.5 mg Fe/kg. rBF maps were generated on a pixel-by-pixel basis from the peak height of a gamma-variate function fitted to the first-pass contrast agent response. The presence of total carotid

occlusion was assessed by MR angiography (MRA). MRA images (3D-GRE, TR/TE 4.4 ms/1.8 ms, voxel size $0.9 \times 0.9 \times 1.5 \text{ mm}^3$) were acquired after contrast administration; utilizing the T1-effect of NC100150 injection. Mean blood pressure and heart rate were measured continuously during the experiment via a catheter placed in the femoral artery. Blood gasses and rectal temperature were measured at regular intervals. At the end of the experiment, the piglets were given an overdose of 150 mg/kg Pentobarbital i.v., and a dissection was performed in order to identify structures identified on the MRI. MR angio (MRA) and volume rendering identified a prominent, highly vascularized, defined structure close to the eyeball, medioventrally. At dissection, when comparing with the MR images, the deep gland of the third eyelid, the Harderian gland, corresponded to this structure both by topography and size. It was thick, brownish and finely lobulated, and extended from the cartilage of the third eyelid towards the origin of the extrinsic eye muscles at the apex of the orbit. In contrast, the lacrimal gland presented as a small, soft and pale structure that was difficult to distinguish from the surrounding connective tissue. At histological examination, the Harderian gland consisted mostly of compact areas of alveolar glands with abundant eosinophilic granules. Wide thin-walled vessels were conspicuous in the interlobular connective tissue and the capsule. The present MRI demonstration of the Harderian gland was an accidental finding during an investigation to assess the extent of brain injury in a hypoxia-ischemia piglet model. In pig, this gland has received little attention in the literature. However, taking its size and blood supply into consideration, it must be of significant importance also in the pig. It is mostly studied in rodents and lower vertebrates and is found to possess various endocrine functions as well as exocrine.

The Influence of Environmental Factors on the Hoof of Horses

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The influence of some environmental factors and some protecting substances were tested in a laboratory trial with healthy coronary horn and compared with specimens of damaged horn from the bearing border, which had been submitted to a faeces-urine medium. Hoof horn of the best quality from the middle zone of coronary horn (3–4 cm distal of the coronary band) was submitted to different organic media or to some hoof horn varnishes or to heat. The horn specimens of defined size were treated for 36 days with urine, faeces and urine, urea, faeces or faeces and biotine and their strength was compared with untreated controls. The mentioned excrements were collected from

horses. The second group comprised some hoof varnishes or hoof-conditioners (Horsepan, Isohoof Protect, Krealit, Pedalo) and were kept for 24 h in these media. The influence of heat was tested after the horn specimens were heated in an oven for 5 min to 125°C. The results of histologically unchanged horn were compared to horn specimens of the bearing border, where frequently histological changes such as microcracks occur. Two groups of 20 specimens from the bearing border were treated with a mixture of faeces and urine for 5 weeks, and their tensile strength was compared to the same number of untreated controls. The tensile strength of unchanged coronary horn remained nearly unchanged or was only slightly diminished when it was treated with the different media. The treatment with heat did not reduce the tensile strength or showed even a slight increase compared to untreated specimens. The tensile strength of bearing border specimens showed a highly significant decrease of tensile strength of about 55% after 5 weeks treatment with faeces-urine in both trials compared to the untreated controls. The results indicate that unchanged horn is much more resistant to environmental factors than bearing border horn with first damages. For prevention of severe changes at the bearing border horses should not stand for a long time in beddings with a high content of faeces and urine.

Regulation of Cellular Proliferation and Differentiation in Claw Epidermis by Epidermal–Dermal Interactions

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Introduction: Epidermal–dermal interactions play an important role in regulation of proliferation and differentiation of keratinocytes and also in disease and regeneration of skin. A double paracrine regulation of keratinocyte growth and differentiation has been postulated and described *in vitro* recently. Interleukin-1 (IL-1) and keratinocyte growth factor (KGF/FGF-7) and their receptors are major mediators in this epidermal–dermal signalling.

Aim: We wanted to generate evidence for a reciprocal regulation of epidermal proliferation and differentiation in the bovine claw involving IL-1 and KGF.

Methods: Tissue samples from claws of slaughtered cows, co-cultures and three-dimensional cultures of bovine hoof keratinocytes and fibroblasts were used. Immunohistochemistry for IL-1 and IL-1 receptor, KGF and KGF receptor were carried out on frozen and paraffin sections from tissue and cell culture. RT-PCR was used to study regulation at the mRNA level in dermis and epidermis (after micro separation) and in fibroblasts and keratinocytes from the cultures respectively.

Results: In claw epidermis Il-1 was accumulated and KGF receptor was detectable. Interleukin-1 receptor and KGF were present in dermal cells. Il-1 stimulated fibroblasts responded with KGF expression *in vitro*. Neutralization of Il-1 reduced or stopped this effect. Keratinocytes co-cultured with fibroblasts showed a significant better growth. Presence of keratinocytes increased KGF expression in the co-cultured fibroblasts.

Conclusion: Our results suggest that Il-1 is leading to stimulation of KGF expression. KGF seems to be an important stimulator of keratinocyte proliferation in our co-culture system. So far our results provide initial evidence for a reciprocal regulatory mechanism present in the bovine claw involving Il-1 produced in the epidermis and KGF originating from fibroblasts. Future *in vitro* studies including mRNA analysis will provide deeper insight into the epidermal-dermal cross-talk and the involved transcriptional/translational regulation in the bovine claw. This work was supported by the European Communities under the Lame cow project QLK5-CT-2002-00969. The authors are solely responsible and the work does not necessarily represent the opinion of the European Communities.

Ultrastructure of Three-dimensional Bovine Hoof Keratinocyte Cultures

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The aim of this study was to characterize the ultrastructure of three-dimensional keratinocyte colonies in relation to *ex vivo* bovine hoof epidermis. Keratinocytes were isolated from the claw and cultured in equal parts of Dulbecco's modified Eagle's medium (Sigma, Taufkirchen) supplemented with 10% fetal bovine serum (Sigma, Taufkirchen) and serum-free MCDB 153 complete medium (Biochrom, Berlin). Cells were allowed to grow until formation of colonies occurred. Then the colonies were isolated by removing the surrounding cells with a cell scraper every 3 days. The colonies were harvested when they reached diameters of 6–8 mm and were fixed in Karnovsky's solution. They were routinely processed for the embedding in Agar 100. Semithin and ultrathin sections were prepared and examined under the light and transmission electron microscope. These colonies revealed the morphological properties and tissue architecture, which are specific for claw epidermis *in vivo*. We detected up to 22 layers of keratinocytes and could discover between three stages of differentiation: a basal cell layer, followed by several cell layers corresponding morphologically to the cells of the stratum spinosum and a few cornified superficial cell layers. Throughout all layers cells were connected by desmosomes, which were linked to the cytoskeleton by dense bundles of filaments. In addition we

observed the formation of desmosomes. The cells of the upper un-cornified layers generated an cornified envelope. In the following layers cell death and subsequent cornification accompanied by the characteristic alterations of the nuclei occurred. The cells at the superficial layers were filled by electron dense, solid horn masses. Additionally we detected intercellular cementing substance in the intercellular space between the upper layers. Our electron microscopical characterization of the colonies provides evidence that the bovine hoof keratinocytes *in vitro* undergo basically a similar pattern of proliferation and differentiation like in the claw *in vivo*. These findings provided a basis for further studies like the long-term culture and differentiation of bovine hoof keratinocytes in perfusion chambers.

In Vitro Infection of Bovine Hoof Cells and Skin Explants with *Treponema Brennaborensis* and *Treponema Denticola*

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The aim of this study was to establish an *in vitro* model that permits *in vitro* infection of bovine skin with *Treponema* spp. and enables to study the role of treponemes in the pathogenesis of digital dermatitis (DD). In all experiments, incubation with *T. denticola* or *T. brennaborensis* was carried out simultaneously. Keratinocytes obtained from the claw were cultivated on cover glasses without antibiotics until they reached sub-confluence. Then they were incubated with OMIZ Pat medium containing treponemes for up to 96 h. Every 24 h two cover glasses were fixed and stained with the DAPI method. Skin explants were obtained from typical sites of DD lesions. First the explants were maintained in medium with antibiotics to eliminate bacterial contamination. Subsequently, they were rinsed thoroughly with medium without antibiotics and incubated with *Treponema* suspensions for 48 h. The treponemes stayed vital under culture conditions for even up to 96 h. They were still showing their typical spiral shaped morphology and adhered to the cultured keratinocytes at all time points. With prolonged incubation time cultured cells began to show morphologic damage and some cells detached from the cover glasses. Light and electron microscopical investigations of the explants revealed that treponemes were adhering to the surface of the epidermis. They were visible in often-enlarged intercellular spaces. In addition treponemes could be detected in deep epidermal layers

and in very high concentrations in the dermis. In periodontal disease, spirochetes were observed in enlarged intercellular spaces. Our results support these findings suggesting that treponemes invade the deeper claw tissue via the intercellular spaces of the epidermis. We suggest that the enlargement of the intercellular spaces can improve an increased infection of deeper tissue layers and facilitates the way for infection with other anaerobic bacteria.

3D-Reconstructions: New Application Fields in Modern Veterinary Anatomy

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Introduction: Today, the claims on the anatomical education have changed with the considerable increase of modern medical imaging. For the analysis of computed tomography (CT) images, magnetic resonance (MR) images or ultrasound, the knowledge of cross sectional anatomy is an important prerequisite for proper diagnosis. Admittedly, it is difficult for students and also for practitioners to imagine three-dimensional objects in space. The possibilities and convenience of 3D-reconstructions were demonstrated by generating virtual casts of frontal sinuses.

Methods: Using a HiSpeed (DX/i Plus, General Electric) scanner transversal CT images of Persian and European Shorthair cats were acquired. The slice thickness and distance were set to 1 mm. With the software system VG Studio MAX 1.1, the obtained volumetric data was used to create volume graphics displaying object surfaces and interiors based on voxel data. Because of density value acquisition, it was possible to identify and label the objects of interest. The extension of the paranasal sinuses was analyzed and demonstrated in relation to the skull. For verification, conventional corrosion casting employing Technovit or Tensolciment, respectively, were carried out on the same specimens.

Results: The frontal sinuses of both feline breeds showed great variations in size, shape and symmetry. With models from the skull based on 3D-datasets including the virtual casts of the paranasal sinuses, it was easy to imagine spatial relations and localizations inside the head. The advantages of the virtually derived casting method are: creating the virtual casts *in vivo*; the

method does not imply any health risks and is repeatedly applicable.

Conclusion: 3D-reconstructions based on CT-datasets are excellently suited for generating spatial sense. Additionally, a single scan series can be applied for different anatomical problems. The resulting digital 3D-objects are readily available and can be employed for educational purposes for better comprehension of the difficult spatial relations of the head.

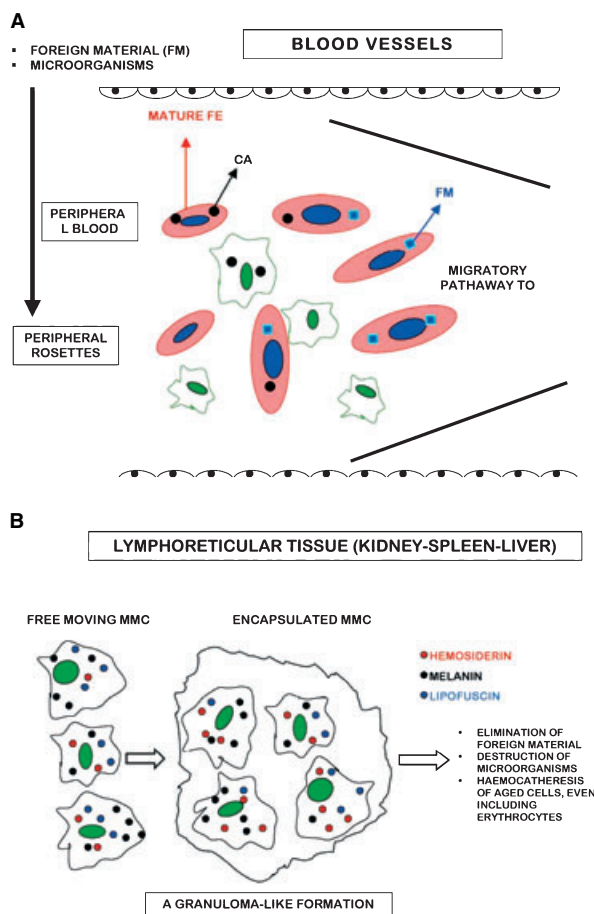
The Origin of Melanomacrophages Centers in Salmo Gairdneri Richardson

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By definition, melanomacrophage centres (MMC) consist of aggregates of macrophage-like pigmented cells located in the kidney and spleen, and to a lesser extent in the liver. In 30 *Salmo gairdneri* Richardson trout, MMC have been identified in various organs. In the kidney, liver and spleen, respectively, macrophages (MØ) are aggregated in clusters, which are next to vessels. MMC have been cytoenzymatically identified with periodic acid Schiff (PAS), α -naphthyl acetate esterase (ANAE), peroxidase (PEROX) reactions. Finally, in order to show pigments in the context of MMC, in particular, we used iron stain to detect iron pigments and Prussian blue reaction to detect melanin and lipofuscin. The presence of iron pigments indicates that MMC function as lymphoreticular centres where fish erythrocytes (FE) are destroyed together with other foreign material. In previous studies, we found that FE are able to bind and engulf *Candida albicans* (CA) without killing this fungus. FE containing engulfed CA form with MØ functional rosettes and, in reason of that MØ are solicited to phagocytose and kill CA. It is likely that rosettes and/or single FE containing engulfed foreign material migrate to distant organs, such as the kidney, the spleen and the liver. Therefore, the MMC may, in part, derive from these peripheral rosettes and pigments result from FE and foreign material digestion and destruction operated by MØ.



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Blood Supply to the Pigeon Wing and Flying Muscles

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In this study the vascularization of the wing and the two main breast muscles of pigeons was investigated using classical anatomical techniques. Three groups of 9-month-old pigeons cadavers were dissected. Each group contained six animals. The cadavers of the group no. 1 were first fixed using 4% formaldehyde solution

and the arteries filled with coloured latex. After coagulation we dissected the pectoral (pectoral and supracoracoid) muscles and the supplying branches of the pectoral trunk. We followed also the subclavian, the axillary, the brachial, the radial and ulnar arteries as they vascularized the wing. The cadavers of group no. 2 were used as fresh cadavers and perfused via the brachial artery in retrograde direction with methyl-methacrylate. After the polymerization, the bodies were placed in concentrated hydrochloric acid so the microvascularization of the great breast muscles and the arteries of the wings could be investigated. The arteries of the group no. 3 were filled with BaSO₄ solution after fixed with formalin and radiographed. Using this method we could represent the relation of the arteries to the bones. The breast muscles that move the wing are mainly vascularized by the pectoral trunk, which is the strongest branch of the subclavian artery and divides into the cranial, middle and the caudal pectoral arteries. Also the sternoclavicular artery is giving strong rami to the supracoracoid muscle by the lateral border of the coracoid. The subcutaneous thoracoabdominal vessels that supply the brood patches are branches of the pectoral trunk too. The brachial artery lies between the biceps and triceps brachii muscles on the medial side of the humerus and ramifies at the level proximal of the elbow joint to the stronger ulnar and weaker radial arteries. The radial lies cranially, covered by the pronator muscles and gives two branches, the deep and superficial radial arteries. The ulnar artery also gives off two: the superficial and deep ulnar arteries, which supply the flexor muscles and vascularize the distal part of the wing.

Expression of Fibroblast Growth Factors (FGF) and FGF Receptor (FGFR) in the Horse Placenta

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Members of the fibroblast growth factor family have been shown to stimulate angiogenesis and cell differentiation during embryogenesis and invasive haemochorial placentation. The non-invasive epitheliochorial horse placenta is characterized by the occurrence of specialized areolae between the microcotyledons, where secretions of endometrial glands are imbibed by the pseudostratified

trophoblast cells throughout gestation. Therefore, we hypothesize that trophoblast cells may be programmed differently depending on their location and function. The influence of FGF family members during placentation in the mare was evaluated in tissues from five mares at days 110, 121, 179, 199, and 309 of gestation. Placental samples were perfusion-fixed and paraffin-embedded, and immunohistochemistry was performed with antibodies against FGF1 and FGF2 (which have been shown to influence cells of mesenchymal origin), FGF7 which acts on epithelial cells, and an antibody against FGFR which recognizes various receptor isoforms. FGF1 immunostaining occurred in most cells of the placenta throughout gestation, with the strongest expression in vascular structures. FGF2 protein expression differed in relation to location and stage of gestation. Trophoblast and endometrial epithelial cells in the microcotyledons showed strong nuclear expression, whereas areolar trophoblast, fetal and maternal endothelial cells, and smooth muscle cells of large allantochorionic blood vessels all showed cytoplasmic localization of FGF2. Near term, on day 309, FGF2 expression ceased in all types of cells. FGF7 immunoreactivity was observed predominantly in the nuclei of endometrial gland cells and the cytoplasm of trophoblast cells, and to a lesser extent in the endometrial epithelium. FGFR immunoreaction occurred almost exclusively in vascular cells, such as the endothelium and smooth muscle cells of large blood vessels, and staining declined towards term. The nuclear localization of FGF2 and FGF7 suggests the presence of high molecular weight isoforms, which have been shown to be involved in specific autocrine transcription processes, like activation of endometrial glands and differentiation of trophoblast cells. In contrast, the cytoplasmic localization of FGFs indicates the presence of low molecular weight isoforms. These would represent the classic FGFR binding isoforms that may be involved in angiogenesis mediated by auto- and paracrine mechanisms.

Real-time Three-dimensional Echocardiography, a New Approach to Cardiac Anatomy in Live Dogs and Cats

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Macroscopic anatomy of the heart is evaluated best in living animals, because postmortal changes and the lack of the normal heart motion lead to misinterpretation. For the last 15 years functional cardiac imaging of animals has been performed by two-dimensional ultrasonography. However, three-dimensional echocardiography has multiple advantages over two-dimensional ultrasonography such as an improved assessment of the endocardial structures and

the possibility of precise determination of the sonographic views for measurement. For these reasons the ultrasound system SONOS 7500 of Philips has been used for examinations since December 2003. These examinations were performed in healthy adult dogs of different sizes as well as in European Short Hair cats. Using a recently developed multi-frequent matrix array probe all regions of interest were directly on-line volume-rendered or represented in two exactly defined orthogonal views. Although the used matrix array probe was large, transthoracic left and right parasternal sonograms could be carried out without any problems in all small animals without sedation. For the first time the cardiac structures in dogs and cats such as the opening and closing of the left atrioventricular and aortic valves were represented within a 60 degrees (15 or 30 degrees pyramidal volume). The morphology and the function of the atrioventricular leaflets could be evaluated exactly. For representation of a larger pyramidal volume or the three-dimensional simultaneous colour flow imaging, careful ECG gating is required. Unfortunately, the distinct respiratory arrhythmia in quiet dogs had a negative effect on the registration of four or eight heart cycles of the same length. Using biplane echocardiography the commonly represented sonographs for measurement were exactly determined. Additionally these techniques enabled the registration of new views for evaluation and measurements, which were not available from two-dimensional ultrasonography. This study has demonstrated for the first time that real time transthoracic three-dimensional echocardiography is feasible in dogs and cats and that it provides comprehensive visualization of the intracardiac anatomy in unique projections in living animals.

Morphological Considerations in the Pathogenesis of Paratuberculosis in Goats

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Paratuberculosis is a chronic granulomatous enteritis and lymphadenitis in ruminants caused by *Mycobacterium avium* subsp. *paratuberculosis*. To investigate the efficacy of assays for early diagnosis, experimental infections were performed in goat kids. Oral doses of *M. a. paratuberculosis* were administered to goat kids over extended periods and the resulting infection was followed for 12–24 months. Complementary experiments were performed on goat kids vaccinated against *M. a. paratuberculosis*. Immunological, pathological and morphological examinations were undertaken in both these experimental systems. The analysis of results from these experiment series provided novel insights into the influence of the biology of gut-associated lymphoid tissue on the progress

of disease, into the character of the tissue granuloma and its dominance of the local tissue response and into the relationship between the local and systemic immune responses.

Secondary or Accessory Glomerular Layer in Mice During Growth

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Most mammals have two different structures in which we found glomerular layers at the same time: the main olfactory bulb (MOB) and the accessory olfactory bulb (AOB). Both bulbs have the same pattern of organization, but there are some differences: although the size is considerably bigger in MOB than in AOB, probably the most important difference is that the principal cells are not differentiated into mitral and tufted cells in the AOB, and are usually described as mitral/tufted cells. We have previously observed that in some mammals, like pigs and sheep, the AOB reaches maturity before birth, but this is not a rule for other species. Surprisingly, mice need several days of life to achieve full stratification of its cellular components. We have studied the chronology of this process, focusing our attention on the glomerular layer, the last to appear. We concluded that there are two critical periods, between E11.5 and E16.5 (migration phase) and between E17.5 and P3.5–7 (true morphological constitution).

Ontogenesis of the Chicken Bronchus Associated Lymphoid Tissue (BALT)

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The avian lung shows fundamental differences to its mammalian counterpart and is characterized by a larger surface area and a thinner gas-blood barrier. These features provide the morphological basis for the enhanced respiratory efficiency of the avian lung. However, they also predispose this organ to an increased risk of injury by pathogenic microorganisms. In order to investigate the host–pathogen interaction during pulmonary infection and vaccine delivery by aerosol immunization a detailed knowledge of the structure and ontogenesis of the bronchus associated lymphoid tissue (BALT) is required. For this reason the BALT of chickens aged from 1 day to 6 months was studied by immunohistochemistry (ICC),

transmission (TEM) and scanning electron microscopy (SEM) as well as by FACS analysis. During the first 2 weeks after hatching no organized lymphatic structures were found. At this time, the defence system was based on the presence of cells of the unspecific innate immune system. In the third week after hatching lymphoid follicles begin to develop at the branching of primary and secondary bronchi. These develop into a highly organized bronchus associated lymphoid tissue (BALT) characterized by high endothelial venules and a follicle associated epithelium (FAE) with ‘M-cell-like’ cells. However, pathogens that have successfully crossed this initial barrier are additionally confronted with large numbers of lymphocytes and macrophages distributed around the parabronchi. Our results imply that the BALT is the first and most important structure encountering antigen and microbial pathogens, thus playing a central role in the initiation of a specific immune response.

Vascularization of the Ciliary Body and Iridocorneal Angle in the Avian Eye

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In experimental ophthalmology avian models play an important role in the investigation of the pathophysiology of glaucomas. One major problem in establishing new experimental designs is a deficiency in basic anatomical data of the avian eye bulb. The aim of this study was to give a detailed description of the vascularization of the ciliary body and the iridocorneal angle in the avian eye. The vascular anatomy of the avian eye was studied in 50 birds of different species and orders (Phasianiformes, Accipitri-formes, Columbiformes) by light microscopy as well as by stereomicroscopy and scanning electron microscopy (SEM) of corrosion cast specimen. The comparison of the different species revealed that independently of the bird order the vessels were orientated in a similar way, thus enabling to establish a schematic diagram. There is a redundancy in the arterial blood supply of the ciliary body and iridocorneal angle, which is supplied by the *circulus arteriosus iridis* and the *circulus arteriosus*. For the first time we were able to create a corrosion cast of the complete avian sinus venosus sclerae by injection of a liquid plastic (Reckli®, Chemiewerkstatt GmbH, Herne, Germany) in the venous system. The *sinus venosus sclerae* is characterized by a double-ring like structure. It drains into several vv. ciliares anteriores reaching the *circulus venosus sclerae*. A clearer understanding of the vascularization of the anterior eye segment, in particular of the sinus venosus sclerae, provides a step forward in experimental ophthalmology, especially regarding glaucoma research based on an avian model.

Histological Study of Lymphatic Nodules and their Distributions Throughout the Cecum of the Caspian Miniature Horse

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The cecum is a great cul-de sac intercalated between the small intestine and the colon. It has a remarkable size in the equine and digestion of more than 30% of total cellulose and production of volatile fatty acids by anaerobic metabolism of cellulose by its micro flora were made on it. It has also a substantial number of lymphatic nodules, which in horses are located near the apex. Topographically, cecum is situated chiefly to the right of the median plane and occupies around the whole volume of there in the abdominal cavity. The histological structure of the lymphatic follicles and their distributions pattern throughout the cecum in five healthy mature miniature horses (an special species of horses lives only in north of Iran) were studied (Fig. 1a). Each cecum was divided into three parts, namely, the base, body and apex, and samples were taken from each part immediately after slaughter and fixed with

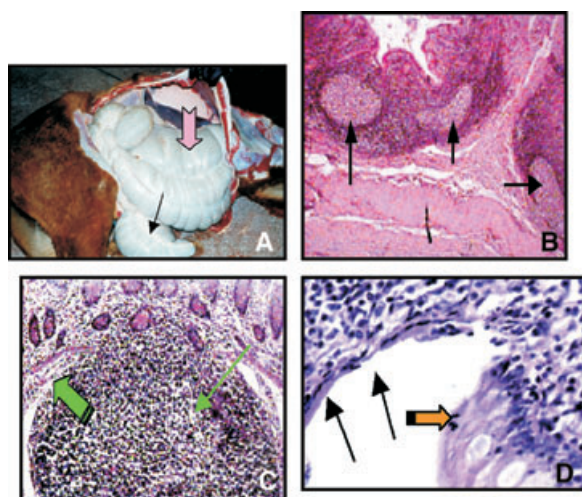


Fig. 1 A: Abdominal viscera of miniature horse: right view. The right lateral wall is removed, right ventral part of colon (thick arrow), cecum (thin arrow). B: Photomicrograph from the apex of cecum in miniature horse, large numbers of lymphatic nodules were seen in lamina propria and sub mucosa (arrows), H&E, 32x. C: Photomicrograph from the base of cecum in miniature horse, lymphatic nodule (thin arrow), discontinues muscularis mocusa (thick arrow), H&E, 100x. D: Photomicrograph from the epithelium covering the base of cecum in miniature horse, normal-epithelia (thick arrow)

10% buffered formalin. Routine histological laboratory techniques were made and 6 μ m sections were cut and stained with haematoxylin and eosin staining and studied under light microscope. Lymphatic nodules were mostly found in the submucosa, but they disrupted the muscularis mucosa and merged into the lamina propria (Fig. 1c). The thickness of the mucosa and submucosa at these sites was smaller, and the epithelium that covered the mucosa was here modified into simple squamous or low cuboidal epithelia with dark and flattened nuclei (Figs 1b,d). The large and small lymphatic nodules were extensively accumulated in the apex, forming a nearly continuous layer in lamina propria and submucosa. However, they were less prominent at the base and least in the body of the cecum.

A Comparison Between Thrombocytes and Monocyte-Macrophages in *Salmo gairdneri* Richardson: A Cytoenzymatic Evaluation

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Fish leukocyte heterogeneity with respect to morphology, morphometry and immunological function has been object of some controversy, mostly because, in lower vertebrates, there are several interspecies differences. Leukocyte enzyme cytochemistry permits cellular differentiation, which is otherwise not possible with routine stains, and may also indicate cell function. In fish, phagocytic performance is principally operated by monocyte-macrophages (MØ) present in tissues and in circulating blood. Fish thrombocytes (FTHR), even if they are the equivalent of platelets in mammals, are complete cells with cytoplasm and nucleus. They vary in shape from round, like small lymphocytes, to spindle-shaped or rod-shaped cells. Several cytochemical methods, as periodic acid Schiff (PAS), naphthol chloroacetate esterase (ASD), α -naphthyl acetate esterase (ANAE), peroxidase (PER-OX), acid phosphatase (Ac. Phos.) with and without tartaric acid (ta) and alkaline phosphatase (Alk. Phos.) have been applied in *Salmo gairdneri* Richardson to investigate the expression of MØ and FTHR enzyme patterns. Here, we compared cytochemistry between fish MØ and FTHR, and between human MØ and platelets/megakaryocytes, respectively, as indicated in the table. It appears that positivity to some enzyme cytochemistry is phylogenetically maintained in MØ, while in FTHR the enzyme pattern has undergone modification during evo-

lution. For instance, this is the case of ANAE, Ac. Phos. and Alk. Phos., which changed in the phylogenesis, while PAS, AS-D, PEROX are still present in the evolution.

	PAS	AS-D	ANAE	Perox	Ac.Phos.-ta	Ac.Phos. + ta	Alk. Phos
Trout MØ	++	+	++	++	+	-	n.d.
Human MØ	+	+	++	+	-	-	-
Trout THR	-/+ +	-	-	-	-/+	-	++
Human platelets/megakaryocytes	+	-	++	-	+	+	-

+, Positive; ++, strongly positive; -, negative; n.d., not determined.

Vitamin D₃ Receptor Immunohistochemistry in Sheep and Goat Intestine

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Disturbed calcium homeostasis during the periparturient period is a major problem in domesticated ruminants. Clinical signs of hypocalcaemia shortly after calving (periparturient paresis) occur in cattle, predominately in elder animals. On the contrary, in sheep and goat, periparturient paresis is mainly evident several weeks before and only in some cases after parturition. In high-yielding dairy goats, however, periparturient paresis also occurs – as in the dairy cow – during the periparturient period (Ruffin et al., 2002). Vitamin D₃ receptors (VDR) participate in the regulation of calcium homeostasis by increasing the efflux of calcium from bone and dietary sources into blood. The nutritional availability of calcium during the periparturient period has been examined in cattle: VDR concentrations in colon decrease with age, being significantly higher during pregnancy and lactation and are temporarily lower at parturition (Goff et al., 1991). However, similar data have not been evaluated for sheep and goat. Thus, tissue specimens of small and large intestines were collected from lambs and non-lactating, non-pregnant adult animals of both species. Formaldehyde fixed and microwave treated paraffin sections were used for VDR immunohistochemistry employing a biotinylated monoclonal rat antibody and streptavidin peroxidase technique. Preliminary results in adult sheep showed that VDR immunoreactivities generally were higher in deep parts of crypts, decreased towards openings and were low in surface epithelium. Staining was most prominent in duodenal mucosa, lower in jejunum and further decreased in ileum and colon. It was lowest in caecum.

Smaller numbers of immunopositive cells and reduced staining intensities resulted in weaker immunoreactions in the latter segments of small and large intestines. In addition to nuclear tinction, cytoplasmic staining was mainly evident in duodenum, at lower levels in jejunum and negligible in ileum, colon and caecum. Immunohistochemistry of goat and lamb intestines are currently performed, thus the effects of age and species are presently unknown. Our preliminary results indicate that VDRs are highly expressed at the site of presumed maximal intestinal calcium resorption.

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Magnetic Resonance Imaging and Cross Section Images of the Cat Thorax

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Accurate interpretation of thoracic magnetic resonance images requires a thorough knowledge of anatomy of this region. The purpose of this communication is to describe the normal cross sectional anatomy of the thoracic cavity of the cat, using MR images, dissections and macroscopic sections. In this study, three cats were used. The animals were anesthetized and positioned in sternal recumbency in the MR scanner. MR imaging was performed at the Special Diagnostic Service of San Roque Clinic of Las Palmas de Gran Canaria with a superconducting magnet operating at a field strength of 1.5 Tesla and a human body coil. Spin echo pulse sequences were used to obtain T1-weighted images in transverse and sagittal planes. At the conclusion of imaging, the cats were euthanatized for medical reasons unrelated to disease of thorax. The cats were frozen and then sectioned using an electric band saw. The cuts were matched as closely as possible to the MR images for identifying the normal planimetric anatomy of the thoracic structures. MR T1-weighted spin echo images provided excellent anatomic appearance of the thorax structures. In MR images the grey scale is directly related to the signal intensity of the thoracic cavity structures. Thus, fat and nerves had higher signal intensity compared with the lower signal intensity of the respiratory system. Bone marrow and muscles had a intermediate signal intensity and appeared gray. The intensity signal of the articular fluid permits a good differentiation of the opposing cartilage surfaces on all MR images. The planimetric or sectional anatomy of the thoracic cavity in the cat allows a correct morphologic and topographic evaluation of the anatomic

structures, being helpful tool for the identification of the MR images. The information presented should serve as an initial reference to evaluate MR images of the feline thorax and to assist interpretation of lesions of this region.

Specialised Vascular Structures in the Pododerma of the Camel Footpad (*Camelus dromedarius*)

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Aims: The even-hoofed, unguligrade tylopods display a well-developed eponymous footpad that absorbs most of the body weight in locomotion and is particularly adapted to the sandy ground surface of the natural environment of dromedaries. Considering the drastic changes in temperature occurring in the desert sand, specialised thermoregulatory vascular and nervous structures are to be expected in the modified pododerma of this footpad. Therefore, the aim of the study was to demonstrate the angioarchitecture of the camelid footpad and to detect specialised vascular structures related to these specific functions.

Material and Methods Corrosion casting of fore and hind feet of dromedaries (*Camelus dromedarius*); scanning electron microscopy of micro-corrosion casts

Results and Discussion The dense angioarchitecture of the dromedary footpad, like that of unmodified skin, can be subdivided into separate plexus arranged at different levels: a subcutaneous plexus, a profound dermal plexus, a superficial dermal plexus and a subepidermal plexus.

The subepidermal blood vessels run very close to the dermo-epidermal borderline and thus the subepidermal angioarchitecture consists of densely vascularised conical papillae comprised of centrally situated papillary arteriole and venule ensheathed in a dense subepidermal capillary network. At the tip of the papillae, the central papillary arteriole and venule form a peripheral capillary loop, acting as a thoroughfare channel. This dermal angioarchitecture of the dromedary footpad is comparable to the papillary system described in the bulbs of the highly specialised digital end organs of many other odd- and even-hoofed mammals, particularly the equine hoof and the bovine claw.

Within the subcutaneous and the deep dermal plexus of the dromedary foot pad, unique glomus-like vascular structures were detectable, presumably associated with cutaneous thermoregulation as described for the unmodified hairy skin.

The Brain of the Dog in Section: a Comprehensive View for Veterinary Students

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Transversal, horizontal and sagittal sections of the brain were stained by the ancient but efficient Mulligan method, a procedure that establishes a clear macroscopic difference between the white and grey substances. Different structures of each section were studied and most of the details were identified and named according to the NAV. All sections were projected onto the whole brain. By means of this easy and basic procedure the students increase their understanding of (1) the size and/or the form and/or the topography of several prominent structures of the brain, (2) the general distribution of the substantia alba and grisea, and they begin to understand the complexity of the brain.

Preliminary Results Concerning the Expression of Carbohydrates in the Olfactory System of the Sturgeon

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Lectins have been used in several areas of biomedicine and are particularly useful for histochemistry. Their ability to map the distributions of various kinds of cell-borne glycan, and thereby to identify particular cell populations, has allowed clarification of a number of issues in neuroscience. In the case of the olfactory system, for example, lectins may be involved in development, in the continual regeneration of olfactory neurons and even in the processing of olfactory information. Whereas in mammals and amphibians lectins have been widely employed for the study of the olfactory and accessory olfactory systems, little information regarding lectin labelling is available in fish. We have used a panel of 11 lectins (UEA, BSI-B₄, DBA, LEA, VVA, SBA, PSA, WGA, WGA-s, ECA, LTA) in paraformaldehyde fixed and paraffin embedded specimens of a Chondrostei fish, the sturgeon (*Acipenser baeri*). Our preliminary findings show that the olfactory receptor cells, the olfactory nerve fibres and their terminals in the olfactory bulbs were labelled with BSI-B₄, DBA, VVA, SBA, PSA, WGA and WGA-s. The presence of glycoproteins, whose terminal sugars are detected by lectin binding, might be related to

the reception of an odour stimulus and its transduction into a nervous signal or to the histogenesis of the olfactory system.

Surface Ultrastructure of Ultimobranchial Remnants in the Thyroid Gland of Buffalo (*Bos bubalis*)

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The remnant of ultimobranchial (UB) body in the thyroid gland of adult buffalo was studied. This remnant appeared in the form of irregular and elongated follicles of various size and shape with folded lumen. These structures occupied a peripheral location beneath the capsule, as well as being embedded mostly in the vascular connective tissue of the thyroid. Calcitonin cells were predominantly localized in the UB follicles. Scanning electron microscopy revealed that surface of most UB follicular cells was almost flat and polyhedral or hexagonal in shape. Other follicles possessed cells with dome shaped apical surface demarcated by a shallow intercellular depression. The luminal surface of the follicular cell lining was studded with microvilli that appeared with various density and length and were numerous at the cellular borders. Interestingly, most of UB follicular cells were provided with single cilium, projecting over the cell surface. Notably some UB follicles presented various stages of apocrine activities.

Perineuronal Nets in the Central Nervous System

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One hundred and twenty years ago, Camillo Golgi described reticular or fenestrated sheaths of extracellular matrix (ECM), enwrapping the cell bodies, axon initial segments and proximal dendrites of certain pyramidal and non-pyramidal neurons in the adult mammalian central

nervous system. Such structures, currently termed perineuronal nets (PNs), are preferentially associated with GABAergic, parvalbumin-containing fast-spiking inhibitory types of cells, and mostly expressing the Kv3.1 subunit of voltage-gated potassium channels. Although some neuroanatomists validated the existence of the PNs on selected neurons in the adult brain, there was a lack of agreement concerning their origin, composition and function. Recent data suggest that PNs result from the visualization of ECM molecules that are confined to the space interposed between glial processes and the nerve cells that they outline. The substance confined to these spaces could be visualized selectively by antibodies directed to glycoproteins, proteoglycans, markers for hyaluronan, some plant lectins recognizing N-acetylglucosamine or by monoclonal antibodies directed to epitopes on unknown molecules. The PNs appear only after birth, which they could serve as recognition molecules between certain neurons and their surrounding cells. Other putative roles include stabilization of synapses, maintenance of cellular relationships in the adult brain, concentration of growth factors around certain neurons, generation of a polyanionic ion-buffering microenvironment, as well as prevention of extracellular-space occlusion and the formation of certain link with the intracellular cytoskeleton.

Surface Ultrastructure of Ultimobranchial Epithelium in the Thyroid Gland of Donkey (*Equus asinus*)

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Surface ultrastructural observation of the thyroid gland in donkey revealed the coexistence of certain unusual follicles or cysts, the ultimobranchial (UB) follicles, which exhibited large diameters and less regular outlines. In addition, several small follicles, cell nests or solid cellular masses were also demonstrated in association with these structures. Surface of most UB follicular cells was almost hexagonal in shape, studded with sparse or dense pleomorphic microvilli, which were much denser at the cellular borders. Each follicular cell bore single cilium that projecting over the cell surface. Interestingly, bleb-like apocrine protrusion of the apical cytoplasm of some follicular cells was frequently demonstrated. These apocrine blebs (or aposomes) were varied in shape, size and showed smooth or irregular surface. In addition to such aposomes, deep to shallow circumscribed or eroded areas of the apical cell surface was frequently demonstrated. These findings indicate that the UB follicular cells in donkeys,

similar to those of camels, exhibit an apocrine activity alongside with the usual merocrine mode of secretion.

Post-natal Development of Perineuronal Nets in the Retrosplenial Cortex of Albino Rat

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The brain extracellular matrix (ECM) has attracted growing interest due to its highly regulated spatiotemporal expression during development and maturation of central nervous system. The present study deals with the post-natal appearance and transformation into adult distribution patterns of the ECM components related to proteoglycans (PGs) and glycoproteins (GPs) in the retrosplenial cortex (RSC) of albino rats at birth (P0), 1 week (P1), P2, P3, P4, P5, P6, P7 and P8. The differentiating PGs and GPs components of the ECM were shown to make their appearance as early as 1–2 weeks post-natally. At this developmental stage, these components of the ECM appeared in association with some neurons and glia cells or diffusely localized at the neutrophill. Interestingly, Golgi complexes of labelled neurons were usually stained with lectin VVA or WFA, and this labelling dramatically disappeared on reaching P4. During P2–3, the pericoated neuronal cells underwent a progressive increment in number, and presented an inside-out pattern of migration and differentiation (toward the V-II cortical layers). On reaching P4, most of the coated neurons appeared distributed into the cortical layer IV and II. At a later stage (P5–8), the overall density and intensity of labelled neurons progressively increased and apparently reached the adult stage of development. They also displayed the usual differential labelling characteristics, after using the cationic iron colloid/lectin staining, for the first time at this juncture. The present findings indicated that the perineuronal ECM components are significantly correlated with age and suggest a possible developmental or biological significance including promotion of migration, as well as functional maturation of the retrosplenial neurons.

The Expression of Connexins during Growth and Development of Skeletal Elements in the Dog

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Efficient and well-regulated intercellular communication is essential to organize the activities of the different cell populations involved during growth and concomitant structural maturation of the skeleton. For direct cell-cell communication gap junctions composed of connexins (Cx) are present. So far 21 Cx types have been identified in mammals, which differ in molecular permeability as well as tissue expression pattern. To characterize the Cx distribution in the developing canine skeleton, samples from humerus, scapula and lumbar vertebrae of Beagle dogs aged 1.5, 3, 4.5 months and 1.5 years were fixed in Bouin's solution, decalcified with 25% EDTA and embedded in paraffin wax. Using immunohistochemistry and in situ hybridization, the expression of Cx 26, 30, 31, 32, 43, and 45 was studied in bone tissue plus associated epiphyseal and growth plate cartilage. Of all bone cells, osteoblasts showed the largest Cx expression repertoire: they were positive for Cx 26, 30, 32 and 43. After transformation into bone lining cells Cx43 expression was lost while in osteocytes only Cx 26 and 32 was detected. A wide variety of Cx reactions was also found in osteoclasts: Cx 26, 30, 32, and 43. In contrast, chondrocytes showed a much more restricted Cx expression pattern. Positive reactions were only seen in hypertrophic chondrocytes of epiphyseal cartilage (Cx 26, 31, 32) and in the proliferative (Cx32) and hypertrophic zone (Cx 26 and 32) of growth plate cartilage. Additionally, Cx31 was strongly expressed in the cellular layer of the periosteum covering areas of intensive bone modelling in the puppies. Age- and microlocation-dependent variations in expression and/or staining intensity were also observed in the other Cx types. These results demonstrated that canine bone cells are capable of synthesizing a wide variety of Cx types during skeletal development, thus propagating different cellular signals via gap junctions in a differentially regulated and requirement orientated manner.

GAL- and VIP-Immunoreactive Nerve Structures in the Ileum and Large Intestine of Pigs with Dysentery

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The morphology, neurochemistry and function of intramural nerve structures in the mammalian gastrointestinal tract are relatively well known, but in normal, healthy individuals. The present study was aimed at investigating the chemical coding of nerve structures in the wall of the ileum and large intestine in normal pigs ($n = 3$) and in pigs undergoing dysentery ($n = 6$). Dysentery was evoked by artificial infection of the clinically healthy animals per os with *Brachyspira hyodysenteriae*. All the animals were deeply anaesthetized and transcardially perfused with 4% paraformaldehyde. The cryostat sections of the intestines were processed for double-labelling immunohistochemistry using antisera against PGP 9.5, GAL and VIP. In the intramural plexuses of the control pigs, the percentage of GAL-immunoreactive (GAL-IR) perykarya varied from 11% (descending colon) to 19% (centrifugal turns of the ascending colon) whereas in the dysenteric pigs, it was distinctly higher, reaching from 28% (ileum) up to 48% (cecum). In the control animals, the percentage of VIP-IR neuronal somata varied from 3% (descending colon) to 19% (ileum). In dysenteric pigs, it was from 6% (descending colon) up to 28% (cecum). In the muscular coat (MC) and mucous membrane (MM) of the normal intestine, very numerous GAL- and VIP-IR nerve fibres were observed. The nerve fibres in the myenteric plexus (MP) were even more numerous than those in the muscular coat while in the outer (OSP) and inner (ISP) submucous plexuses, they were less abundant. In the dysenteric pigs, the nerve fibres found in MC, MP and OSP were less numerous, whereas those observed in ISP and MM were more abundant than those in the control animals. The present results suggest that GAL and VIP are involved in the regulation of inflammatory processes developing in the porcine gastrointestinal tract during dysentery.

Fractures of Ossified Rib Cartilages in the Horse – a Concern of Veterinarians?

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In connection with an osteological study of more than one hundred complete, disarticulated skeletons of the Nor-

wegian Fjord and Døle horse breeds in the collection of the University Museum of Bergen, attention was drawn to the ossified parts of the rib cartilages because of frequent healed fractures transverse to the cartilages. Unossified cartilage had been removed, so the ossified cartilages were available as dry specimens. The skeletons derive from pedigreed stallions as well as some few mares, so both name, year of birth and age are known. Most often the age was between 15 and 30 years, but some few horses were younger, the youngest 5-year-old. Ocular inspection of each skeleton revealed that healed fractures occurred in ossified rib cartilages of every horse, varying from some few instances to > 30 . Skeletons of horses of other races in the museum collection corroborated this. Healing of the fractures was obvious. Four kinds were observed: The first, and probably least sincere fracture appeared as a visible transverse ring, the second displayed callus elevation around the fracture, the third showed dislocation and callus formation, and the fourth concerns pseudoarthrosis, when fractured parts fail to unite during healing. Even a fifth kind may be considered, with circumferential callus formation leaving the interior healed but unfused. X-ray displays the fractures as white, cross-sectional bands of bone repair. However, such fractures appear not to be considered in contemporary veterinary pathology, but the large number of fractures observed in some cases, callus formation and dislocation of fractures in others, indicates that some period following these fractures is likely to have been traumatic to the horse, so that fracture of ossified rib cartilages should not be ruled out concerning horses showing unexplained symptoms of pain.

Teaching of Practical Anatomy in the University of Veterinary Science

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Morphology subjects taught by our department are very important parts of the veterinary medical studies. Functional anatomy, concluded with a final exam, provides knowledge of classical anatomy and is taught for three semesters. The lectures are followed by the plenary sessions where the dissection is done under a camcorder, which shows the students the methods used. Topographical anatomy is taught in the fourth semester – mostly demonstrated on live animals – the visible, palpable structures and the projections of the inner organs on the surface of the body. Introduction to biology summarizes the study of cell biology and also the general knowledge of developmental anatomy. Histology is also given in the second and third semesters, ending with a final exam. However, for the 25 years I have been teaching the anatomy and the subjects taught by our department, I always had the feeling that the students do not pay enough attention or show much appreciation of these studies after the initial enthusiasm.

Anatomical knowledge should also be well known during the later year studies at our university. Until recently our teaching program lacked the repetition of systemic anatomical studies that is mostly for applied science. The very recently introduced elective courses, such as radiological anatomy, surgical anatomy, organogenesis and the anatomy of birds are all scheduled for the late evening hours, but the lecture hall are still always full with students of all grades. The enquirers could find the needed anatomical information in special books and atlases until now. However, practising vets say it takes a lot of time and making uncomfortable difficulties to analyse the books. Because of forgetting the details that are less important for the vet's everyday work it can be hard to find solutions for small but urgent problems. That is why we decided to picturise and make a summary in an edition for demonstrating different elements of certain species of animals. The multimedia CD-ROM was made based on clinical view providing the opportunity for the vet to analysing and studying the part that is the actually subject of his work. After 30 years long break, today there is no graduate or continuing education program without teaching anatomy, which means to give the knowledge to our colleagues applying to their needs also to their everyday work.

The Biology of the Development of the Genital Organs. A Multimedia Teaching Program

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In my presentation, I review the sexual differentiation from the genetic sex until the appearance of the external genitalia and the developmental anomalies to use an animated cartoon. The first critical stage of sexual differentiation occurs at the moment of fertilization, when the genetic sex of the zygote is determined by the nature of the sex chromosome contributed by the sperm. Although an XY zygote is destined to become a male, no distinctive differences between the early development of male and female embryos have been noted. This is accomplished after migration of the primordial germ cell into the early gonad. Because of the early commonality of genital structures, anomalies are the result of abnormal retention or loss of appropriate genital structures. Therefore, most genital anomalies are some form of intersex. During the early differentiation of the

gonads, while the mesonephros is still the dominant excretory organ, the gonads arise as ridge like thickenings (gonadal ridge) on its ventromedial face. Differentiation of the indifferent gonads into ovaries or testes occurs after the arrival of the primordial germ cells. The primordial germ cells arise from the endodermal cells of the yolk. The principal function of the Y chromosome is to direct the differentiation of the presented indifferent gonad into a testis from the sixth week, while two X chromosome are presented the ovaries start to develop, from the 12th week. The next and most obvious phase in sexual differentiation of the embryo is the differentiation of the somatic sex. The early embryo develops a dual set of potential genital ducts, one is the original mesonephric (Wolff) ducts, which persists after degeneration of the mesonephros as an excretory organ, and the another is newly formed pair of ducts called the paramesonephric (Müllerian) ducts. Under the influence of testosterone secreted by the testes, the mesonephric ducts develop into the duct system through which the spermatozoa are conveyed from the testes to the urethra. The potentially female paramesonephric ducts regress under the influence of another product of the embryonic testes, the Müllerian inhibitory factor, a glycoprotein secreted by the Sertoli cells. In genetically female embryos, neither testosterone nor Müllerian inhibitory factor are secreted by the gonads. In the absence of testosterone the mesonephric ducts regress and lack of Müllerian inhibitory factor permits the paramesonephric ducts to develop into oviducts, the uterus and part of the vagina. The next stage is the development of the external genitalia. In very young embryos, a vaguely outlined elevation known as the genital eminence can be seen in the midline, just cephalic to the proctodeal depression. This is soon differentiated into a central prominence (genital tubercle) closely flanked by a pair of folds (genital folds) extending toward the proctodeum. Somewhat farther to either side are rounded elevation known as the genital swellings. From this common starting point the external genitalia of both sex differentiate. If the individual is to develop into a male the genital tubercle, under the influence of dihydrotestosterone, becomes greatly elongated to form the penis and the genital swellings become enlarged to form the scrotal pouches. During the growth of the penis a groove develops along the entire length of its caudal face and is continuous with the slit-like opening of the urogenital sinus. This groove later becomes closed over by a ventral fusion of the genital folds, establishing the penile portion of the urethra. The portion of the urogenital sinus between the neck of the bladder and the original opening of the urogenital sinus becomes the prosthetic urethra. In the female, the genital tubercle becomes the clitoris, the genital folds become the labia mi-

nora, and the genital swellings become the labia majora. The urethra in the female is derived from the urogenital sinus, being homologous with the prostatic portion of the male urethra.

Expression of the PACAP-immunoreactivity in the Lymphoid Organs of the Duck

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Introduction: The interactions occurring between nervous and immune systems are well documented. These interactions involve several types of chemical messengers including hormones, cytokines, classic neurotransmitters and neuropeptides. It has been observed that the lymphoid organs receive a dense peptidergic innervation and immune cells produce neuropeptides. Neuropeptides, in turn, are involved in the regulation of the inflammatory processes and in the maturation of the lymphoid organs. Several studies have demonstrated that the immunomodulatory neuropeptides and their receptors are expressed in the thymus and bursa of fabricius. PACAP is a glucagon/VIP/secretin family peptide. It was originally isolated from the ovine hypothalamus and then it was found in the autonomic nervous system. PACAP is involved in the regulation of the hypothalamic-pituitary function, neurotransmission and neuromodulation. In the immune system, PACAP is expressed in lymphoid tissues of the rat and in the lymphocytes of the duck GALT. PACAP, therefore, could be a messenger of the dialogue between nervous and immune system. It may have a role in the regulation of the inflammatory processes by stimulating histamine and serotonin and modulating the production of the cytokines in immune cells.

Methods: Immunohistochemistry on paraffin-embedded sections of thymus and bursa of fabricius of the duck of different ages by using an antibody anti-PACAP38.

Results and Discussion: In the thymus, PACAP-immunoreactivity was found in lymphoid cells and, with a lesser extent, in epithelial reticular cells. The immunoreactive lymphocytes were primarily observed in the interlobular septa in close vicinity to the interlobular veins. The number of positive lymphocytes increased with ageing. In the bursa of fabricius, PACAP-IR was found in nerve fibres and in a few lymphoid cells. These results suggest that PACAP could play a role in the maturation and involution of these organs and in the immune functions.

Characterization of the Acute Immune Response in the Retina of Borna Disease Virus-infected Lewis Rats

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Borna disease virus (BDV) causes a non-purulent meningoencephalitis and retinitis in a variety of species. In Lewis rats infected intracerebrally with the highly neurotropic BDV the retina is one of the most severely affected central nervous system (CNS) structures. While BDV-induced damage in the brain has previously been shown to be caused by a T-cell-dependent process, the immunopathological mechanisms leading to BDV-induced retinitis, remain to be elucidated. RNA samples from retinæ were subjected to RNase protection assays to detect transcripts of proinflammatory cytokines and chemokines known to be involved in the recruitment of T-cells and macrophages in the CNS. The observed expression profile of proinflammatory cytokines and chemokines, as well as the immunohistochemical detection of $\alpha\beta$ TCR-positive and CD8-positive T-cells in the BDV-infected retinæ, is reminiscent of the situation observed in the brains of Lewis rats during the acute phase of Borna disease. This suggests, that similar immunopathological mechanisms are operating in retinæ and brains of infected rats. This research was supported by grants from the Deutsche Forschungsgemeinschaft (J.S.-PA 615/1-1).

Distinct Fibro-vascular Arrangements in the Equine Periodontal Ligament

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The morphological pre-requisites to sustain pressure loads and tractive forces during mastication are examined histologically in the periodontal ligament (PDL) of cheek teeth from 17 horses. The upper and lower jaws were dissected and perfused via the infraorbital and the inferior alveolar arteries with Bouin's solution or, alternatively, with Karnovsky's solution. After perfusion, the jaws were subdivided with a steel band saw and subsequently with a

diamond saw to obtain PDL specimens from the first molar tooth. After decalcification, specimens were processed for (1) collagen labelling with fluorescein, (2) oxytalan-fibre staining, (3) immunostaining of vascular smooth muscle actin, and (4) TEM examinations. The periodontal ligament (PDL) of equine cheek teeth is composed of two distinct functional units: (1) a collagen-oxytalan-fibre apparatus, (2) a specific vascular system. These elements are associated with each other in characteristic ways: (1) extremely wide, 'ballooned', venules are accompanied by collagen fibres that run parallel to the cemental surface. Plenty of oxytalan fibres are packed between the venous walls and the dental cementum. This is a typical feature for regions of the PDL that are supposed to receive compression (related data concerning entheses, osteoclasts, osteoblasts have been presented by Staszuk and Gasse, 2003, 2004*). (2) Regions that are defined to sustain tractive loads – instead of pressure – are distinguished by the so-called 'anchored' venules. The vessels are 'embraced' bilaterally by thick bundles of dense collagen fibres, which are orientated in an alveolo dental direction. The fibre bundles are directly linked to the venous walls by means of a complex of oxytalan fibres, veil cells, and smooth muscle cells. These different fibrovascular arrangements (a,b) represent morphological adaptations that may be suitable to withstand either compression or traction. The first (a) provides the prerequisites for a 'hemodynamic cushion' to receive pressure forces; the latter (b) is suggested to receive tractive forces by means of an elaborate 'lateral compression' mechanism that shall be presented in detail.

Myosin Heavy Chain Transitions in Dog Skeletal Muscle Fibres during Post-natal Development

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Myosin heavy chain (MHC) isoform expression was established in the masseter, axial and limb skeletal muscles in 18 young dogs from late prenatal to 6 months of age. Results obtained by immunohistochemical procedure using 11 monoclonal antibodies, SDS-PAGE electrophoresis, Western-blotting and enzyme-histochemical reactions were compared to those found in skeletal muscles of adult dogs. In the skeletal muscles of both fetuses in the last stages of development and neonatal dogs, the developmental isoforms MHC-emb and MHC-neo were prevalent. During the first 10 days after birth, the level of embryonic isoform expression declined, as did that of the neonatal isoform

during the second and third weeks. Correspondingly, the expression of MHC-IIa, and later, of MHC-I increased. By 6 weeks of age the presence of MHC-IIx became obvious. This transitional pattern was similar in all muscles except in the rhomboideus, where secondary slow fibres were present before birth. In the other muscles, only one primary, centrally located, slow fibre was present in a fascicle and the secondary slow fibres evolved from developmentally fast fibres between the second and third weeks of age. Our results indicate that timing of muscle maturation depends on its activity right after birth. The fastest developing muscles were the masseter muscle and the diaphragm. They were followed by others, postural and locomotory muscles. A pronounced swap from developmental to adult isoforms was noted between the 4th and 6th week of age. At this age puppies are weaned and begin playing intensively and investigating their surroundings. The results of our study indicate that dog skeletal muscles exhibit an immature morphology at birth and develop relatively slowly compared to those of other domestic animals.

Investigation of Senescence in Canine Fibroblasts

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Introduction and Aims: Primary and secondary cell cultures of canine dermal fibroblasts were used to test the suitability of several senescence-linked cellular markers to characterize the chronological age of dogs. In further studies these markers will be used to test the hypothesis that senescence-linked changes at the cellular level are representative for the observed inverse relationship of body mass and life expectancy in different dog breeds.

Methods: In this study standardized skin samples of Beagles were used to analyse the cell morphology and several senescence-linked parameters as Cumulative Population Doubling Level, Highest Population Doubling Level, Culture Doubling Time and Life Span Completed to determine the replicative potential of fibroblasts. The samples were divided by the age of the donors into three groups ('young', 'middle' and 'old').

Results: There are notable differences between the replicative potential of young and old Beagles at the cellular level. Preliminary results show that in fibroblast cultures obtained from the young group the highest population doubling level is higher and the culture doubling time is shorter than in cultures of the old group. Cultures from the young and middle-aged group cannot be distinguished by the measured parameters. For final results the complete statistical analysis of sampled data has to be carried out.

Conclusions: Our results demonstrate a relationship between the chronological age of the skin donor and the replicative potential of the dermal fibroblasts. Therefore

cellular senescence-linked markers could serve as suitable tools to compare the age of individual dogs of different breeds.

The Prenatal Development of the Canine Claw

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Introduction: The mammalian digital end organ has developed in three basic functional forms, i.e., claw, hoof and nail. Whereas detailed information on the ontogeny and phylogeny of the equine and bovine hooves, and the human nail, respectively, are on hand, such data are still not available for the canine claw. Because phylogenetically the carnivore claw is considered as a primary form of digital end organ, data on the organogenesis of the canine claw will complement current knowledge on onto- and phylogeny of the carnivore claw, and digital end organs in general.

Materials and Methods: Light and electron microscopy of 18 canine fetuses of 40–150 mm crown-rump-length and five puppies aged 1 day–4 weeks post-partum.

Results: The development of the canine claw can be divided into four periods. The first stage is initiated by the differentiation of the connective tissue layers of the claw anlage. The second stage is characterized by formation of the specific claw shape, and the third stage by further regional function-related modifications resulting in the development of the five different segments of the digital end organ. The fourth stage proceeds post-natally when the influence of increasing body weight and weight bearing leads to the final functional adaptation of the papillary body.

Conclusion: The organogenesis of the canine claw resembles that of the feline claw in many aspects, but also shows some specific deviations. In particular, the development of the dorsal ridge – characteristic for the carnivore claw – differs between both species. The results of our study therefore raise the question whether the current hypothesis on allocation of the different segments of the canine claw has to be reconsidered.

Comparative Morphology of Trigeminal Ganglion in Some Species of Vertebrates

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The reported examinations were carried out on turtles, Japanese quails and chinchillas of either sex. The trigeminal

nerve was studied with the modified thiocholine method and routine histological techniques. The trigeminal ganglion of the turtle is a single, compact structure connected with the trigeminal nerve. Analysis of consecutive sections showed a presence of ganglionic neurocytes only at the medial side of the trigeminal nerve. A small number of nerve cells and large number of fibres is characteristic for the investigated structure. The trigeminal ganglion of Japanese quail is a structure connected with two branches (ophthalmic and mandibular) of the trigeminal nerve. Both parts of the ganglion distinctly differ in their morphology. The part connected with the ophthalmic branch – V₁ forms a thickening of the nerve, which becomes narrow approaching the orbital cavity. The second part connected with the mandibular branch – V₂ is of regular shape. This part of the ganglion is located in the vicinity of a delicate maxillary branch. Histological investigations allow us to observe a characteristic arrangement of neurocytes with only one layer of nerve cells on the surface of the ganglion. The trigeminal ganglion in chinchilla is a single, irregular cluster of neurocytes closely connected with trigeminal nerve where this nerve divides into ophthalmic and maxillary nerves. Agglomerations of neurocytes allow us to distinguish two parts of trigeminal ganglion. From the medial aspect, the nasal part of ganglion is connected with this part of the trigeminal nerve, which is not distinctly divided into ophthalmic and maxillary branches. The caudal part of the trigeminal ganglion is connected with the mandibular branch of trigeminal nerve. These two parts are separated from each other by the motor radix of the trigeminal nerve. From the lateral side all parts of the ganglion form one structure and it is difficult to distinguish precisely different cell aggregations corresponding to the particular branches of trigeminal nerve.

Computed Tomography (Ct) of the Liver of the Dog by a Six-generation CT Scanner and Intravenous Contrast Medium

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By computed tomography the liver has an intermediate degree of attenuation, with a CT value (HU) between 70 and 75. However, this number may vary according to the generation of the apparatus, the use of contrasts medium, the efficiency of the detector, etc. The purpose of the present work was to evaluate the liver of the dog and

related structures by means of a six-generation spiral CT scanner (Toshiba Ex Vision [[Au: please provide manufacturers details for 'Toshiba Ex Vision': city, state code (if USA) and Country]]), and the previous injection of intravenous contrast medium via cephalic vein. Four half-breed mature dogs (4–6 years, 15–20 kg) were used, two males and two females. The dogs were anaesthetized and kept over a stretcher in sternal recumbency. To evaluate the liver two different CT windows with different window width and window level were applied: soft tissue window (WW 652; WL -34) and mediastinal-vascular window (WW 352; WL 220). The CT images obtained allowed an excellent resolution at the level of the hepatic lobules, the gall bladder, the porta vein, the hepatic artery, the caudal cava vein, etc. The attenuation shades of these structures were very similar in the two used windows, however a better contrast was obtained with the soft tissue window.

Canine Endothelial Remodelling During the Estrus Cycle: an Overview

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The present study in 58 dogs aimed to verify the changes of the canine endometrial surface epithelium during the estrous cycle using electron microscopy and to investigate the roles of apoptosis, cell proliferation and matrix metalloproteinases in canine endometrial remodelling using immunohistochemistry. Possible correlations between these phenomena and serum levels of estradiol-17 β and progesterone were assessed. Prominent scanning electron microscopic changes of the canine endometrium were observed during early metestrus and included a variable height and number of microvilli of the surface epithelium. Using transmission electron microscopy, pronounced features of the surface epithelial cells were noticed in late metestrus and anestrus, during which a highly vacuolized cytoplasm and deformed nuclei due to fat accumulation were present. In these stages no prominent signs of cell death were noticed and also very low apoptotic numbers were observed using immunohistochemistry. Cell proliferation in the endometrial surface epithelium seemed to be in balance with apoptosis. Most prominent features of apoptosis and proliferation were found in the basal glands and to a lesser extent also in the crypts. In contrast, only low amounts of apoptosis and proliferation were observed in the stroma, in which high collagenase-1 scores coincided with a decreased amount of collagen in the deep part of the endometrium and a massive increase of glandular cells during early metestrus. No cyclic differences were noticed for the expression of TIMP-1, the collagenase-1 inhibitor. Finally, no or only low correlations were found between apoptosis, cell proliferation and MMP-1 expression in the

canine endometrium, and the serum levels of estradiol-17 β and progesterone. In conclusion, these findings indicate that there is no desquamation of the canine endometrial surface epithelium at the end of the estrous cycle, but a regular turnover of cells throughout the estrous cycle. Furthermore, our data also suggest a role for MMP-1 in cellular proliferation. This study was supported by the BOF (#011B4101) of Ghent University.

The Ovine Tonsilla Lingualis: Scattered Lymphocyte Aggregations Rather Than a Well-defined Tonsillar Structure

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Recently, it has been hypothesized that the BSE-causing agent PrPSc can infect sheep, making these animals a possible source of infection for man. Furthermore, experimental studies in inoculated sheep have demonstrated that PrPSc is not only present in neuronal tissue, but also in lymphoid tissue. Therefore, ovine tonsils are considered as high-risk material and consequently have to be removed after slaughter. The present study is part of an elaborate project in which anatomical, histological and immunological aspects of ovine tonsils are studied within the framework of BSE/TSE research. The tongues of two sheep of 9-months-old were collected immediately after slaughter. The lingual root between the base of the epiglottis and the torus linguae was dissected and the 4.5 cm wide and 8 cm long sample was cut into 24 pieces which were subsequently fixed in 3.5% formaldehyde and embedded in paraffin. From each tissue block, ten 8- μ m-thick sections with 1000 μ m interval were cut and stained with eosin–haematoxylin. Microscopic analysis revealed that lymphocytes are mainly present at the lateral side of the tongue forming numerous small aggregates, while in the more central part the amount of aggregates was lower. In both lateral and central parts, the aggregations were located in an area ranging from 1–2 cm till 6 cm rostral to the epiglottis. These lymphatic clusters were in general smaller than the lymphatic nodules seen in the other pharyngeal lymphoid tissue, and a differentiation into secondary nodules was never observed. The epithelium covering these lymphocyte aggregations was a stratified squamous epithelium without infiltration of lymphocytes. It was concluded that the tonsilla lingualis in sheep consists of scattered lymphocyte aggregations without forming a well-defined tonsillar organization. Further studies are planned to examine the tongues of more sheep and to determine the presence of dendritic cells, follicular dendritic cells and different B- and T-cell populations. This

study was supported by the BOF project grant #011B4101 of Ghent University.

Dissociated and Explant Cultures of Myenteric Neurons of Pig Small Intestine

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The gastrointestinal tract of pig is a good and available (slaughterhouse material) model to study the enteric nervous system in relation to nutrition, inflammation, development and disease in general. In order to investigate the responses of the enteric nervous network to a variety of stimuli, e.g. growth factors, hormones, extracellular matrix components, but also noxious compounds in a controlled manner, an *in vitro* experimental set-up is most appropriate. Methods to obtain *in vitro* cultures of enteric neurons of rodents, chicken and human are well described. This study attempted to use these methods on pig material. The muscle layer containing the myenteric plexus was dissected from pieces of fetal, neonatal and adult pig jejunum. They were rinsed in Hanks basal salt solution (BSS) in which antibiotics were added. Pieces of $\pm 25 \text{ mm}^2$ were transferred to BSS with 1 mg/ml collagenase and 1 mg/ml trypsin inhibitor and incubated for 60 min (5% CO₂, 90% RH). Subsequently, they were vortexed for 20 s and ganglia were selected. The procedure was repeated ± 4 times. Myenteric ganglia could then be plated or further dissociated in 1 mg/ml trypsin in BSS for 30 min in the incubator. Afterwards, the dissociated ganglia were centrifuged (5 min 1500 rpm) and the trypsin-solution was replaced with Dulbecco's minimal essential medium (DMEM). The explants or dissociated myenteric ganglia were transferred on non-coated or coated (poly-L-lysine, laminin, fibronectin, collagen or extracellular matrix gel) cover slips. After incubating for 45 min they were topped with 1 ml of DMEM with antibiotics and with or without fetal calf serum (5%). Medium was replaced twice a week. Using immunohistochemistry, both neurons (PGP9.5) and glial cells (S100) could be identified in both culture types. When cultivated under harsh conditions, the dissociated cultures gave rise to neurosphere-like bodies, containing neurons and glial cells. Thus, the digestion and dissociation technique is applicable to pig material.

Assessment and Development Centre and E-Learning in the Teaching of Anatomy

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Today's students belong to an interactive generation and receive information through multiple channels. Curricula in veterinary medicine are changing due to trends such as 'student centred education' and 'competence based learning'. Taking in account these changes, we were encouraged to rethink the way in which we teach and assess the knowledge of veterinary anatomy. The amount of dissection (90 h) and lecture hours (90 h) has not changed, but the way they are used and how the 'knowledge' is assessed has been revised. First, goals were defined: students have to acquire knowledge and insight in the structure, function and relationships of anatomical structures of various species. They have to be able to observe, palpate and expose the structures properly. Additionally, they must attain general skills, e.g. communication. Secondly a new learning environment was developed so students could accomplish these goals. The three main components, constituting this new environment were: the introduction of an assessment and development centre (ADC) as a means of assessing students as well as giving them feedback, the incorporation of e-learning and the raise of student-lecturer interactions during lecture hours. During the ADC the students have to go through three set-ups: they demonstrate structures, they name structures and they 'solve' a problem (e.g. cut tendon). They report their findings both verbally and written. Using checklists the student's practical skills are assessed during the dissections and during the ADC. Students receive feedback at four different times. An E-learning environment was devised, giving the students the possibility to consult photographs of the dissections, to discuss the course on a discussion board and to take self-assessment tests to monitor their progress. During the student-lecturer interactions, photographs, clinical examples, are discussed using various teaching methods, resulting in a more 'active' approach to the learning content. Students are enthusiastic about this way of teaching although they experience it as difficult. Student scores seem not different from previously. In the future the practical checklists and the ADC will be optimized.

Culture of Differentiated Adult Rabbit Auricular Chondrocytes

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Chondrocytes dedifferentiate to a fibroblast-like phenotype on plastic surfaces. Dedifferentiation is reversible if these cells are then cultured embedded in gels as alginate, agarose or collagen. Chondrocytes cultured in suspension on a non-adherent surface are also known to form aggregates of differentiated cells. The knowledge of chondrocyte behavior in culture is relevant for tissue engineering purposes. In this report we describe a simple method to culture differentiated or redifferentiated rabbit auricular chondrocytes on plastic surfaces with a stable phenotype. When chondrocyte aggregates formed in suspension are next seeded on plastic surfaces, most of them attach to the plastic as round or polygonal cells, and this morphological differentiation, confirmed by the presence of type II collagen, is stable for long culture periods. We also report that the addition of aggregates to monolayer cultures of dedifferentiated chondrocytes results in their redifferentiation, as is shown by their morphological changes and the synthesis of type II collagen. Therefore, this simple method can be useful for the study of chondrocyte behavior on plastic surfaces and for redifferentiating previously proliferated chondrocytes in tissue engineering techniques. Furthermore, these results demonstrate that, in addition to culture conditions such as cell isolation method or cell-density, chondrocyte behavior on plastic depends on the presence or absence of aggregates resulting from the dissociation process.

Histamine and VIP-positive Mast Cells in Renal Blood Vessels of Domestic Swine

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Introduction: Histamine and vasoactive intestinal polypeptide (VIP) are well known biogenic amines (substances), which rapidly elute from the mast cell granules under physiological conditions. They play an important role in the smooth muscle cells motility.

Aim: The aim of the study was to investigate histamine- and VIP-positive mast cells in the wall of pig kidney blood vessels in concern to their role of motility of smooth muscle cells and renal haemodynamics, respectively.

Material and Methods: Extra- and intraorgan renal blood vessels taken immediately after slaughtering of eight 6–8 months pigs were studied. Tissue samples were immersion fixed in 0.05 M cacodylate buffer, pH 7.2, containing 2% paraformaldehyde and 0.2% glutaraldehyde for 5 days at 4°C. After fixation, tissue blocks were rinsed in 30% sucrose in distilled water at 4°C for 1–3 days. Cryostat sections (20–30 µm thick) were prepared and thawed in 0.05 M cacodylate buffer, pH 2, overnight. Immunocytochemical procedure started for free-floating sections which were incubated in 1.2% hydrogen peroxide in methanol for 30 min, and rinsed in 0.1 M phosphate buffered saline (PBS), pH 7.4, for 15 min. Afterwards, sections were incubated with the primary antibodies in a dilution of 1:100 (histamine) and in vasoactive intestinal polypeptide (VIP) 1:1000 for 24 h at room temperature (RT), then rinsed in 0.1 M PBS, pH 7.4. Incubation succeeded: first, with biotinylated anti-rabbit (histamine) and anti-mouse VIP antibodies in a dilution of 1:20 for 4 h, and rinses in 0.1 M PBS, pH 7.4, and in 0.05 M Tris-HCl buffer, pH 7.5, for 10 min. Peroxidase activity was localized by using a mixture of 3 mg 3,3' diaminobenzidine (DAB), in 15 ml 0.005 M Tris-HCl buffer, pH 7.5, and 36 ml 1 % hydrogen peroxide for 10–20 min. Finally, a rinse in 0.01 M PBS, pH 7.4, was undertaken.

Immunochemicals: The antibodies used were rabbit anti-histamine (H7403) and rabbit anti-vasoactive intestinal peptide (VIP), or (PEPA41) (Sigma, Saint Louis, Missouri, USA).

Results and Discussion: It was established that VIP-positive mast cells were located in the media and in the adventitia of large arterial vessels and renal vein, as well. Histamine positive mast cells were found mostly in the intima of renal artery and in the other layers of the wall. Both VIP and histamine mast cells were observed next to small arteries and veins in the fat tissue, around extraorgan blood vessels. Our finding suggest to mast cells participation in the smooth muscle cells motility and the maintenance of local homeostasis, as well.

Characteristics of Claw Horn Microstructure and the Effect of Flooring Systems on Horn Quality

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Introduction: Interactions between horn quality and flooring systems significantly impact on claw health. Poor horn

quality is associated with reduced milk yield, painful lesions or slaughtering. Horn quality is determined by horn cells, intercellular cement and hoof horn architecture, i.e. the structure and arrangement of the horn tubules. Soft and elastic flooring systems have been reported to have a positive influence on the microstructure of hoof horn.

Aims: We wanted to correlate the microstructure of claw horn to the horn quality in a large number of samples and detect influences of the flooring system on hoof horn microstructure.

Methods: Samples were obtained from two different studies: 40 cows from Hohenheim/Germany kept on slatted floor and subsequently on rubber mats for 4 months; 1000 cows from 11 farms in Luxembourg, kept on concrete floor. From all samples 7 μ m-sections were cut and stained with the PAS-reaction. Morphometry was done with the LuciaG[®]-software (Nikon GmbH, Düsseldorf, Germany) and statistical analysis was performed with SPSS[®] (SPSS Inc., Chicago, IL, USA). Parameters measured were area, minimal and maximal diameter of both medulla and cortex of horn tubules. Statistical analysis included: tubules per visual field, area per visual field, total medulla area, total cortex area and the limiting values of diameters.

Results: Luxembourg samples showed consistency of tubule-parameters between both dates; all farms showed an average number ranging from 13.36 to 14.79 tubules per visual field. Hohenheim horn from the hard flooring showed 19.9 and horn of the same animals kept on rubber floor showed 20.5 tubules per visual field. Maximum diameter of hard-floor horn is 55 μ m whereas 45 μ m in animals kept on rubber floor.

Conclusions: The results show that genetic/individual differences take smaller influence on horn quality than housing systems. These outcomes provide evidence for the economical and animal welfare benefit of soft and elastic flooring systems. Further research will focus on correlation of biodata, such as physiological state, lactation, age, and claw horn microstructure. This work was supported by Biopsytec/Rheinbach.

Vascularization of the Fleshy Comb in the Domestic Chicken

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Up to now little is known about the vascularization of the chicken fleshy comb (crista carnea). In order to evaluate the vascularization of the crista carnea of the cock (breed White Leghorn), corrosion casts were created by injecting Plastoid into the internal carotid as described by Schummer (1951). Specimens were investigated by stereomicroscopy and scanning electron microscopy (SEM). Generally the dermis is highly vascularized by two capillary networks, which are localized beneath the epithelium

and beneath the dermal papillae. The dense subepithelial network is characterized by the presence of sinusoid vessels. In the subcutaneous plexus numerous arteriovenous anastomoses of different types occur. Additionally there are arteriovenous anastomoses between the main vessels reaching the indentations of the comb. Our results show the presence of superficial and dense capillary networks and arteriovenous anastomoses are the anatomical basis for the functions of the chicken comb in mating behaviour and thermoregulation.

Reference

Schummer, A. 1951: Simplified method for plastoid corrosion. *Anat. Anz.* **98**, 288–290.

Occurrence and Distribution of Lactate Dehydrogenase in Knee Joint Cartilage in Healthy and Osteoarthrotic Dogs

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According to clinical studies, degenerative diseases of canine joints lead to higher lactate dehydrogenase (LDH) levels in synovial fluid. The goal of the present study was to examine the intraarticular distribution of LDH in healthy and osteoarthrotic knee joints in order to identify possible sources of LDH in synovial fluid. As synovial LDH concentrations neither correlate with the number of leukocytes nor with synovitis, our investigation focused on the articular cartilage. Samples from healthy and osteoarthrotic knee joints were fixed and processed for transmission electron microscopy (TEM), immunohistochemistry (IHC), and immunocytochemistry (ICC). In addition, fresh cartilage samples were investigated cytochemically by the tetrazolium-formazan reaction. Analyses of blood and synovial fluid samples were used to confirm the absence of inflammatory disease. Morphology of articular cartilage was assessed macroscopically and by means of TEM. IHC revealed highest levels of LDH in chondrons and a diffuse labelling of the matrix with a distinctive decrease in signal from superficial to deeper cartilage layers. Ultrastructural localization by ICC showed LDH to be present in the cytoplasm of all chondrocytes and confirmed the density gradient in the matrix. Labelling was absent from nuclei and from pericellular rims. Cytochemistry confirmed the distribution pattern and, thus, expanded our findings beyond immunological evidence by providing proof of enzymatic activity of LDH in articular cartilage. The present results indicate that LDH is transferred from chondrocytes to the

cartilaginous matrix. We suggest, therefore, that LDH found in synovial fluid originates from the articular cartilage and that osteoarthrotic processes promote LDH release from the cartilaginous matrix.

Structure of the Seminal Pathway in the European Chub (*Leuciscus cephalus*)

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The European chub, *Leuciscus cephalus* (Cyprinidae), is a fish species employed in bioindication for the effects of endocrine disruption chemicals (EDC) upon the reproductive system. In applied histopathology, a useful tool in aquatic ecotoxicology, detailed knowledge on the intact structural integration of an organism is an indispensable prerequisite for the diagnosis of pathological alterations. This is particularly evident for gonadal structures in bioindication for the effects of EDCs interacting with the hypothalamus-pituitary-gonadal-liver axis. However, little is known about the histology of *L. cephalus* until now, especially the extratesticular duct system is widely disregarded. Therefore, in the present study the testicular efferent duct system was described for three phases of testicular development (pre-spawning, spawning, post-spawning). Testicular main ducts were analyzed by means of histology and transmission electron microscopy. Additional techniques applied were lectin histochemistry to determine the secretory activity of epithelial cells, as well as immunohistochemistry to demonstrate cell proliferation activity and muscle actin to show distribution and amount of contractile cells. Remarkable variations of the extratesticular ducts mainly concerning the epithelium, were observed during the reproductive phases. Contribution of the main ducts' epithelia (secretion and cellular detachment) and of degenerating spermatocytes, as sources of carbohydrates and phospholipids, to the seminal fluid were confirmed. Epithelial cell proliferation and secretory activity were dependent on the spawning phase of the specimen. Lectin binding experiments in the present study demonstrated sugar residues (mannose, glucose, galactose, N-acetyl-glucosamine) in the epithelial cell

cytoplasm and apical glycocalyx, which is important in cell recognition. In the stroma of the testicular efferent ducts, a high number of cells expressed muscle actin and tropomyosin indicating the ducts' contractile potential for the transportation of seminal fluid towards release. Adjacent to these contractile cells, numerous nerve fibres were found indicating neuronal control of sperm fluid flow.

Something Different: the Articulation Genus of African Elephants (*Loxodonta africana*)

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In elephants, the limb posture in general and the angle of the knee joint in particular differs to those of most mammals. Although previous studies on the osteology of elephants are available, knowledge of the structure of their joints is sparse. The aim of this study was to provide an anatomical description of the African elephant's knee joint. Hindlimbs of six juvenile individuals of *Loxodonta africana* were dissected at the Department of Anatomy and Physiology, University of Pretoria (Onderstepoort, South Africa). All animals had lived in the Kruger National Park and were shot due to the regular culling program during the 1990s. The tight joint capsule encloses both the *articulatio femorotibialis* and the *articulatio femoropatellaris*. The articular surfaces of the tibial and femoral condyli show a high grade of congruency. The *meniscus lateralis* and especially the *meniscus medialis* are very narrow merely resembling folds of the joint capsule. A *ligamentum meniscofemorale* is present. The *Ligamentum cruciatum craniale* attaches to the caudomedial surface of the *condylus lateralis femoris*, passes the proximally flattened surface of the *eminentia intercondylaris tibiae* and inserts on a craniolateral impression of this eminentia. The *ligamentum cruciatum caudale* courses far caudally from the lateral surface of the *condylus medialis femoris* to the area *intercondylaris caudalis*. The tendon of origin of the *m. popliteus* passes caudoproximally over the *meniscus lateralis* and a recess of the joint capsule lies between this muscle and the *condylus lateralis tibiae* or the *caput fibulae* respectively. The *ligamentum collaterale mediale* and the smaller laterale are embedded within the thick fascial layers surrounding the knee. Retinacula patellae are not discernible. The *ligamentum patellae*

attaches to the tuberositas tibiae and the distal tendon of the *m. rectus femoris* lies cranial to this ligament. It is likely that the structural peculiarities of the elephant's knee stand in close relationship with limb posture and weight bearing.

'Sonobasics', a Computer-based Program for Self-Education on Sonographic Anatomy of the Urinary Bladder and Kidneys in Small Animals

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While the macroscopic anatomy of the urinary bladder and the kidneys is well known, the ultrasound anatomy of these organs still remains difficult for physicians. The objective was to develop a multimedia computer-based program that helps to reduce the time necessary for learning ultrasonography in dogs and cats. Therefore, the program contains many ultrasound films and images, which are clearly marked. Additionally, colourful diagrams explain the ultrasound examination. Although the urinary bladder is comparatively easy to scan the surrounding structures are often confusing. Many of these structures such as the neighbouring fat are not especially mentioned in macroscopic anatomy. The program clearly demonstrates the advantages of the ultrasound method. For example, the physiological urinary jet into the urinary bladder is compared to urinary jet after application of a diuretic agent (furosemide). The ultrasound anatomy of the kidneys and the related structures are described in detail and illustrated by anatomic preparations and sketches. The comprehensive visual and audio representation of the Doppler ultrasonography of renal vessels is intended as an advanced training for already experienced physicians. The structure of the program is designed with regard to didactic and ergonomic aspects. The user can go through the program step by step or move freely using a navigation bar, visual links (viewer) or one of the various links. Nevertheless, all pages are clearly marked to keep track. The structure of the program is similar in all sections, starting with the anatomical topography and the patient preparation. After a detailed description of the scanning procedure the resulting ultrasound images are demonstrated. The last part includes different techniques of Doppler ultrasonography. The self-explaining multimedia structure of the computer-based program enables an autodidactic training of the ultrasound b-mode and the Doppler ultrasonography of the urinary tract and the kidneys for physicians with little experience as well as for advanced investigators.

Immunohistochemical Demonstration of Lumbar Intervertebral Disc Innervation in the Dog

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Low back pain is a common ailment in dogs, particularly in specific breeds as for instance the German shepherd dog. A number of structures such as facet joint capsules, ligaments, dorsal root ganglia, periosteum, vertebral endplate, and meninges have been associated with this condition. However, in a substantial proportion of all cases, the origin of pain remains obscure. A further structure often being involved in vertebral column disorders is the intervertebral disc. However, the presence of nerves is a precondition for pain sensation and, consequently, structures lacking sensory innervation can be left out of consideration as a cause for low back pain. Nerve fibres have been demonstrated at the periphery of intervertebral discs in man, rabbit and rat. With regard to the dog, however, extent of intervertebral disc innervation still is disputed. The goal of the present study, therefore, was to assess previous findings based on silver impregnation. Protein gene product (PGP) 9.5 was used as a marker for immunohistochemical examination of serial transversal and sagittal paraffin sections of lumbar discs from adult dogs. This technique revealed nerve fibres to be confined to the periphery of the intervertebral discs. Thus, even limited pathological processes affecting the outer layers of the intervertebral disc are prone to cause low back pain.

A New Look on the Origin of the Gonad and the Müllerian Duct: the Sturgeon (Acipenser) as a Model for Vertebrate Urogenital Development

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The origin of the vertebrate gonad and the Müllerian duct are still a matter of debate. According to the majority of recent textbooks, the gonad is a product of the proliferating coelothelium and therefore derived from the mesoderm of the lateral plate region. The Müllerian duct grows parallel to the Wolffian duct, but it is not clear to what extent the latter contributes actively to the development of the former. In the last decade, we reinvestigated early gonadogenesis and Müllerian duct development in a number of vertebrate model species using various mor-

phological techniques (TEM, SEM, immunohistochemistry). The conclusion of our studies is that rudimentary or regressing nephrostomial tubules, particularly cells of their nephrostomes, must be regarded as the immediate precursors of the somatic cells of the gonadal crest and the Müllerian infundibular field. According to this concept, both structures are derivatives of the intermediate mesoderm. Nephrostomial tubules are regular components of the primitive pro- and mesonephros. They connect the nephric tubule or the nephric corpuscle to the coelomic cavity and open into the latter by means of a funnel-like mouth, the nephrostome (coelomostome). In the larval sterlet, *Acipenser ruthenus*, short, segmentally arranged nephrostomial tubules with well-developed nephrostomes are present in the region of the cranial opisthonephros. Cells of the medial nephrostomial lips proliferate, surround the germ cells that have accumulated in this location and form a continuous gonadal crest. Cells of the lateral nephrostomial lips proliferate also, spread out on the coelomic surface, replace the original flat mesothelial cells over the Wolffian duct and the cranial opisthonephros and form the Müllerian infundibular field. At about 28 days, a flat pocket begins to invaginate the infundibular field. This pocket is the primordium of the Müllerian ostium abdominale. The findings in *Acipenser* can be generalized and transferred to other vertebrates.

The Isolated Haemoperfused Distal Cow Limb as an Experimental Model for Studies on the Pathomechanisms of *Bovine laminitis*

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Introduction: The pathomechanisms of bovine laminitis are still poorly understood. Isolated perfused organ models are promising tools to study the biological mechanisms that lie behind diseases. They are preferable for ethical and financial reasons but also for experimental necessities.

Aims: To develop an isolated haemoperfused distal cow limb model, basing on an already existing porcine limb model used in experiments studying the effects of bioactive molecules.

Methods: Isolated limbs and blood were obtained from routinely slaughtered cows and subsequently perfused in the laboratory of a biotech company (Vitro-Tec, Berlin) with expertise in organ perfusion. A standard perfusion apparatus enables perfusion for up to six hours under close to physiological conditions. Perfusion pressure and flow were calculated based on values available for horses and pigs. The oxygen saturation was kept at 100% and

the pH between 7.35 and 7.45. Glucose was added as nutrient. The lactate concentration was measured as a parameter for oxygen supply and free haemoglobin as an indicator for haemolysis. Potassium content and pressure served as vitality indicators. Additionally, morphological criteria of tissue integrity were assessed using light and electron microscopy.

Results: A constant flow of 150–200 ml/min was optimal. The resulting pressure ranged from 30 to 117 mmHg. Light microscopy revealed, with exception of a few thromboses, no cellular damage after 4 h. The perfusion was successfully extended to 5 h. Then first pressure related alterations including vascular dilatations and cellular degeneration occurred in the dermal lamellae but not in the other regions. Potassium never exceeded physiological values.

Conclusion: The model is now available for experiments. Equipment and technical implementation have been standardized and described in a standard operating procedure. The model will be challenged by bioactive molecules such as lactate, neuropeptides and chemokines. Ischaemic conditions will be simulated and studied. This work was supported by the European Communities under the Lame Cow Project QLK5-CT-2002-00969. The authors are solely responsible and the work does not necessarily represent the opinion of the European Communities.

The Cartilage Canals of the Epiphyseal Growth Cartilage and their Role in the Formation of Lesions of Osteochondrosis

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Cartilage canals are temporary vascular structures within the growth cartilage. They were described as early as 1736 and the first perfusion study performed already in 1743. In its most complex form, the vascular structure consists of an arteriole, originating in the perichondrial plexus and branching out into an anastomosing network of capillaries. The capillaries circle back along the arteriole and rejoin into a venule that follows the course of the arteriole back to the plexus. The canals regress with age in a process designated as chondrification, where the content of the canals is replaced by cartilage. The process of chondrification is considered physiological; however, premature regression has been associated with the formation of lesions of osteochondrosis. The purpose of the present study was to gain further insight into the nature of chondrification

and the initial steps in the pathogenesis of osteochondrosis and the relationship between these events and age, growth rate, weight and femoral shape of the individual animals. This involved studies of the distribution and pattern of regression of the cartilage canals. The articular-epiphyseal cartilage complexes of the distal femur of 48 half-siblings of the Norwegian Landrace pig breed were studied, combining techniques of perfusion and tissue clearing with histological and immunohistological methods. The results of this study indicate that the physiological regression of the cartilage canals is an orderly and time-dependent process in which the endothelial cells at the termination of the vessels die by apoptosis, while perivascular cells differentiate into chondrocytes that fill in the cartilage canal with matrix. However, as the ossification front advances, the deepest portion of some of the canals form anastomoses with vessels of the subchondral bone, giving the axial portion of the growth cartilage a new source of blood supply. Lesions of osteochondrosis, typically occurring in the axial growth cartilage, are characterized by necrosis of the portion of the affected cartilage canal distal to a point of interruption, with subsequent necrosis of adjacent resting zone growth cartilage. This interruption occurs at the junction between cartilage and bone where anastomoses are formed between cartilage canal vessels and subchondral vessels.

Bovine Trophoblast Cells: Establishment of Cell Culture System and Comparison to *In Vivo* Qualities

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Cell culture offers a specifically good understanding of single factors during complex signal transduction pathways, and cell-cell interactions can be examined in small scale. In the bovine synepitheliochorial placenta

the peculiarity of 'restricted' trophoblast invasion is found as non-polarized trophoblast giant cells migrating into the uterine epithelium. Specific expression patterns of integrin receptors in trophoblast cells have already been shown in relation to the differentiation of the haemochorial placenta of humans and rodents. For evaluation of integrin function during the precisely controlled trophoblast invasion in cow, it is important to develop a model system, where primary cell cultures from cotyledons are compared to *in vivo* trophoblast cells. Uteri of pregnant cows were gathered during routine slaughtering. In parallel, placentomes were snap-frozen for the production of cryosections, and other placentomes were manually separated into fetal cotyledon and maternal caruncle. Dissociation of the cotyledonary cells was accomplished with collagenase I, and cells were cultivated in Quantum 286 (PAA Laboratories GmbH, Linz), a special medium for epithelial cells, containing 10% fetal calf serum. A population of mainly epitheloid cells was achieved by controlled trypsin/EDTA treatment, because of different adhesive forces of fibroblasts and epithelial cells to the substrate. Both, confluent growing cells and tissue sections, were investigated by indirect immunofluorescence with antibodies against cytokeratin, integrin subunits α_6 , β_1 , α and β , and the extracellular matrix (ECM) components laminin and fibronectin. The epithelial character of the propagated cells was proven by their distinct immunostaining for cytokeratin. *In vivo* expressed integrin subunits α_6 and β_1 were found colocalized along cell-cell borders and towards the flask ground. The corresponding ECM ligand laminin occurred in both, cells and tissue. In tissue sections fibronectin and subunits α and β_3 were not detected in trophoblast cells, however, *in vitro* these cells displayed a weak immunoreactivity. Summarizing, a method for the cultivation of bovine trophoblast cells was successfully established. The comparison of *in vivo* and *in vitro* expressed integrins and ECM showed that the cells in part maintained their *in vivo* attributes, and therefore are suitable for the study of cell-cell interactions.

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